Centrality Measures

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February 4, 2020

Network

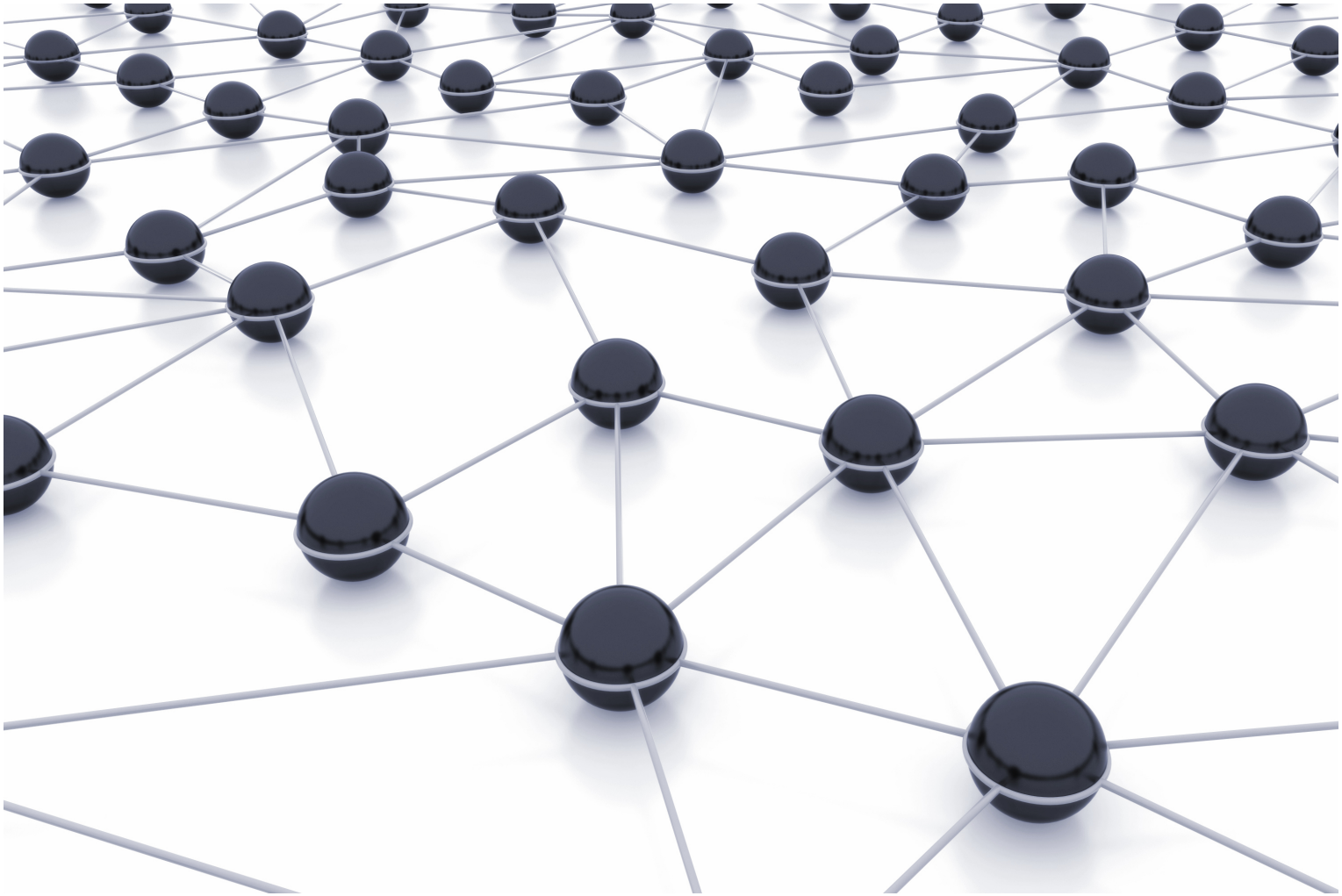


Figure 1: Network (Source :- Google Search)[1]

[1] http://www.imt-ase.com/imt-ase-co-ltd-information-technology/

Biological Networks

(a) Brain (Neural) Network

(b) Circulatory Network : Veins and Arteries

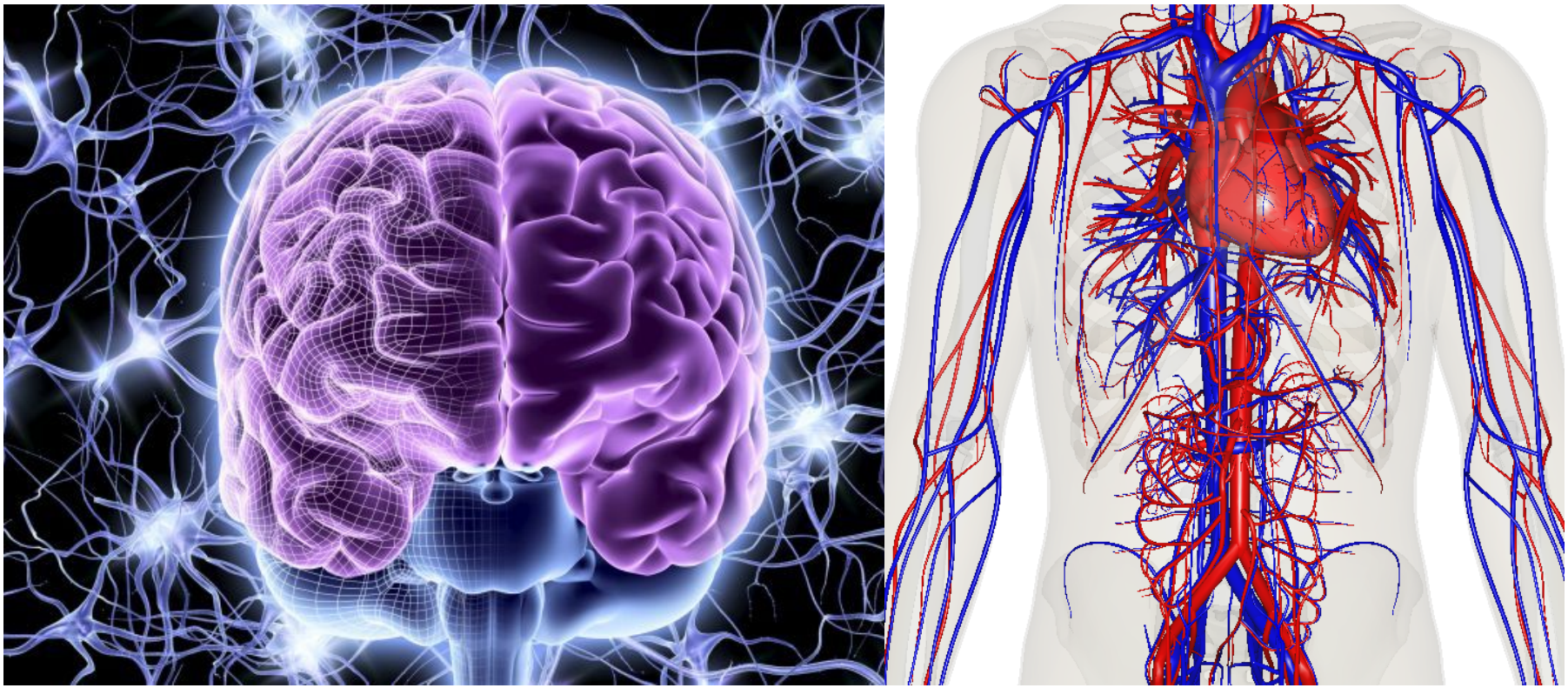


Figure 2: Two biological networks (Source :- Google Search)[2]

[2] http://www.dailymail.co.uk/sciencetech/article-2344398

https://www.nhtworld.com/health-of-the-cardiovascular-system

Chemical Network

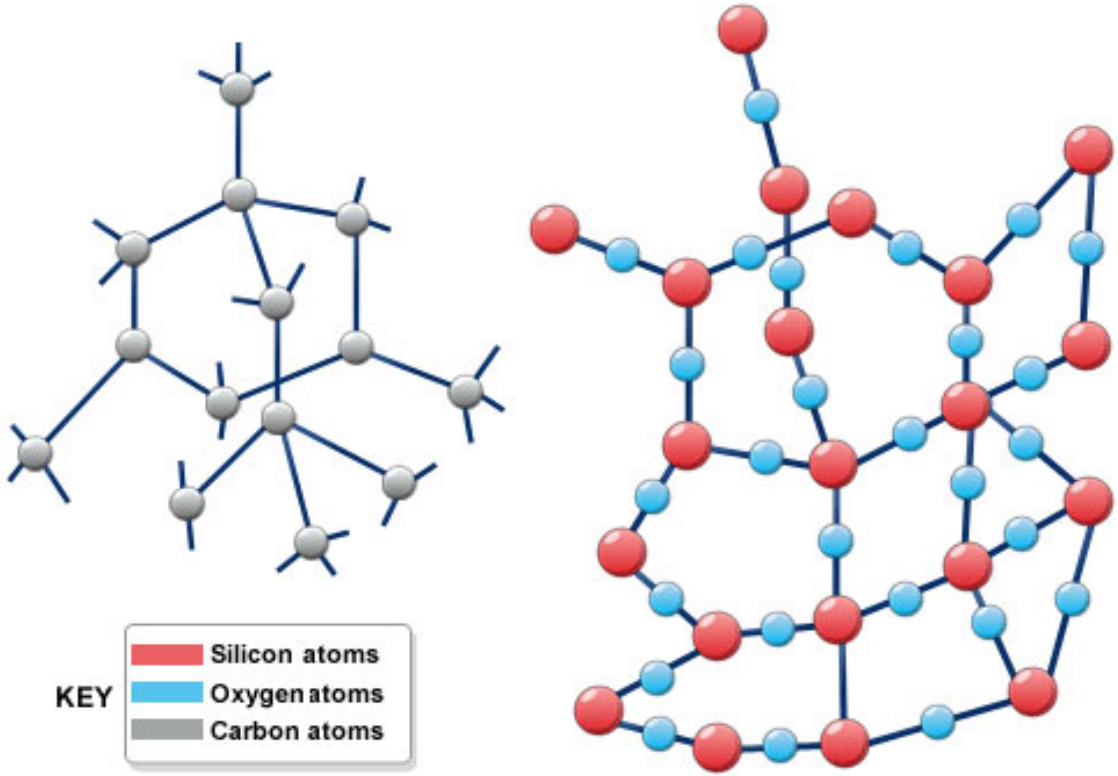


Figure 3: Covalent Network (Source :- Google Search)[3]

[3] http://www.bbc.co.uk/

Electrical Networks

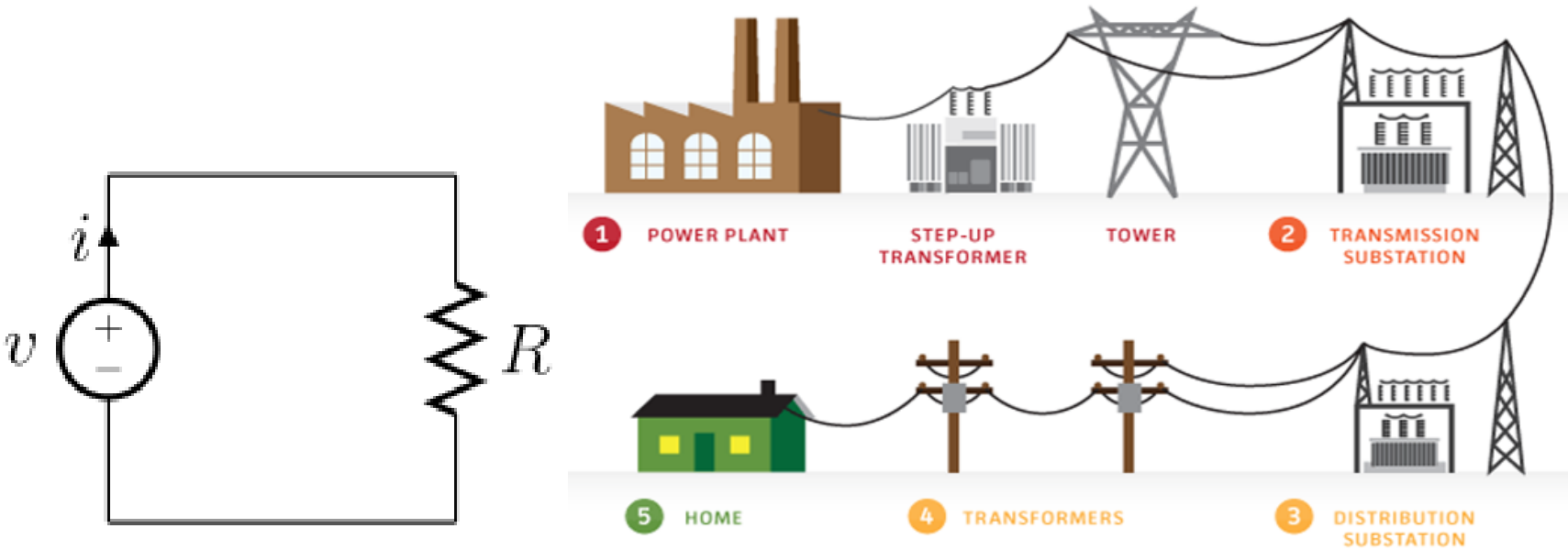


Figure 4: Electrical Networks (Source :- Google Search)[4]

[4] https://en.wikipedia.org/wiki/Dependent_source
http://nsong.org/Pages/ContentPageLink1.aspx

Friendship Network



Figure 5: A Social Network (Source :- Google Search)[5]

[5] http://peopleinsight.co.uk

Mobile Network



Figure 6: Mobile Network (Source :- Google Search)[6]

[6] http://www.dreamstime.com

Computer Network



Figure 7: Computer Network (Source :- Google Search)[7]

[7] http://business.globe.com.ph/solutions/connectivity

Road Network



Figure 8: Indian Road Network (Source :- Google Search)[8]

[8] http://www.newikis.com/en/commons/File:India_major_road_network_map.png

First Network Problem



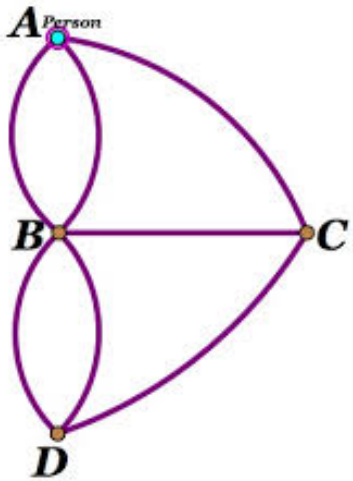


Figure 9: Seven Bridges of Konigsberg problem[9] (Source :- Google Search)[10]

[9] Leonhard Euler, 1741

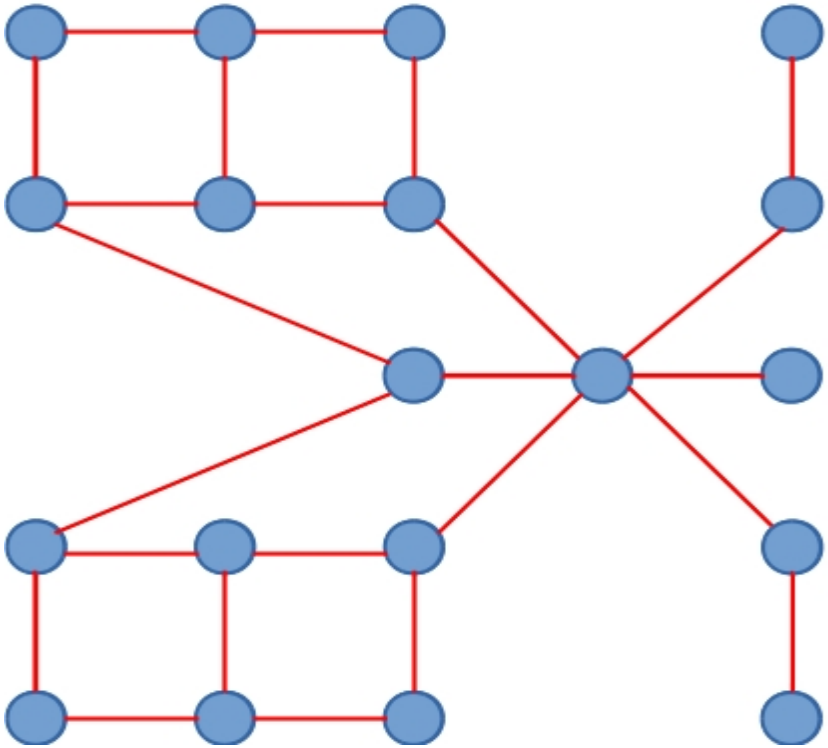
[10] https://007x7.wordpress.com/
http://jwilson.coe.uga.edu/EMAT6680Fa2012/Faircloth/Essay2alf/KonigsbergBridge.html

Categorization of Network Problems

1. ***Network Analysis*** : mining network
   - Centrality measures
   - Community detection
   - Information diffusion etc.

2. ***Network Optimization*** : maximize or minimize an objective function.
   - Minimum spanning tree
   - Maximum flow in network
   - Facility location problem
   - Vehicle routing problem

Where to install a facility?

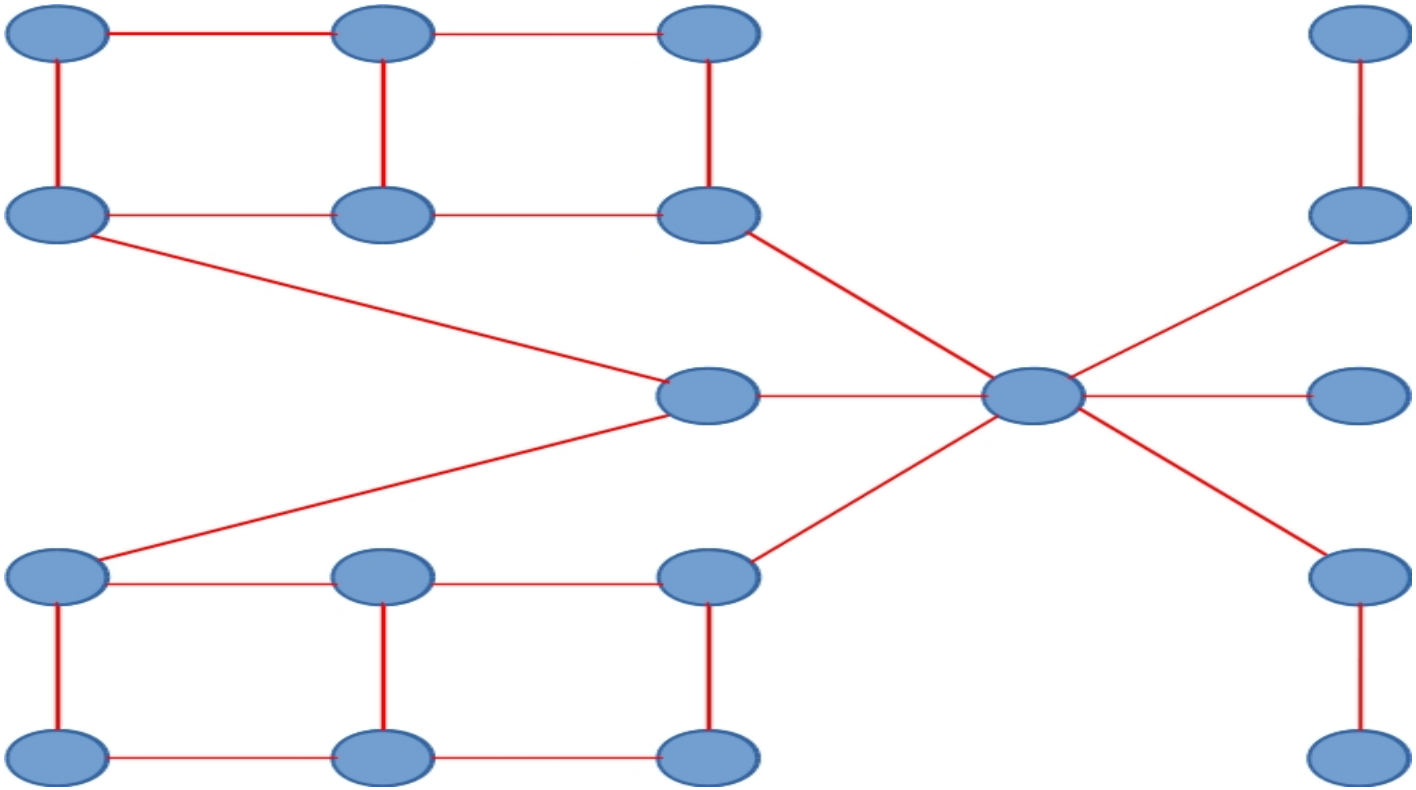




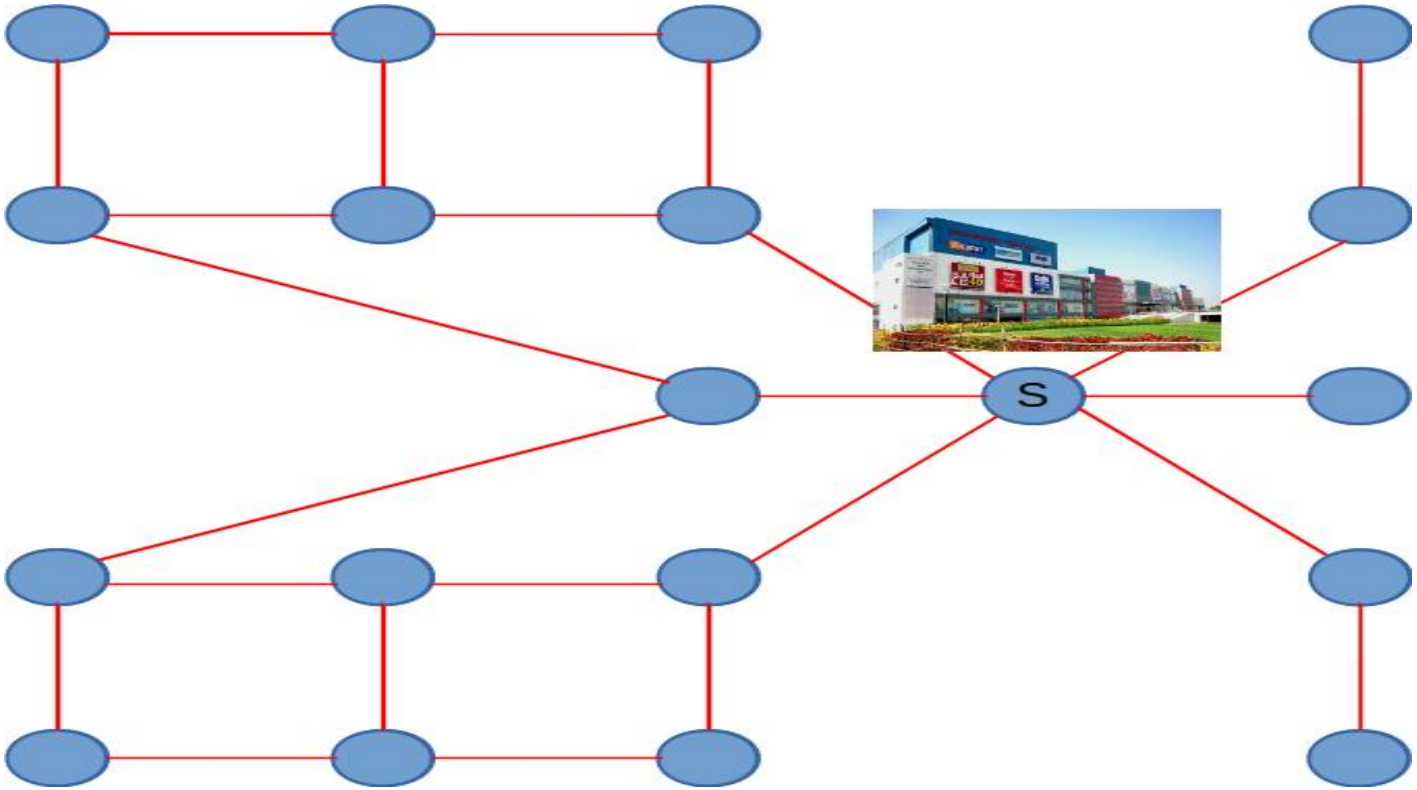




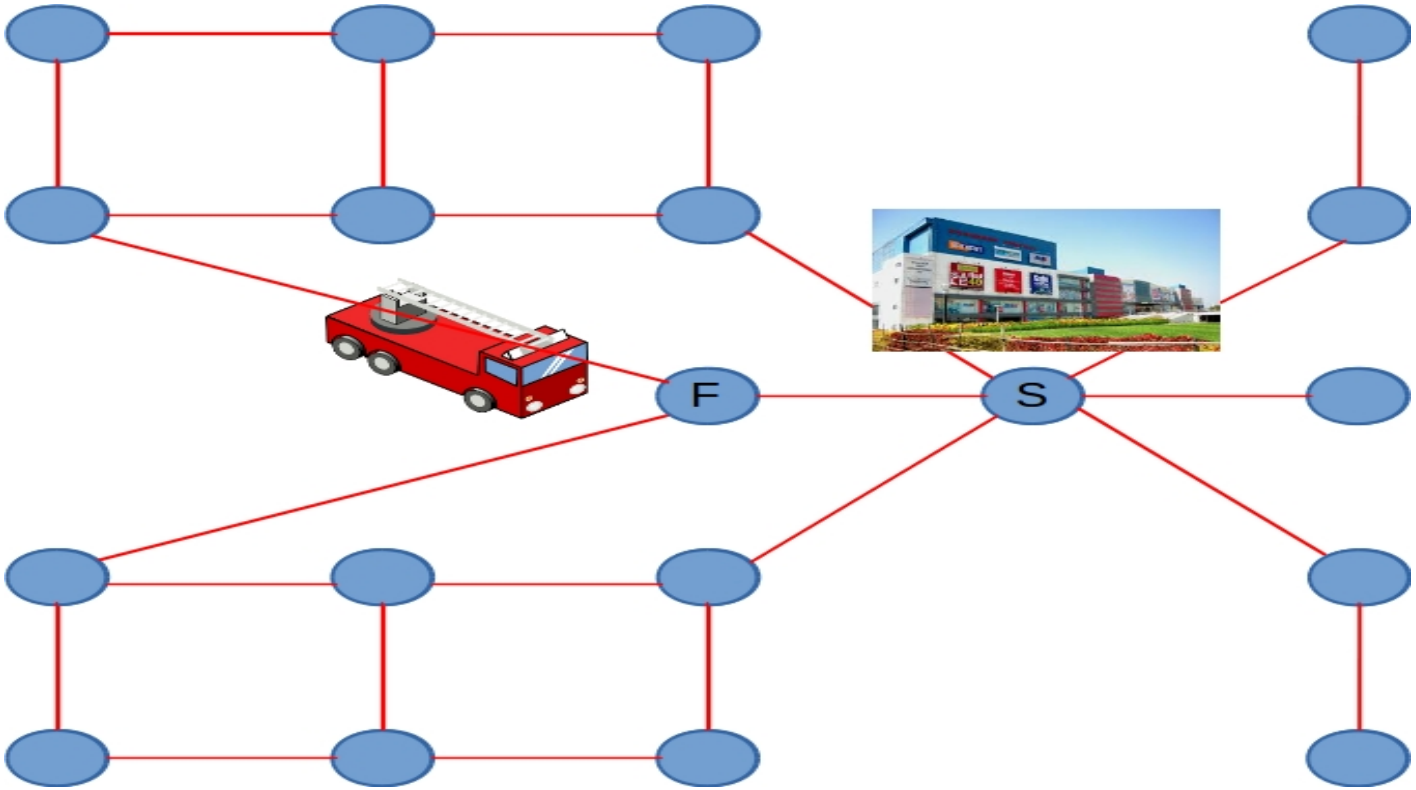
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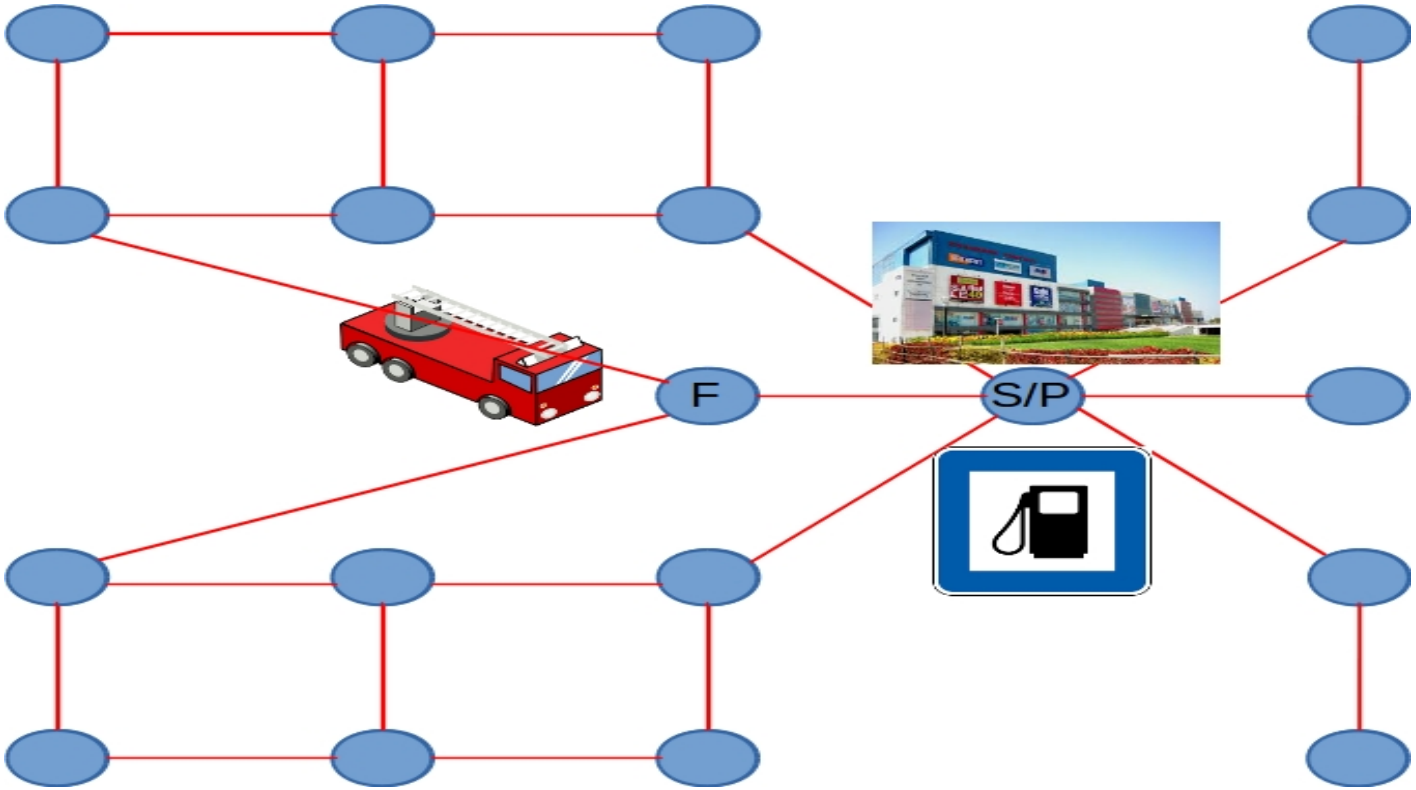
Where to install a facility?



Where to install a facility?



Where to install a facility?



Guess who will win?

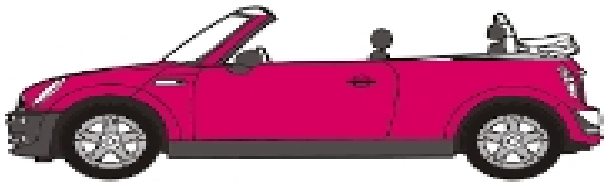


RAVI

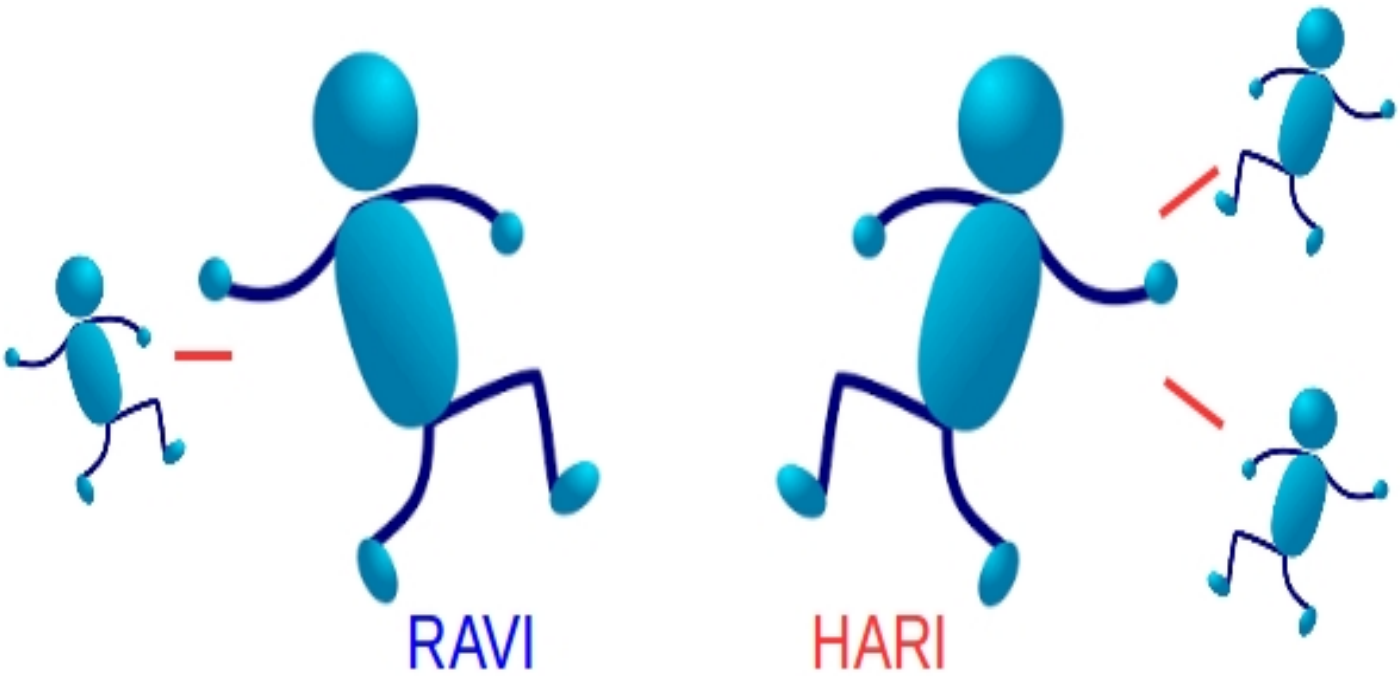


HARI

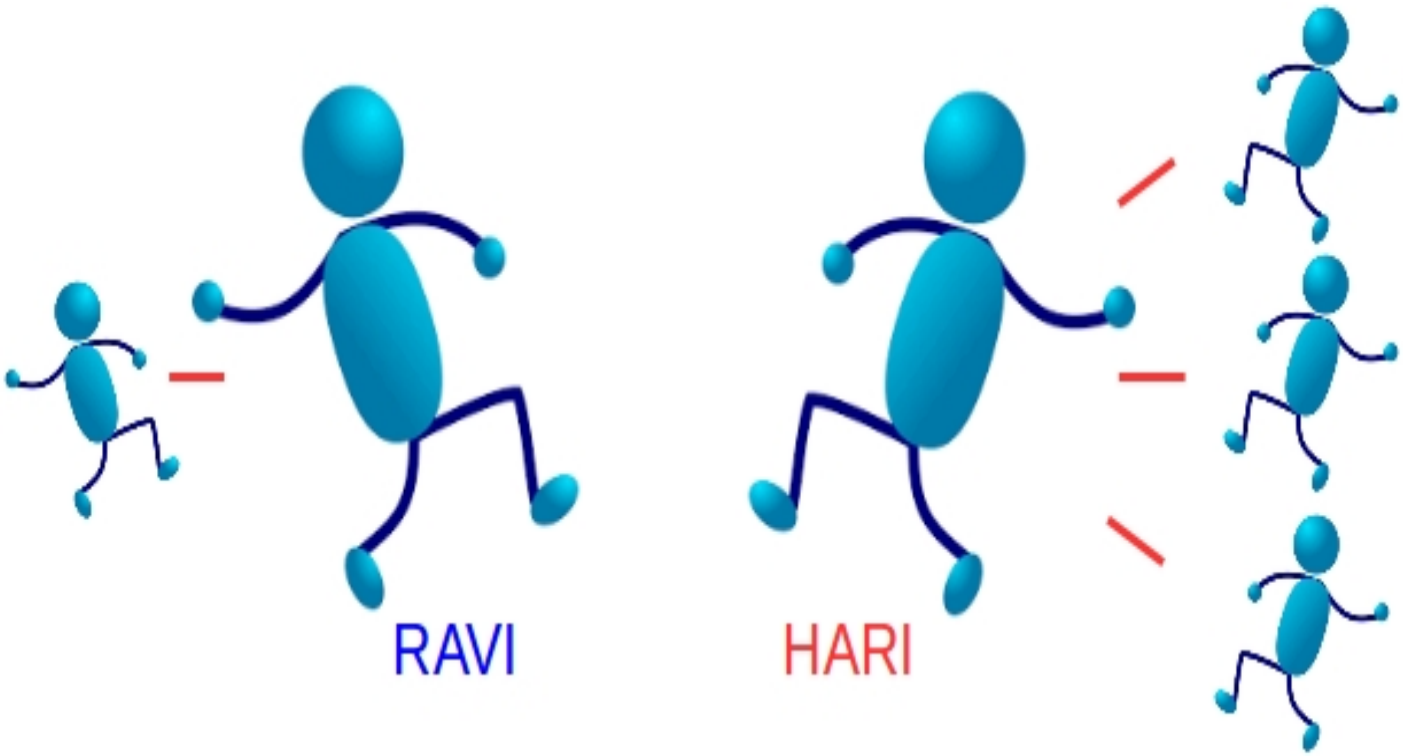




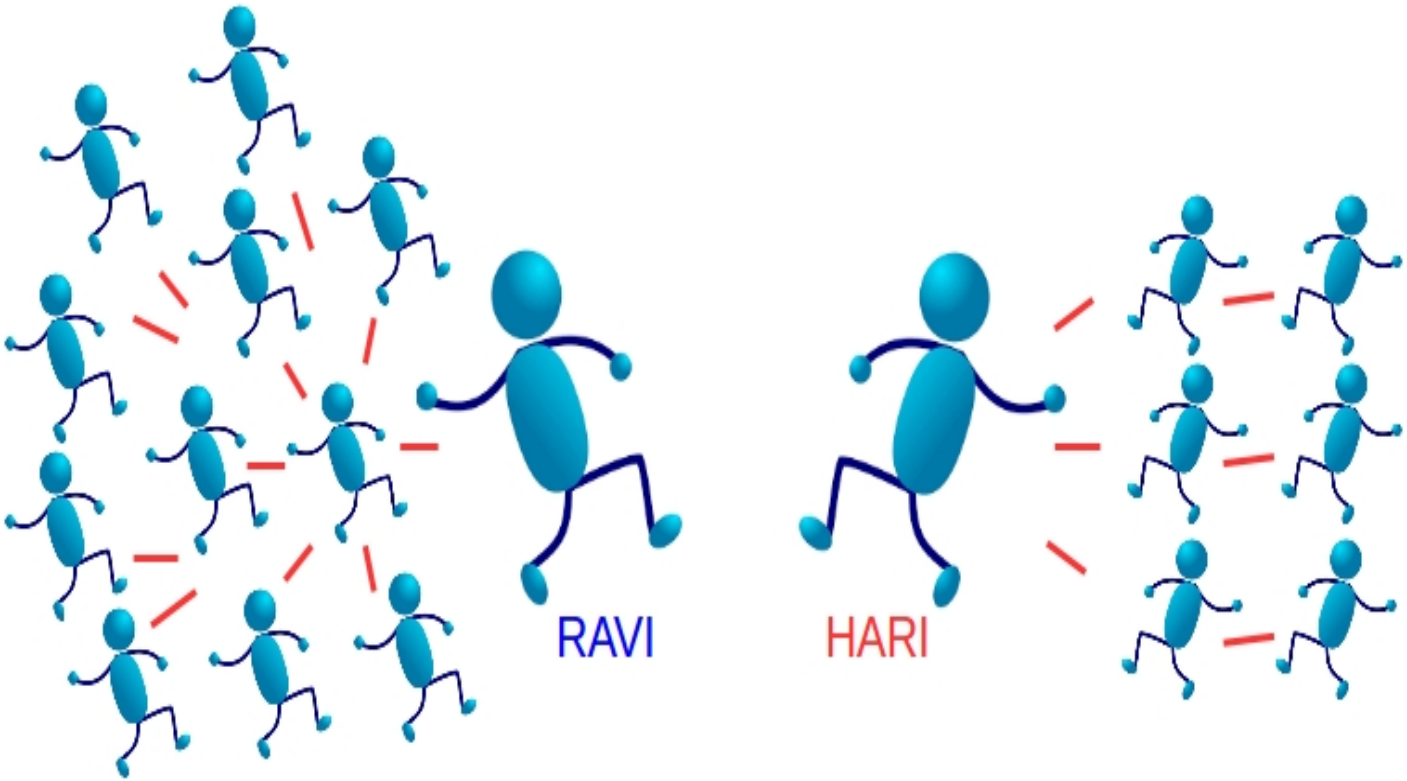
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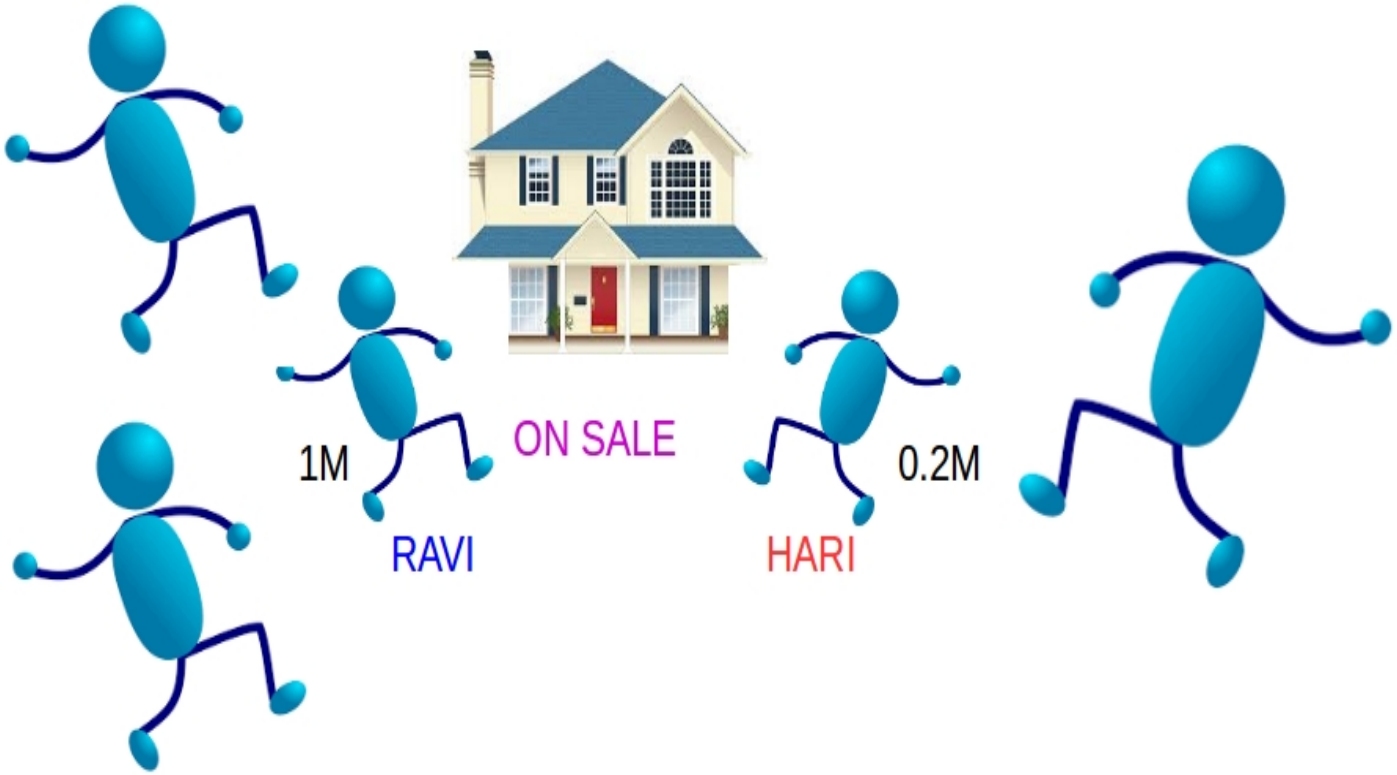
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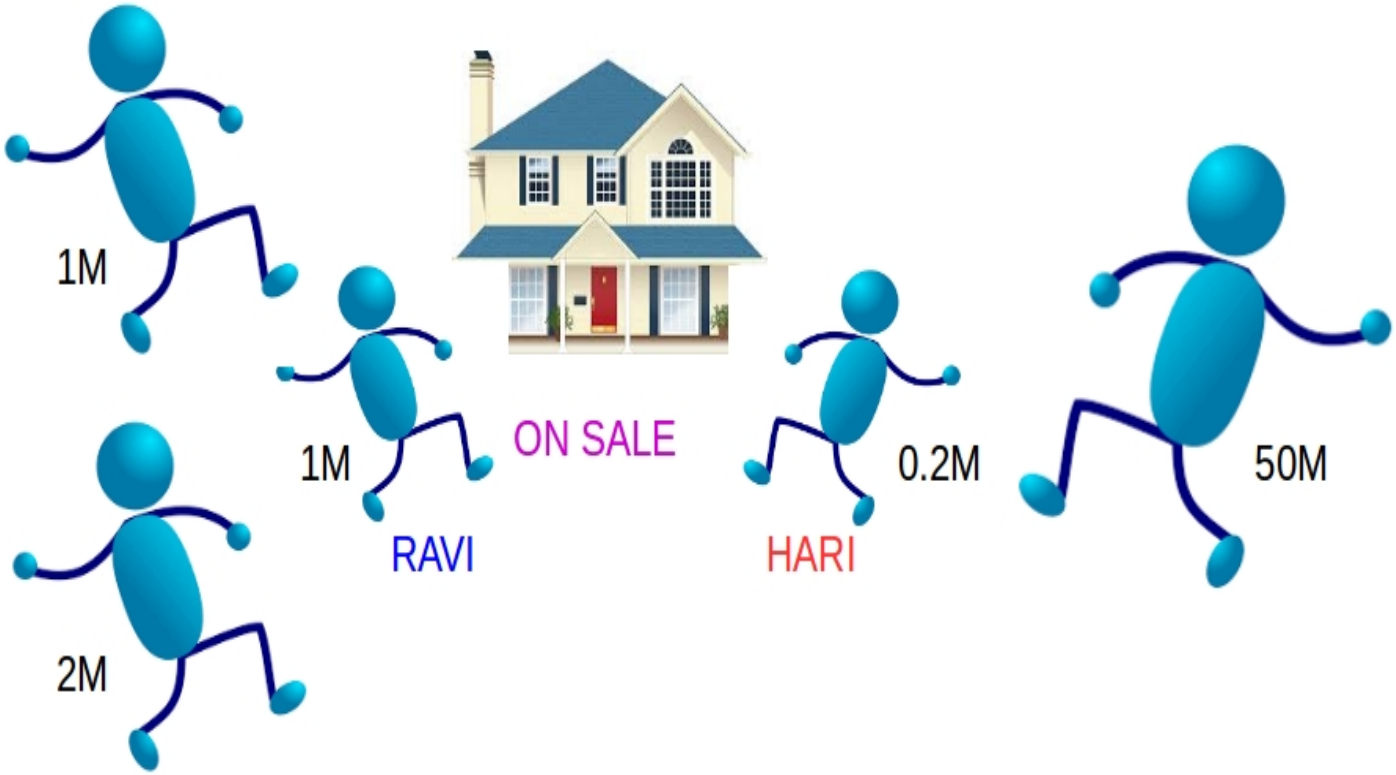
Guess who will get the house?



Guess who will get the house?



Guess who will get the house?



Questions

- Which actors in a social network is the most influential?
- Which road is the busiest among all road?
- Which protein is vital for a cell to live?
- Which actor in a social network will hear any news on an average first?
- Which paper has higher prestige?
- Which web page is more important?

The General Question

"Which is the most ***important*** or ***central*** node in a given network?"

Centrality

- What is centrality?

  In network analysis, centrality of a vertex is the quantification of the intuitive notion of importance of a node in a network.



Figure 10: (Source :- Google Search)[11]

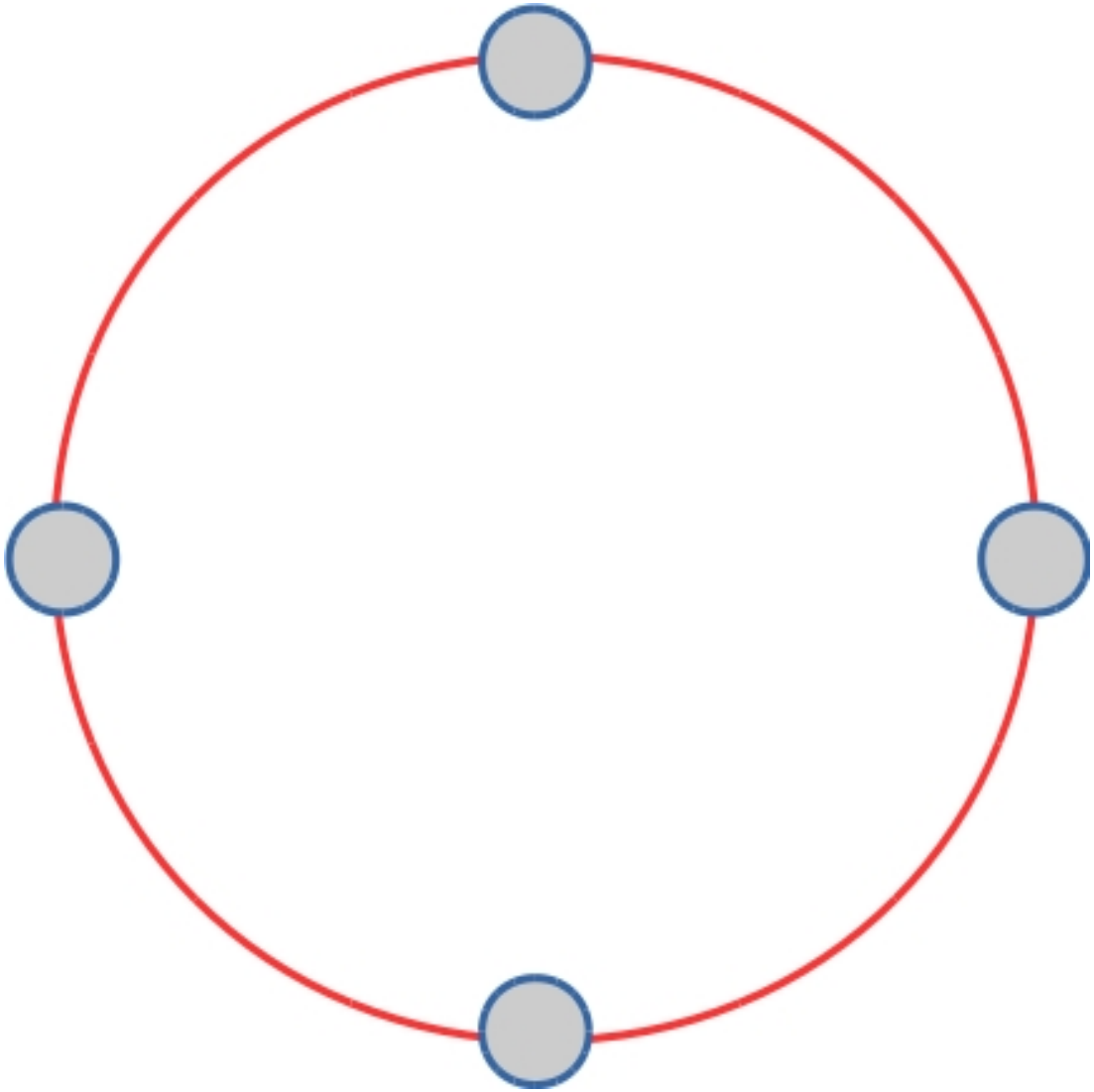
[11] http://www.livescience.com/43368-top-10-rules-facebook.html

Centrality

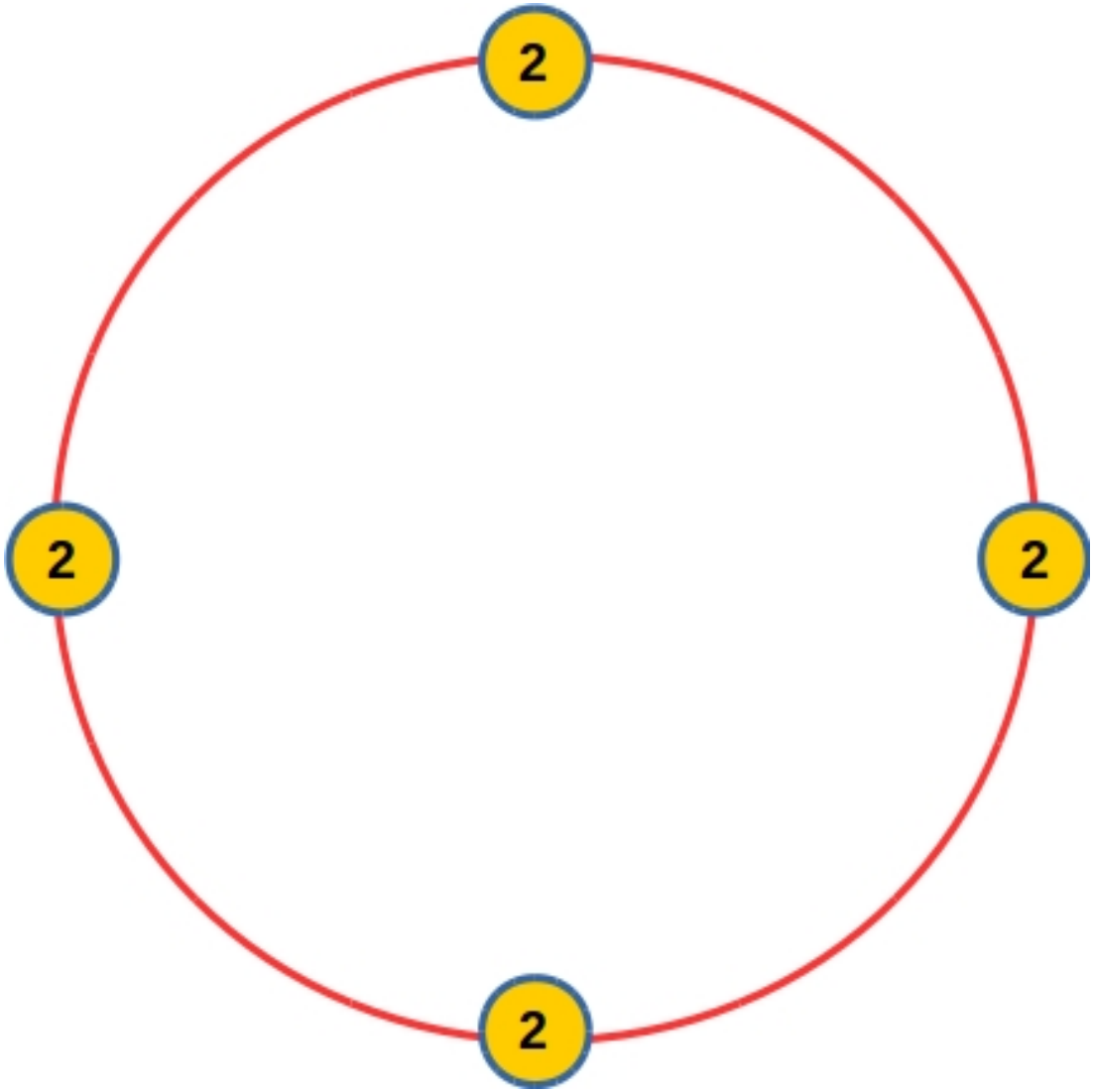
- Different centrality measures have been coined in the literature for application specific reasons. Traditional ones are:
  - Degree Centrality
  - Closeness Centrality
  - Betweenness Centrality
  - Eigenvector Centrality

Degree Centrality

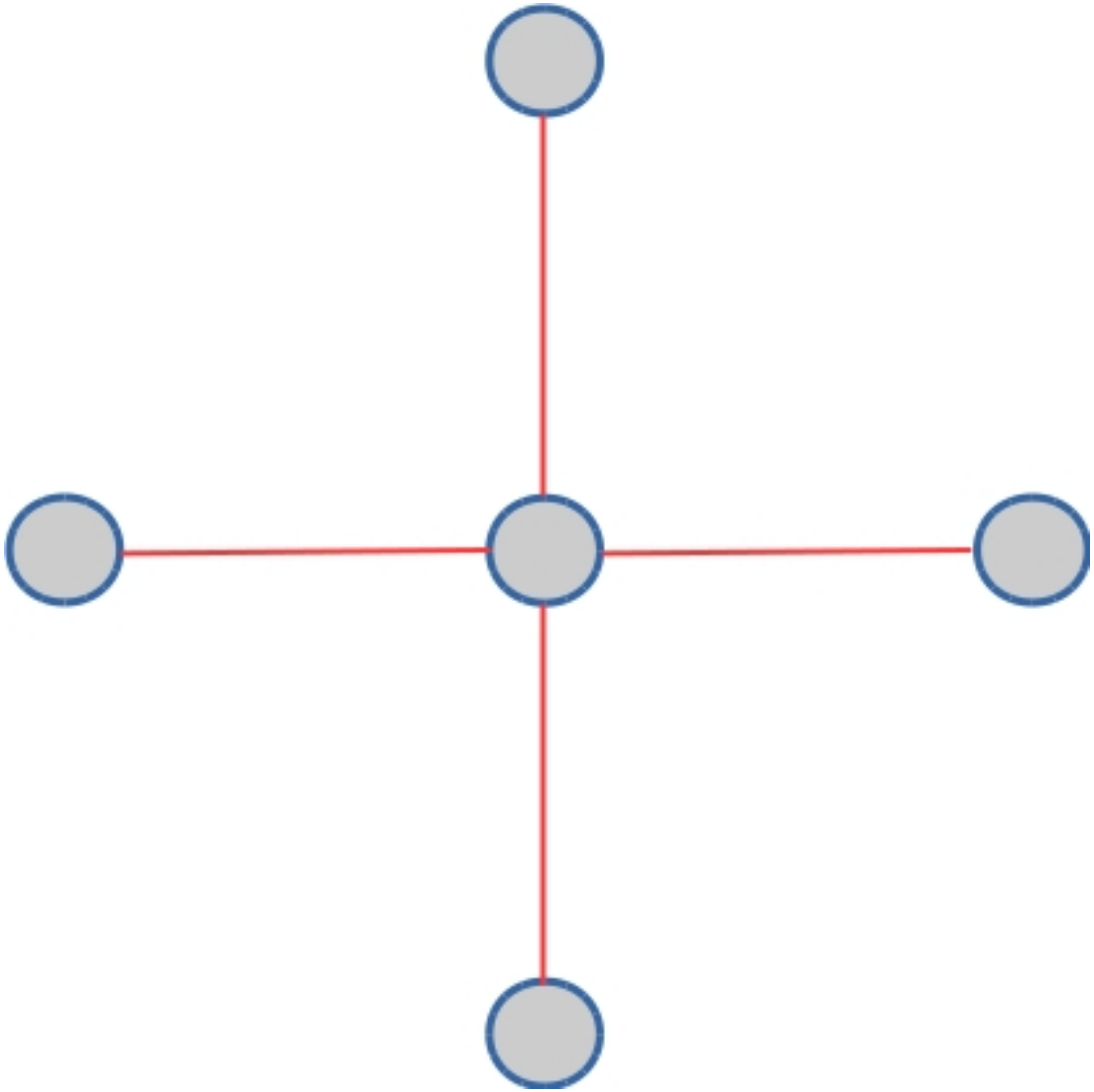
Degree Centrality: Example



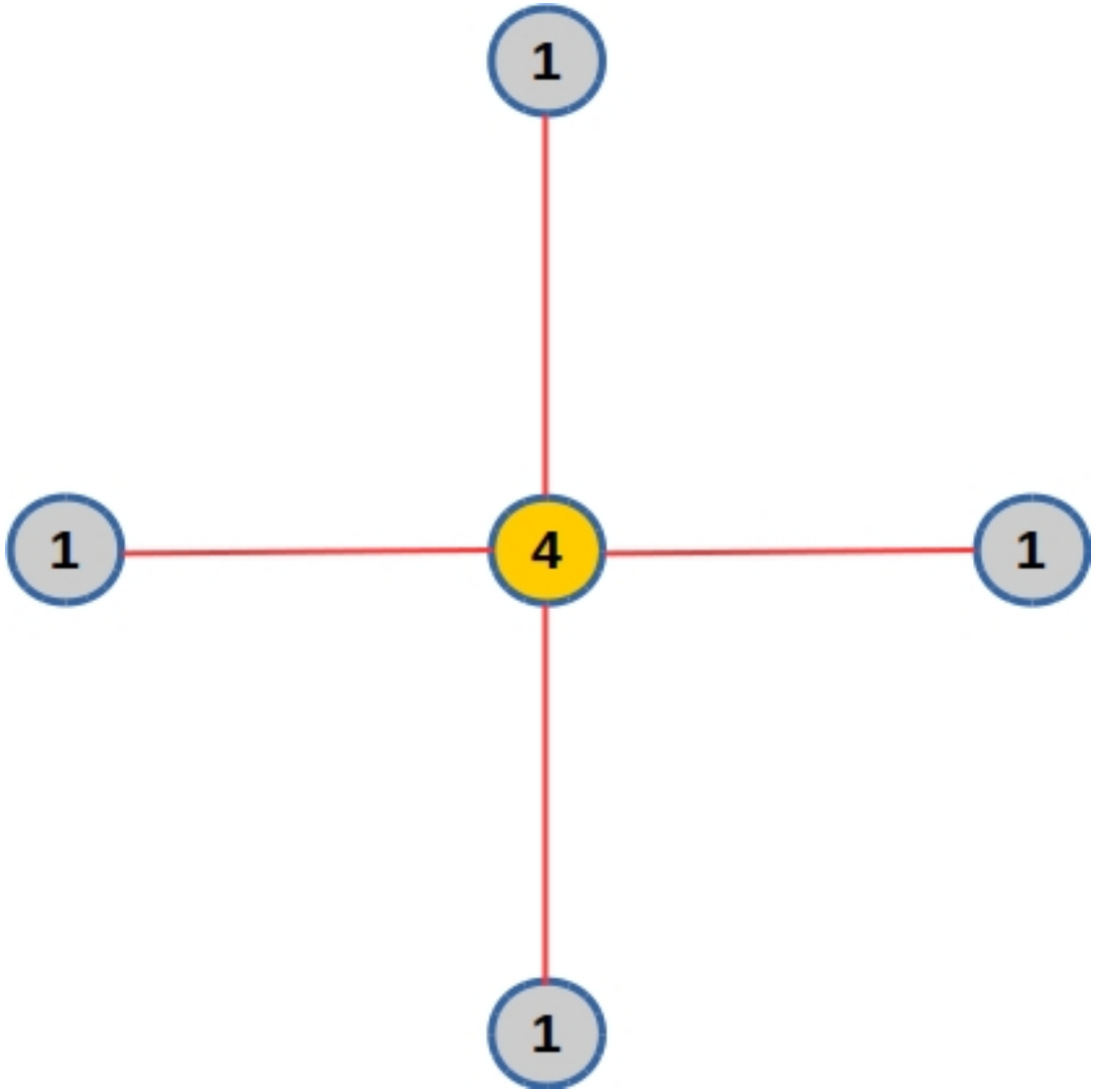
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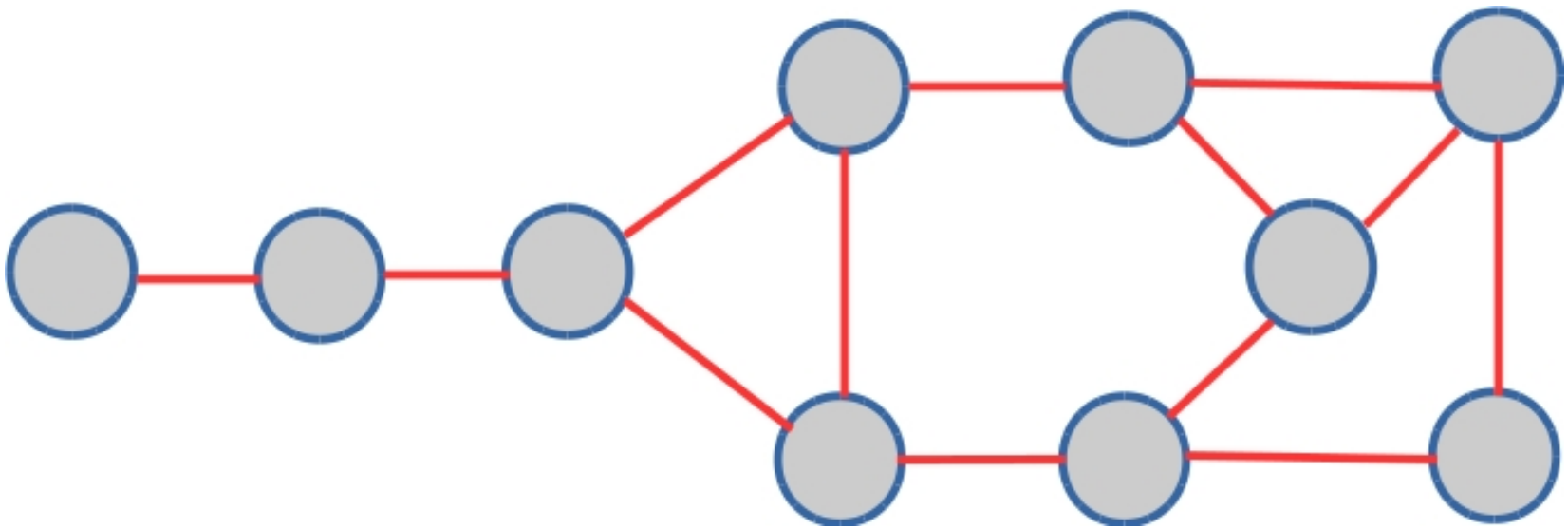
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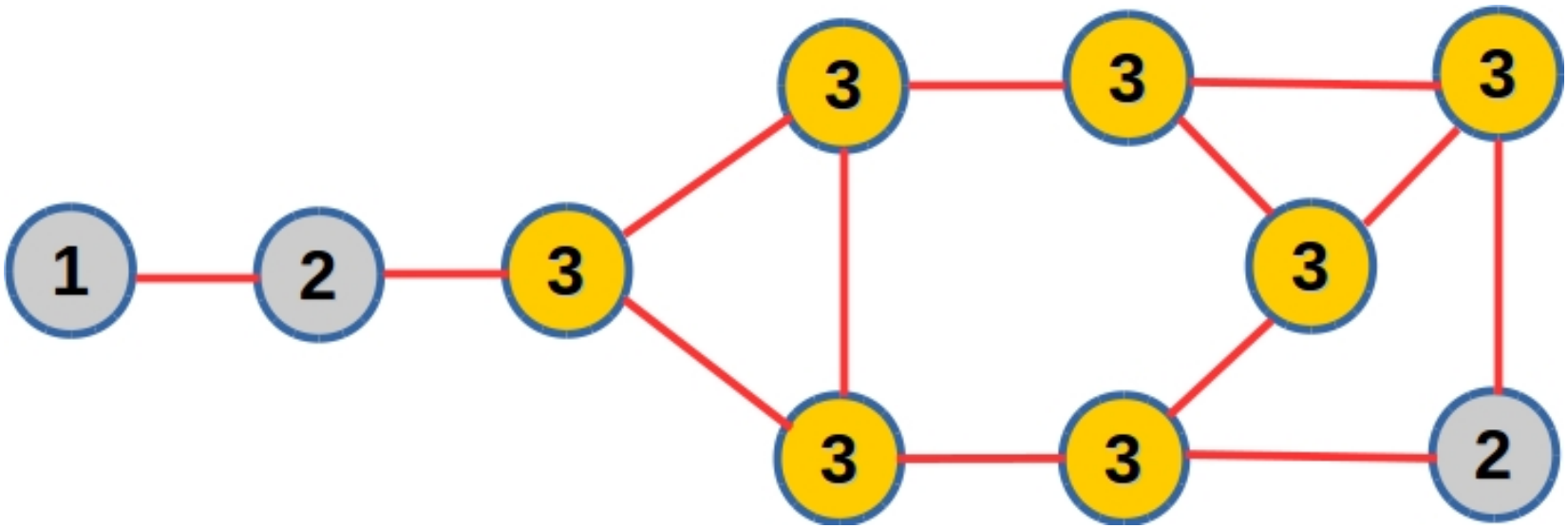
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Degree Centrality in Social Networks

- Degree centrality $\mapsto$ popularity
- Higher degree node has many followers/friends $\mapsto$ Higher strength.
- Lowest degree nodes are nearly isolated from the population and are least popular
- Shortcoming: Local measure

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Application: Degree Centrality

- To hypothesize the amount of emotional support available to a person.
- To find influential people
- Highly visible actors
- Highly exposed nodes to a flowing infection/information

Closeness Centrality

Closeness Centrality

- Introduced by Freeman in 1979[12].
- Called "Status" earlier by Harary in 1959[13].
- Node's importance is inversely proportional to the sum of its distance to other nodes.
- Mathematically:

$$CC(u) = \frac{1}{\sum_{v \in V \setminus \{u\}} d_{uv}}$$

- Normalized by multiplying with $n-1$ i.e

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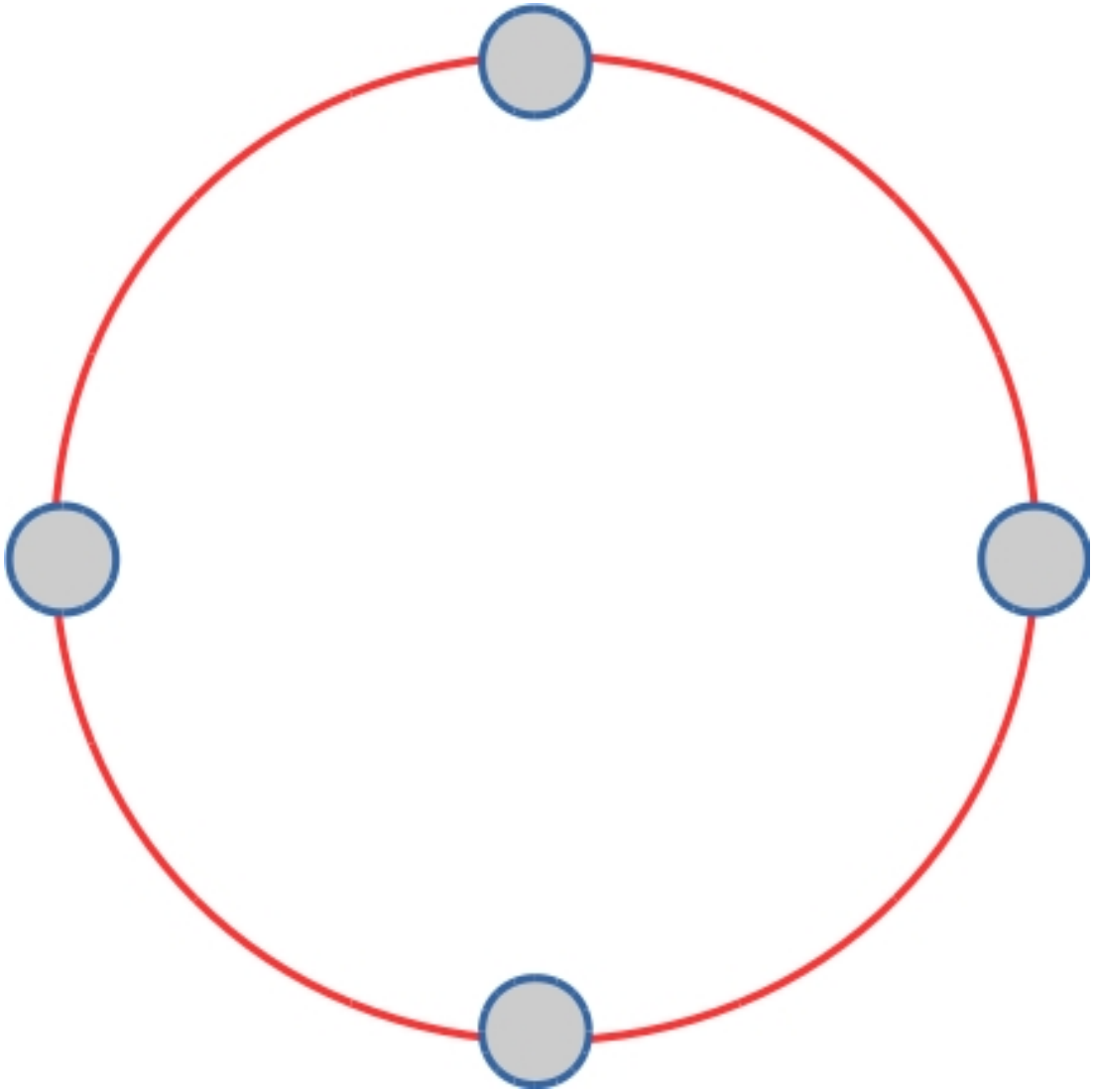
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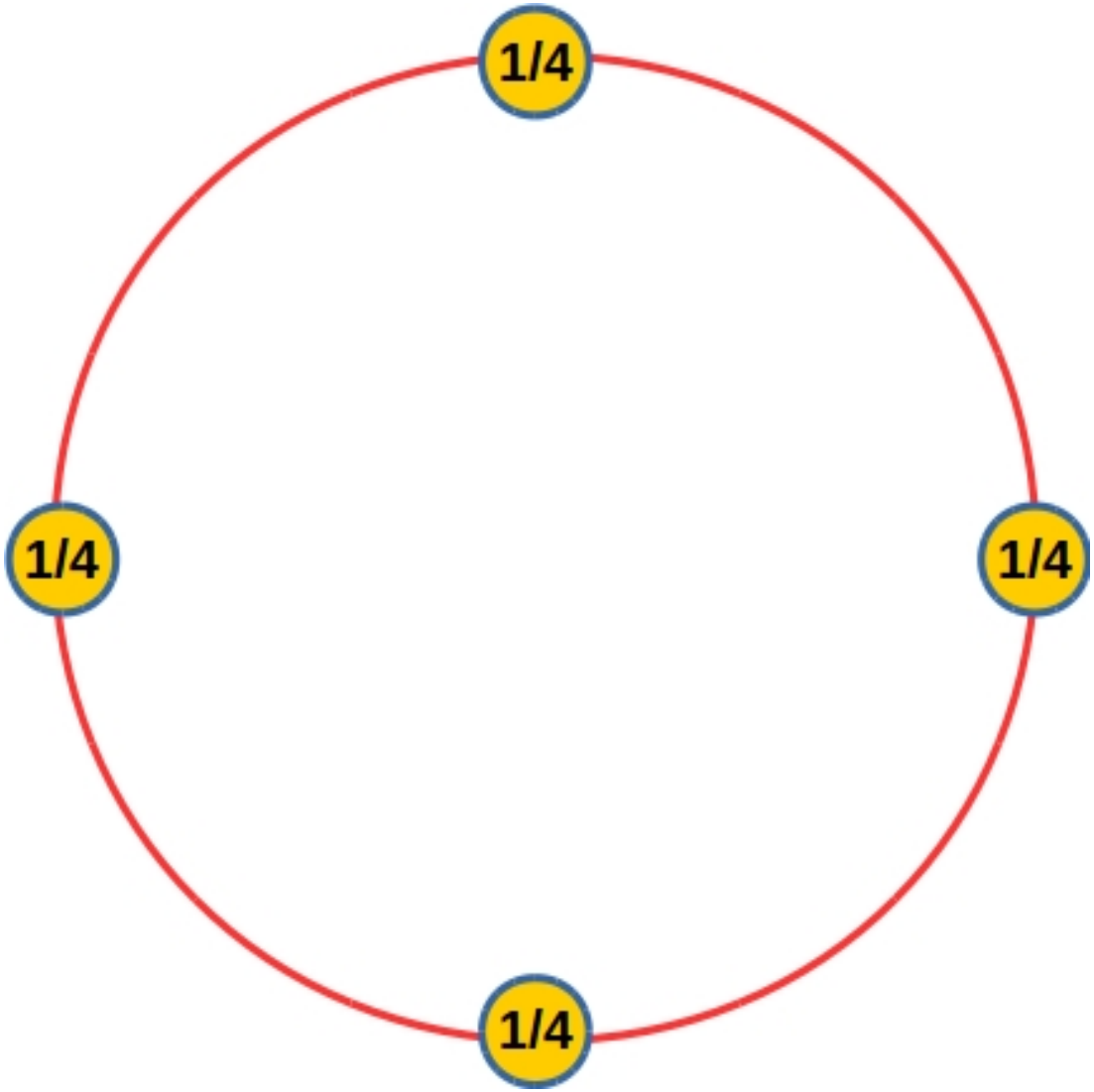
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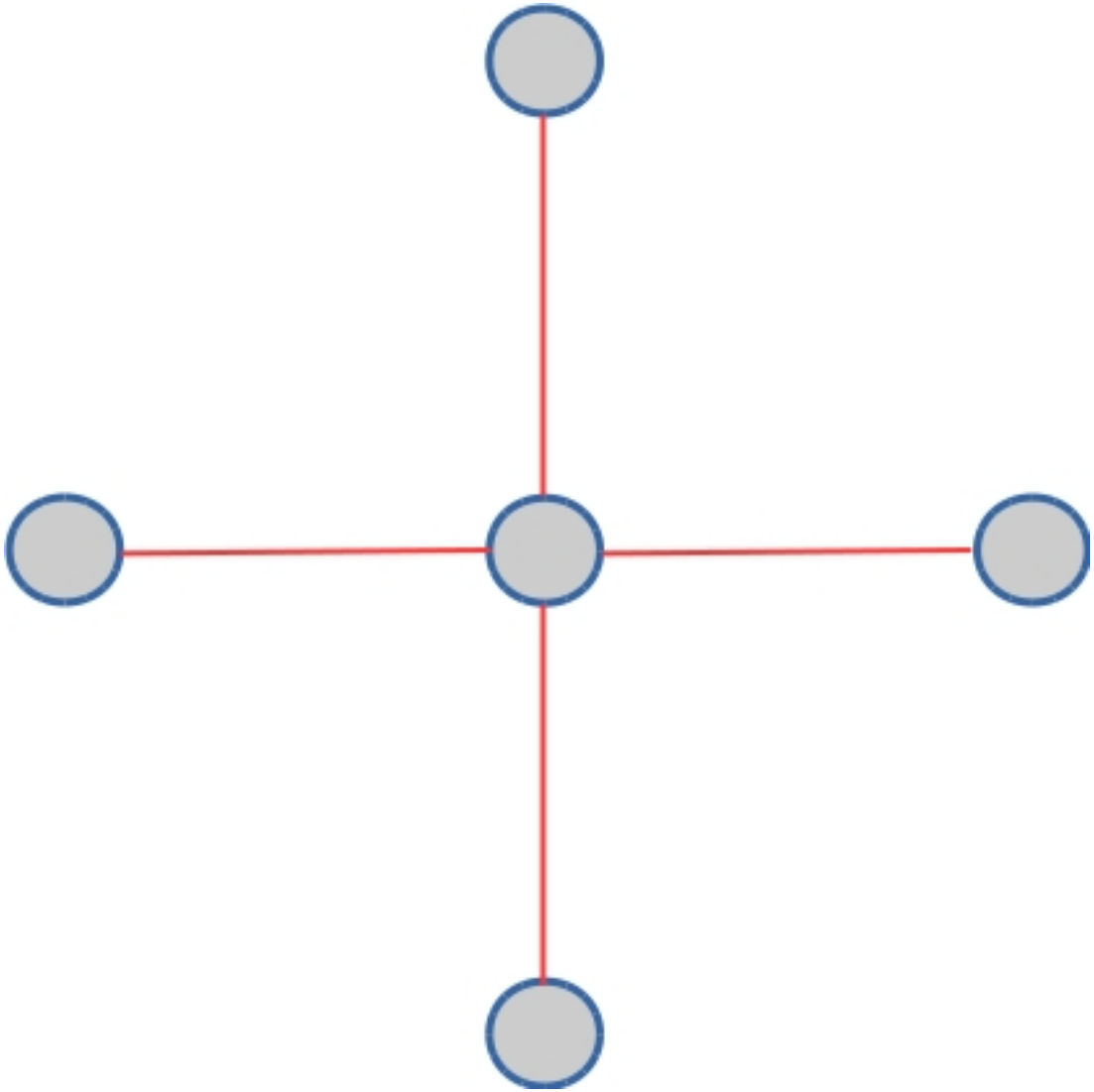
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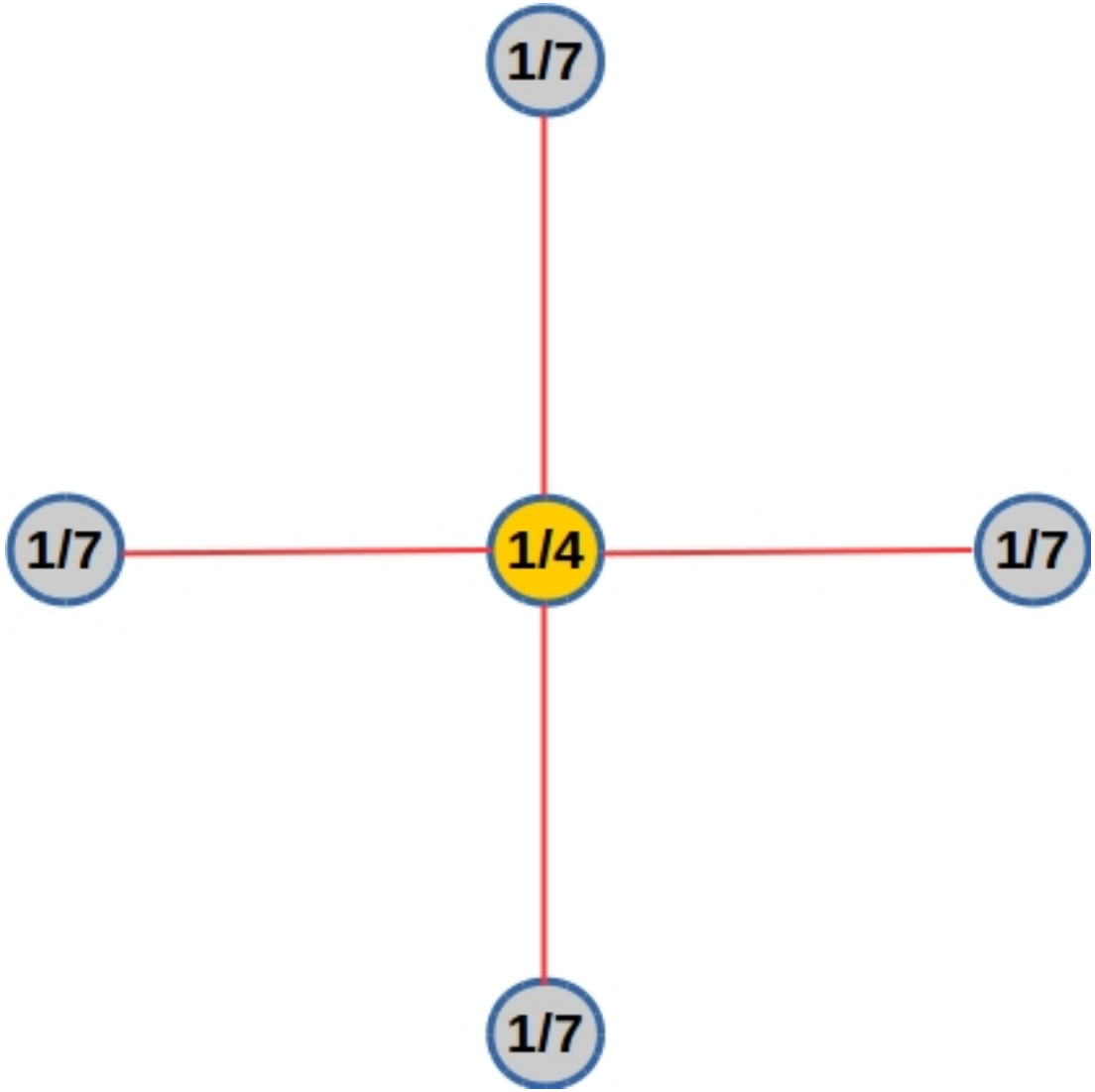
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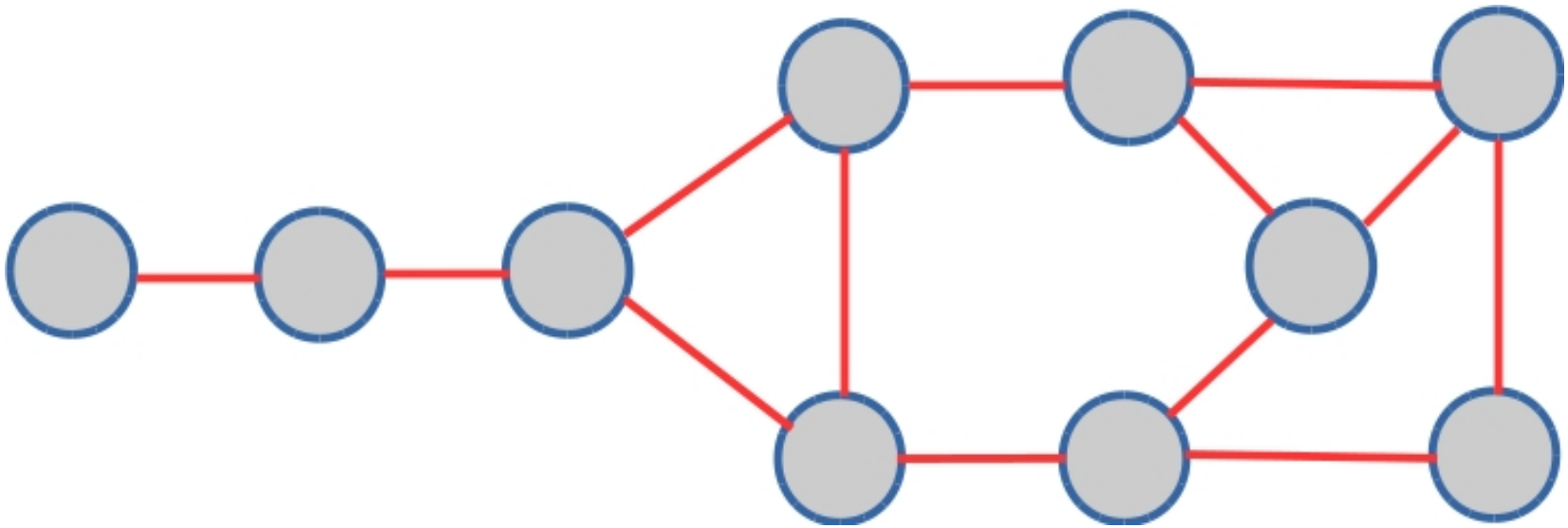
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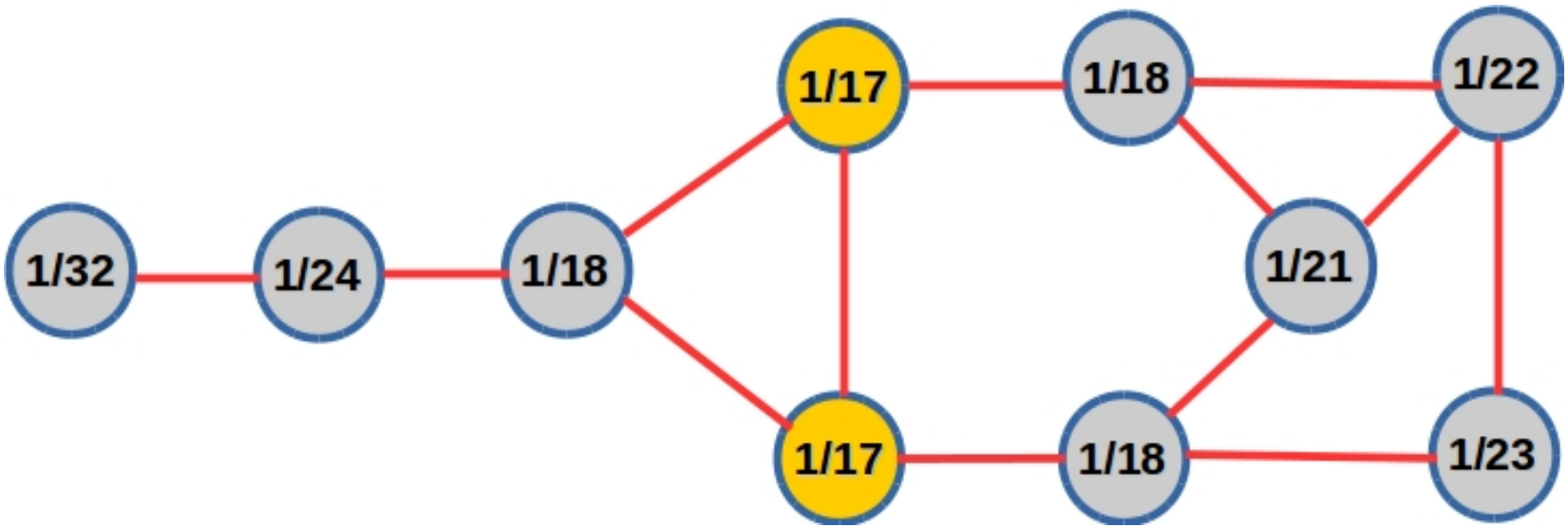
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Closeness Centrality: Example



Closeness Centrality : Some Key Points

- Closeness centrality of $u \mapsto$ the expected time for an information originated at a random node to reach u or vice-versa.
- Higher closeness $\mapsto$ Node gets updated earlier with an information flowing in the network and vice-versa.
- Does not work in disconnected networks.

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Another measure like Closeness Centrality: Harmonic

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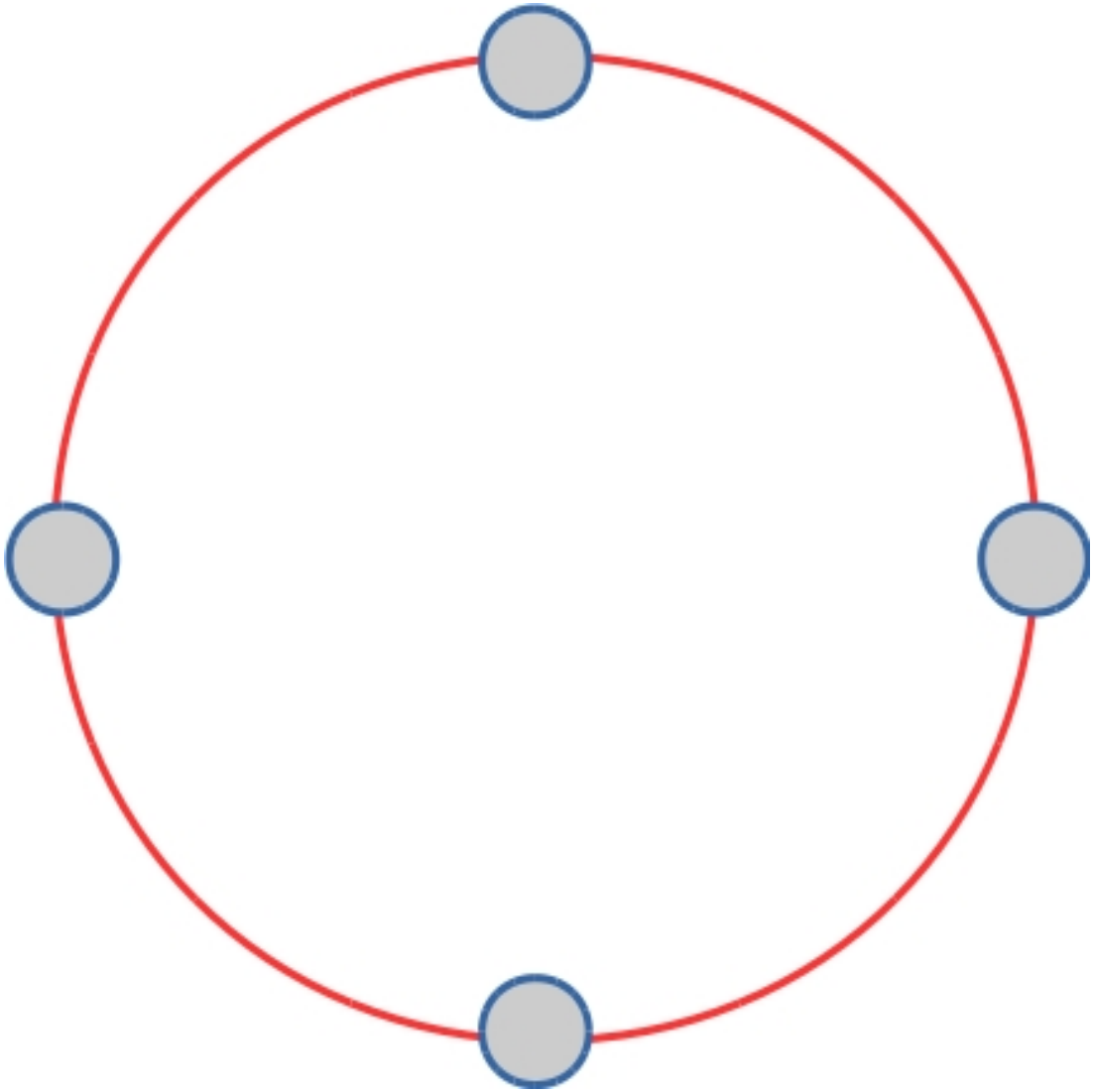
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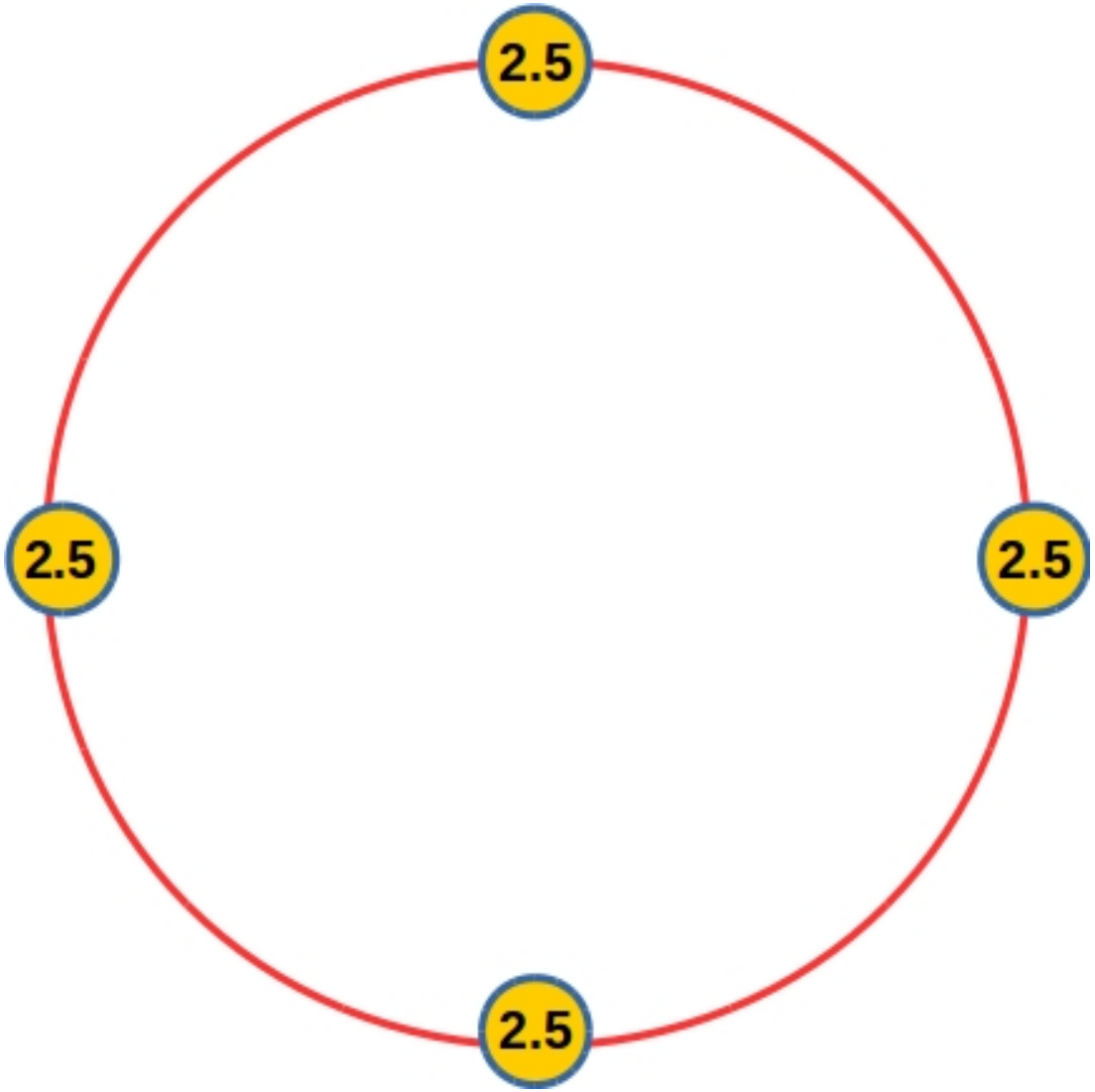
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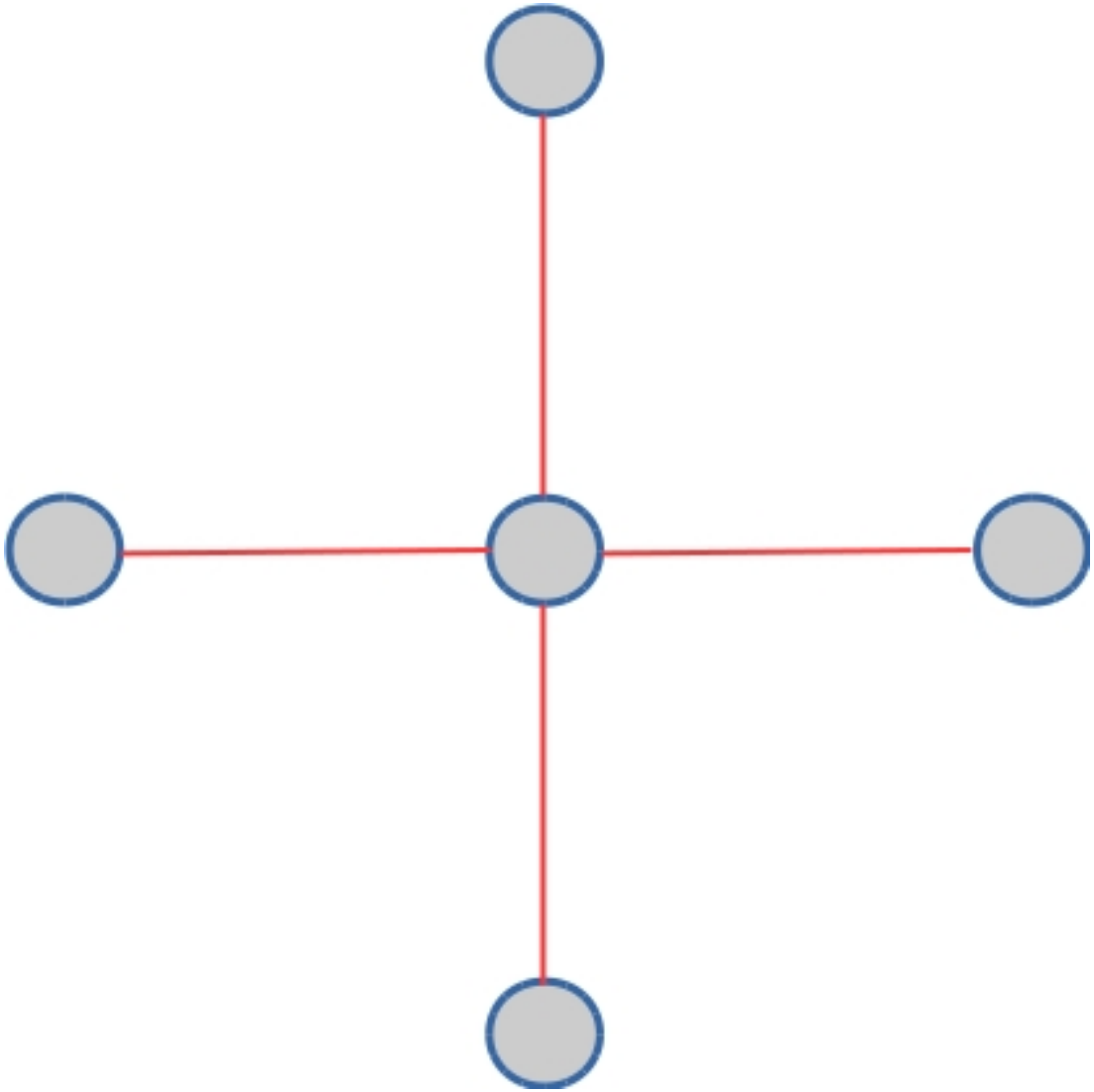
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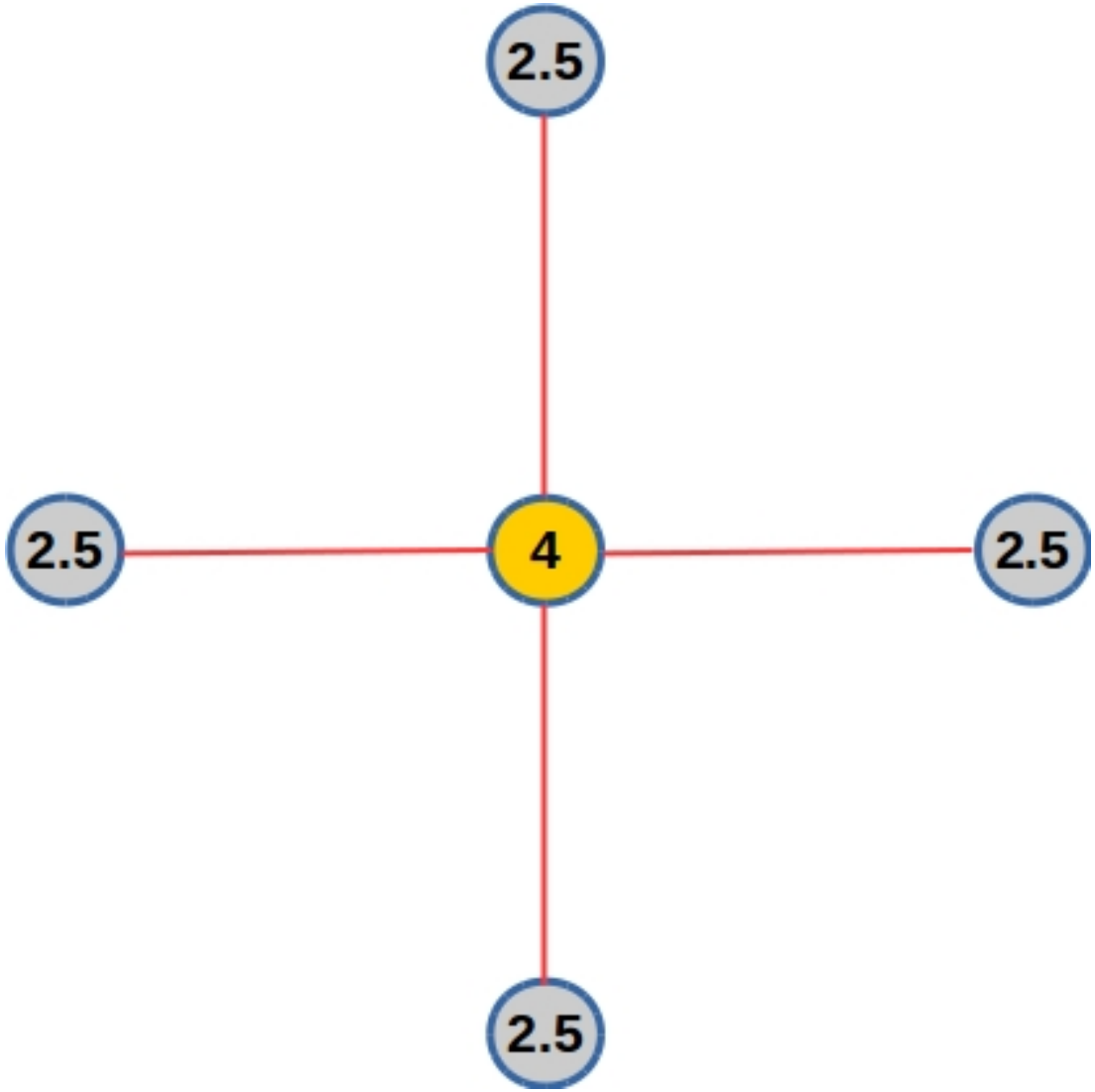
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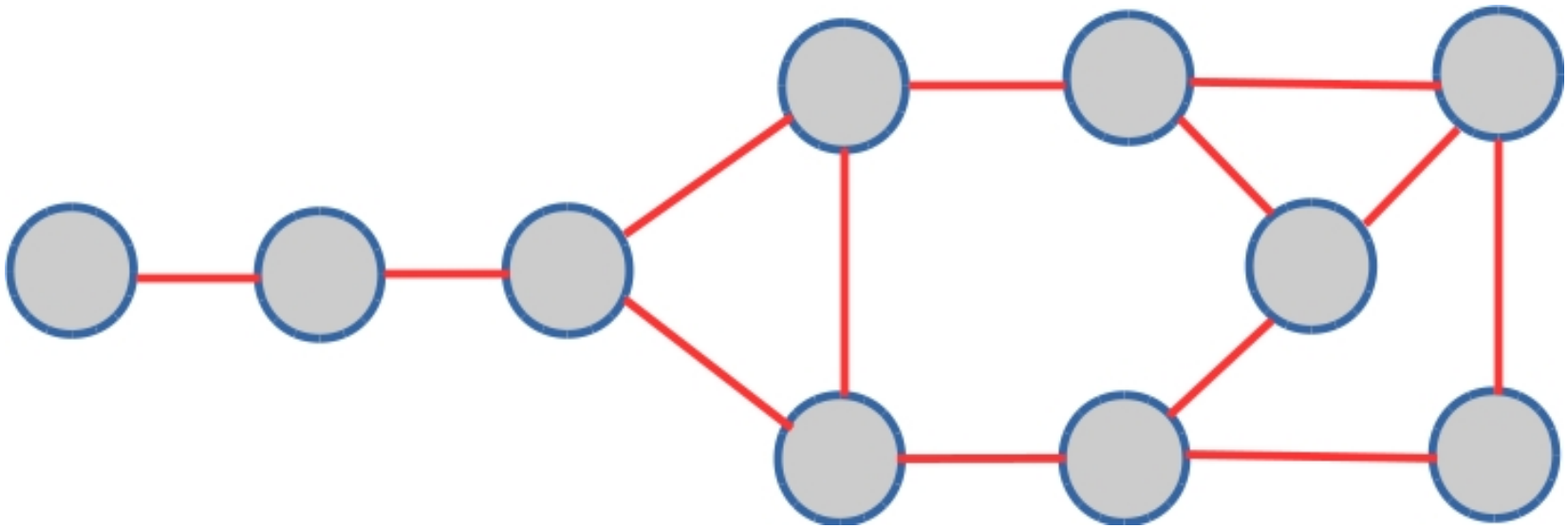
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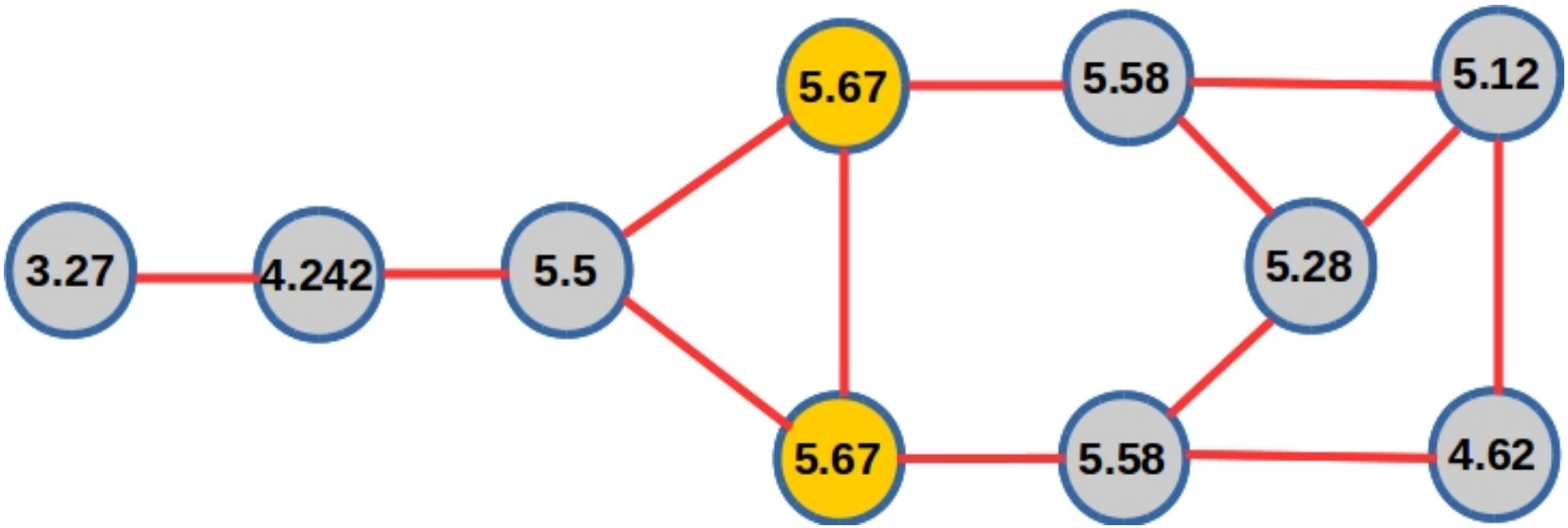
Harmonic Centrality: Example



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Harmonic Centrality: Example



Application: Closeness/Harmonic centrality

- Nodes receiving information of higher fidelity on an average.
- Nodes for installing facilities that needs to be used regularly by the whole network.
- Nodes prone to get exposed to infection/information quickly.

Eccentricity

Another measure based on geodesic: Eccentricity

- Node's importance is inversely proportional to the max of its distance to other nodes.
- Mathematically:

$$E(u) = \frac{1}{\max_{v \in V \setminus \{u\}} d_{uv}}$$

- Already normalized.

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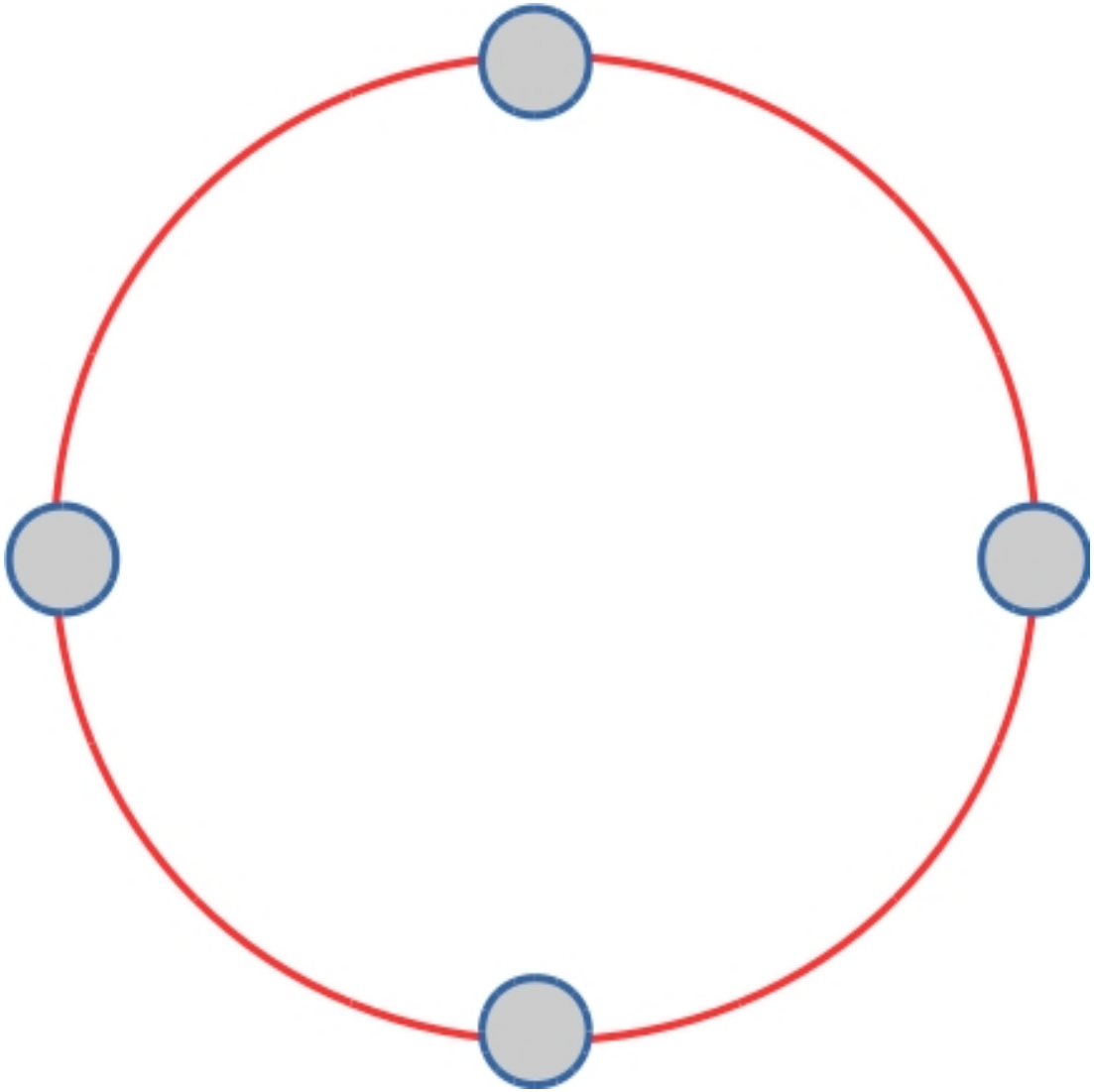
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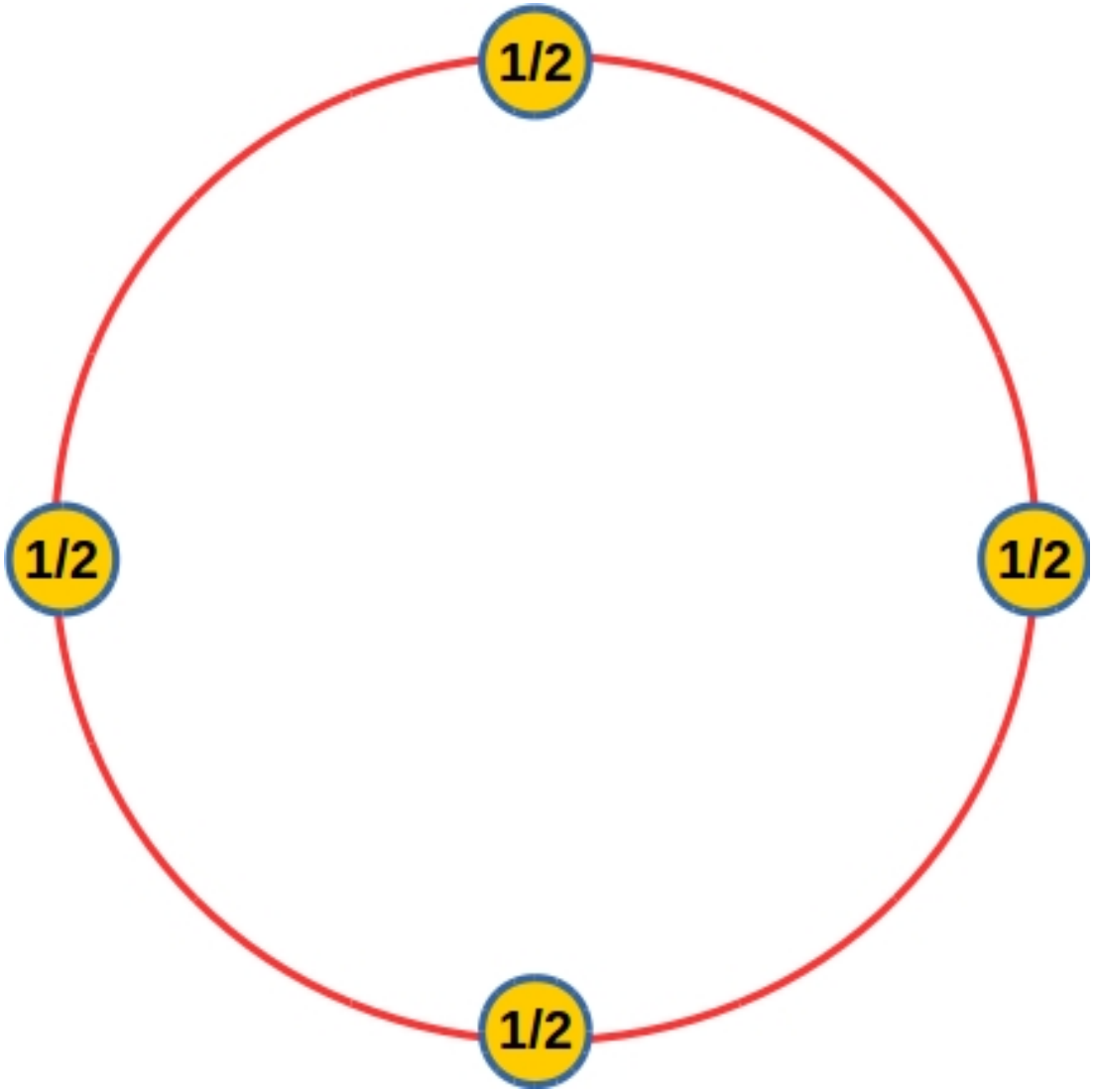
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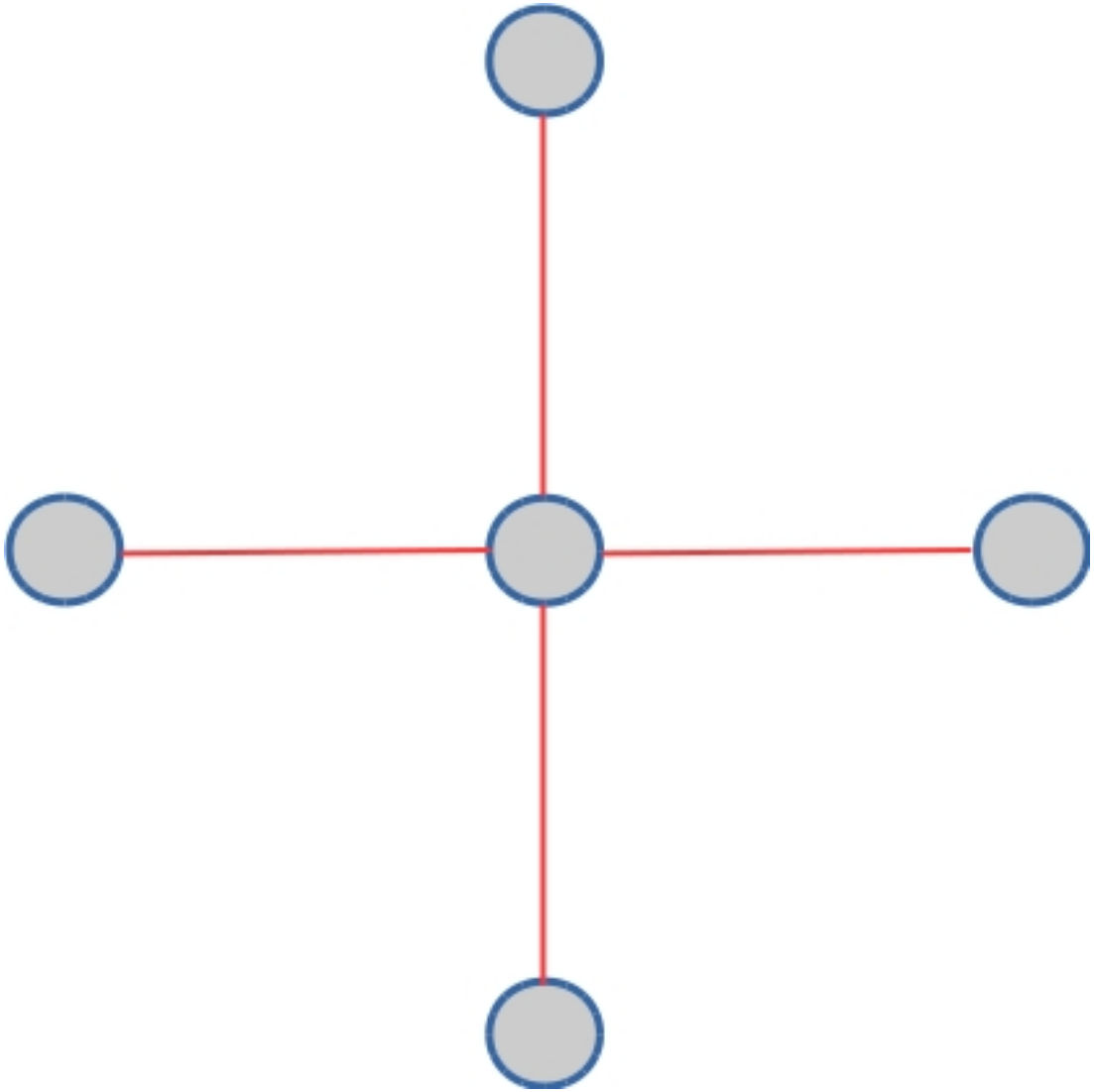
Eccentricity: Example



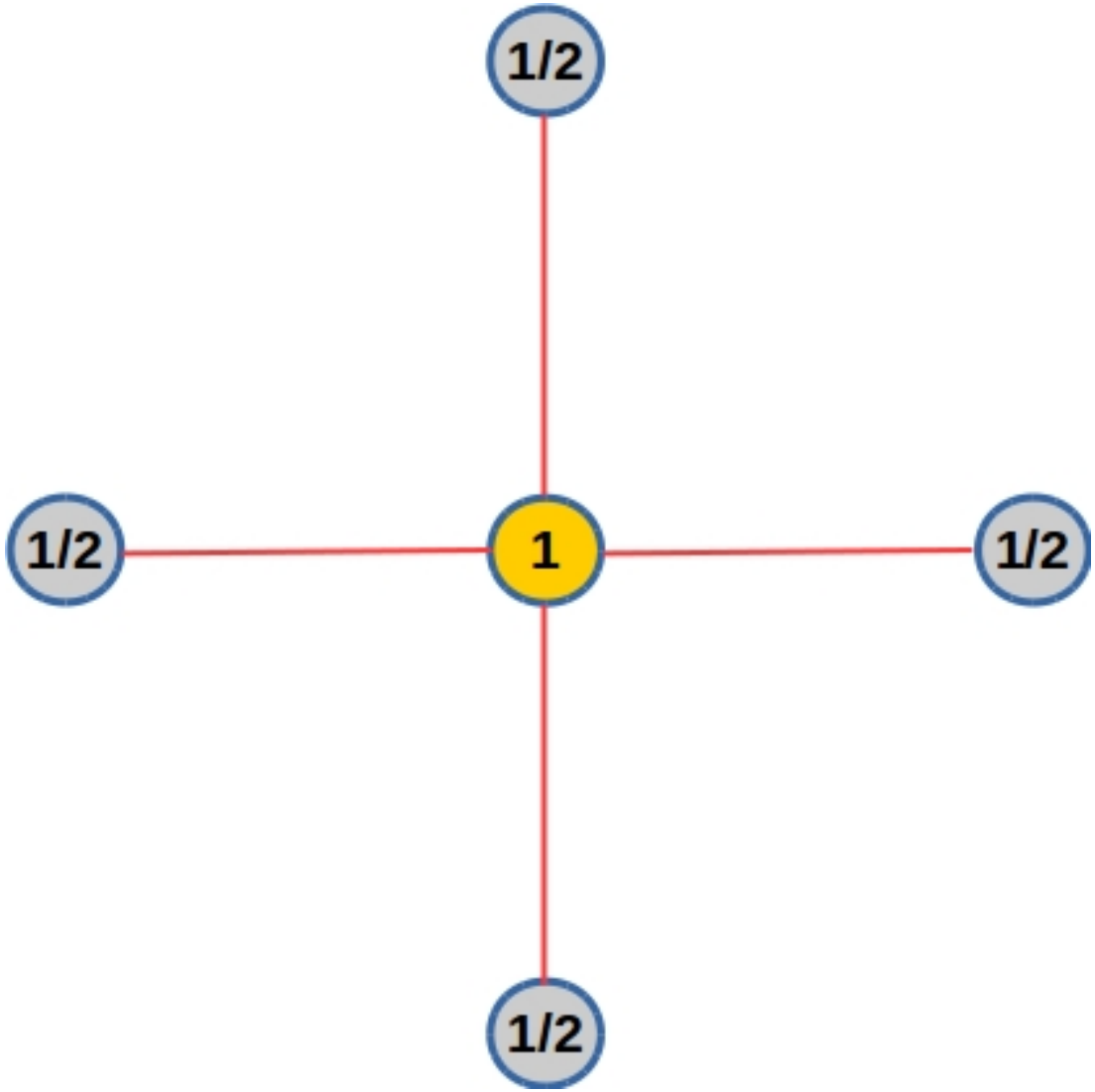
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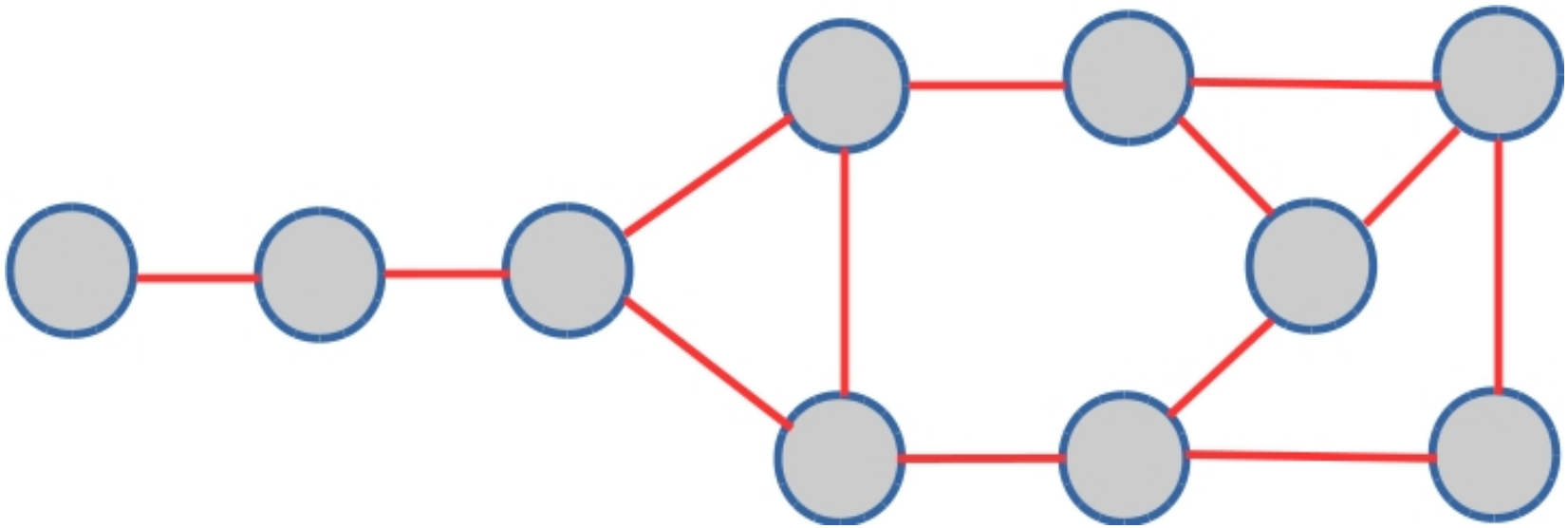
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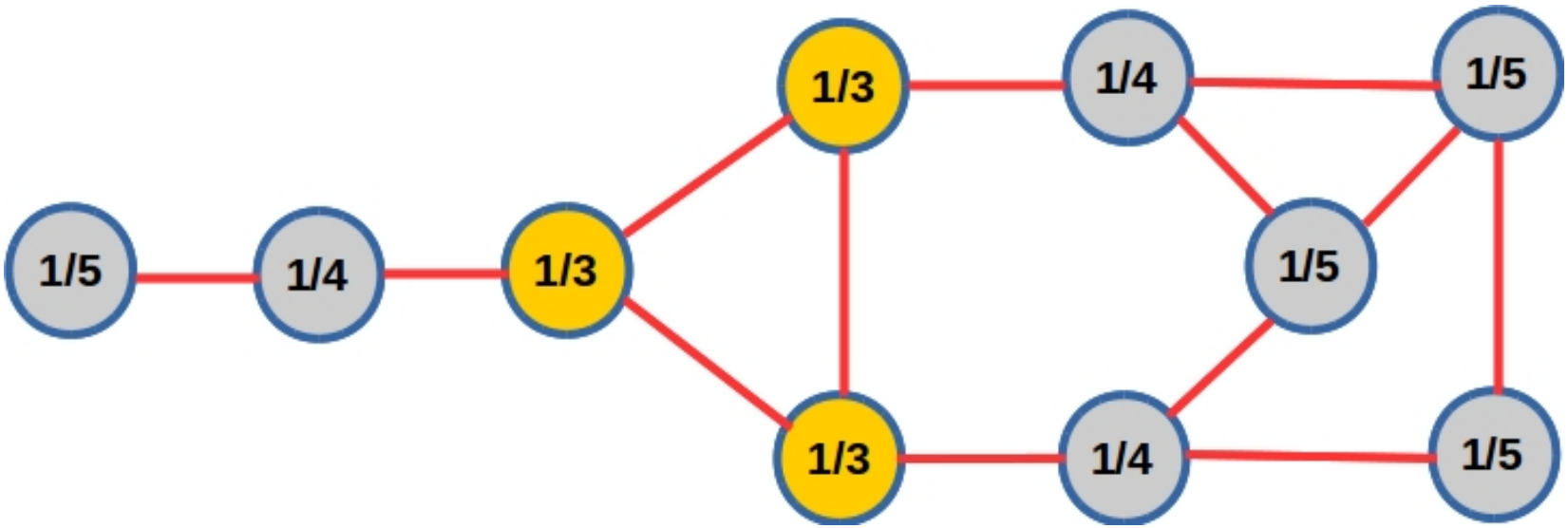
Eccentricity: Example



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Eccentricity: Example



Betweenness Centrality

Betweenness concept in *LAKSHYA* Movie



Figure 11: (Source :- Lakhsya Movie)

Betweenness Centrality

- The idea of betweenness centrality was proposed by Freeman[16].
- Mathematical notation for betweenness centrality:

$$BC(v) = \sum_{s \neq t \neq v \, \epsilon \, V} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where σ_{st} is the number of shortest paths from vertex s to vertex t and $\sigma_{st}(v)$ is the number of shortest paths between s and t that pass through vertex v.

- Normalized by dividing with $(n-1)(n-2)/2$ i.e

$$BC(v) = \frac{2}{(n-1)(n-2)} \sum_{s \neq t \neq v \, \epsilon \, V} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

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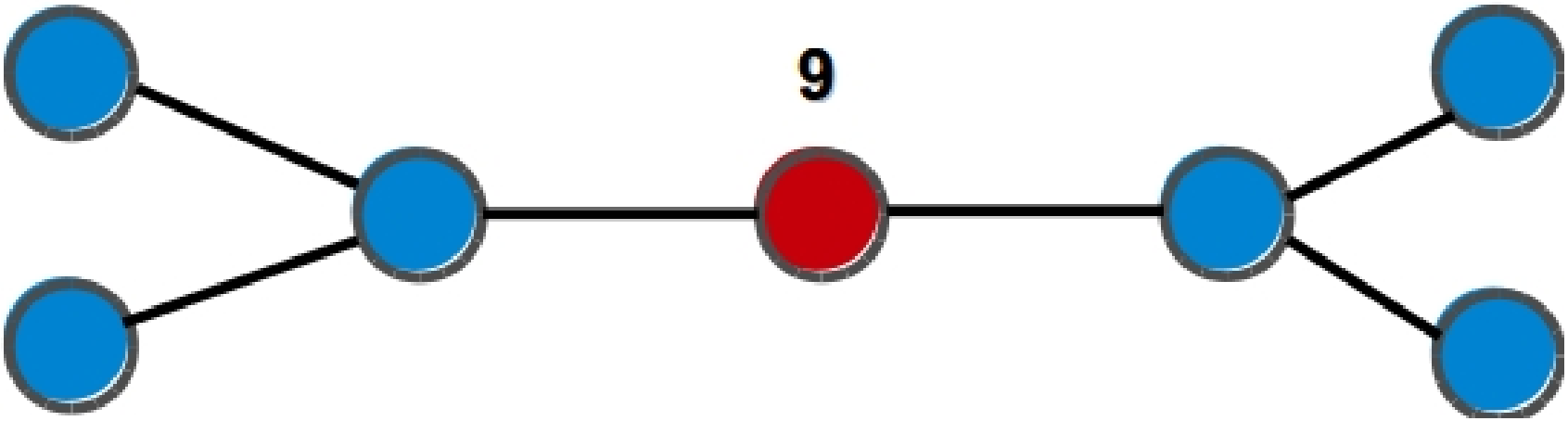
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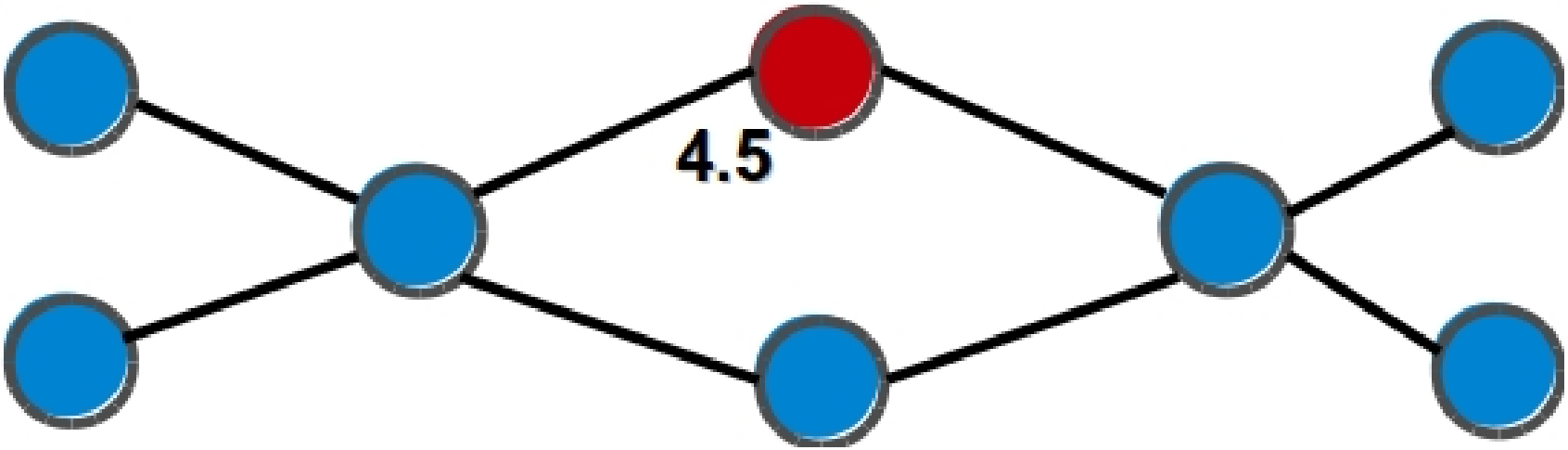
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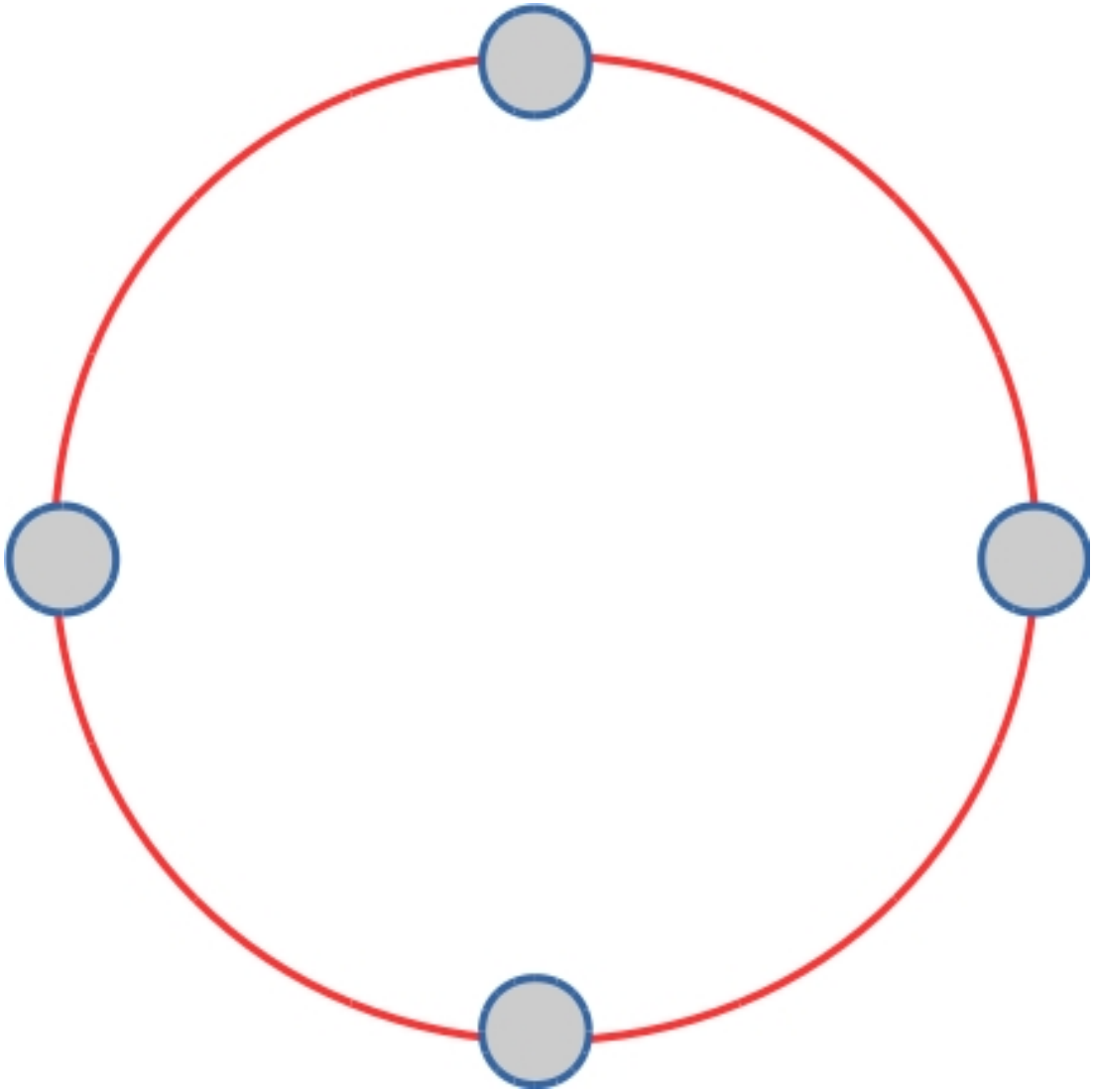
Betweenness Centrality : Example



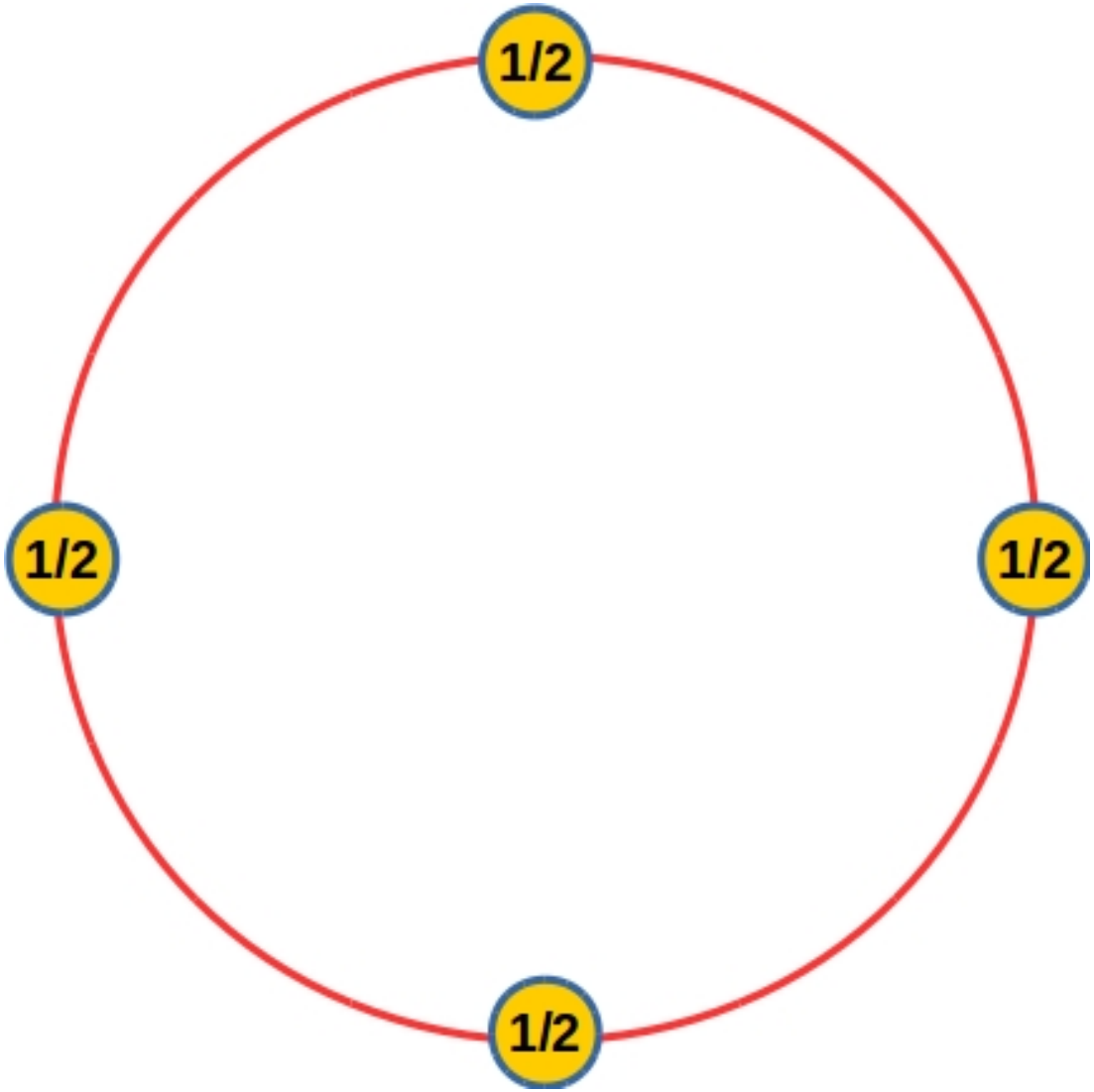
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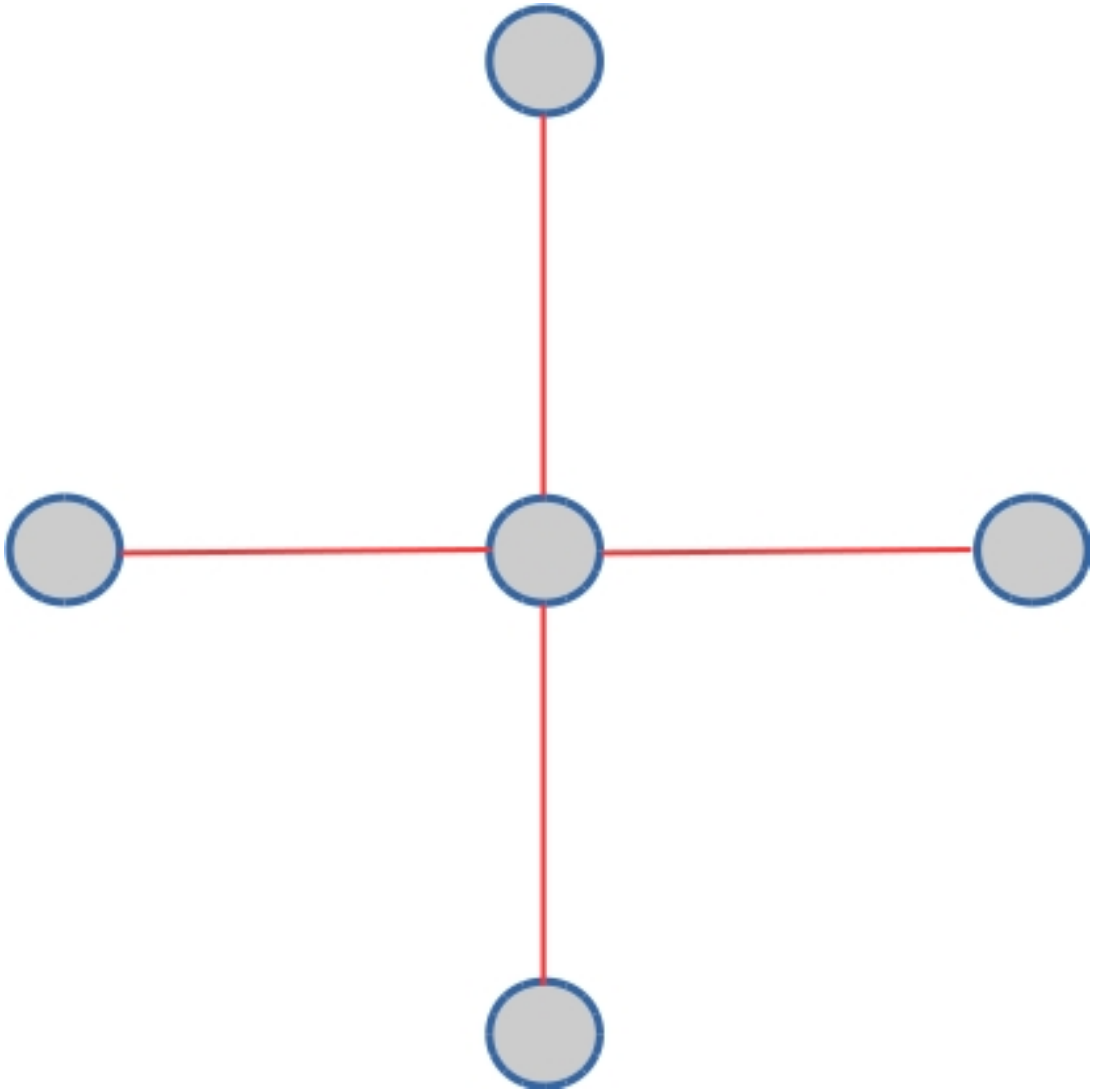
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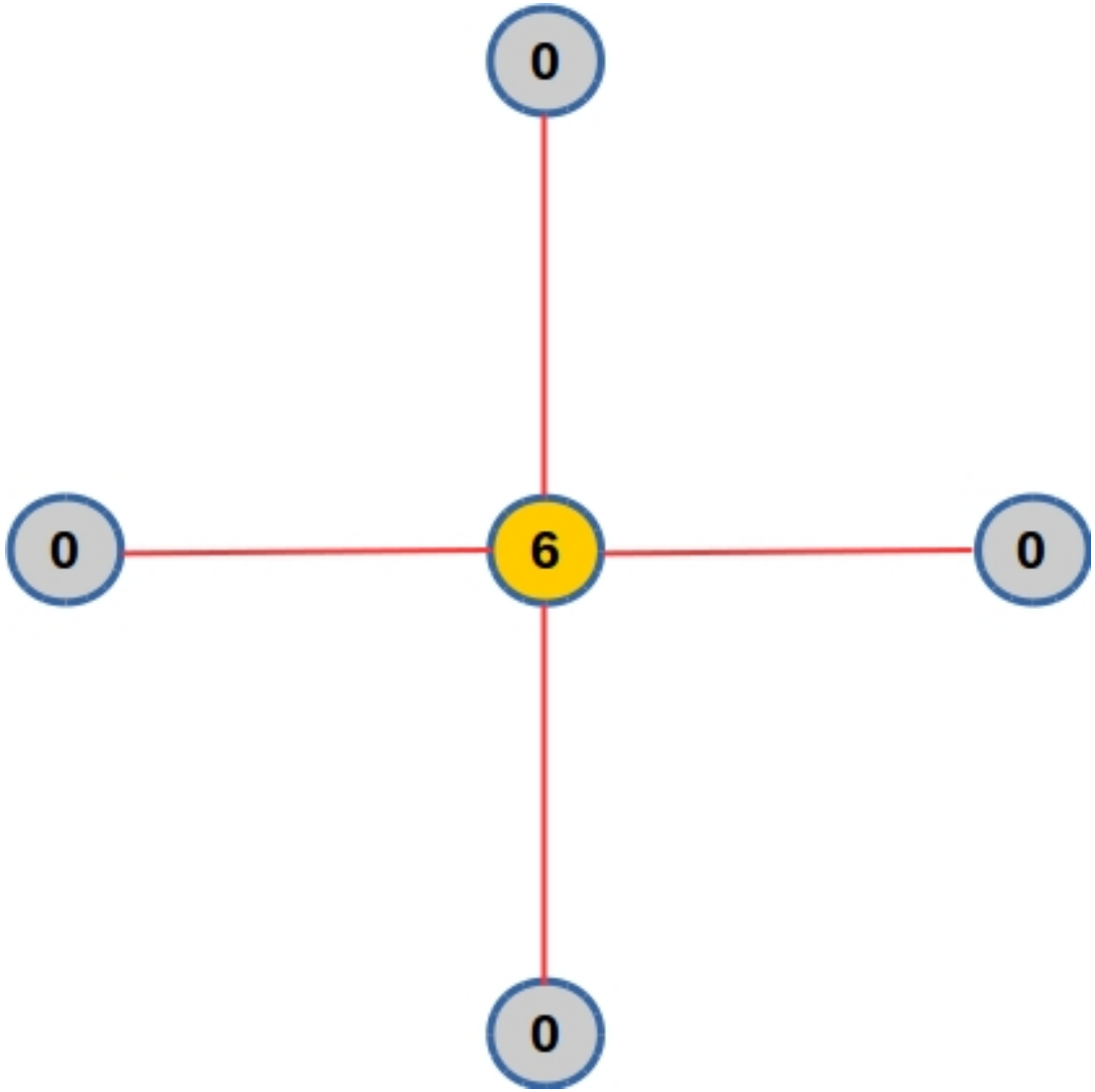
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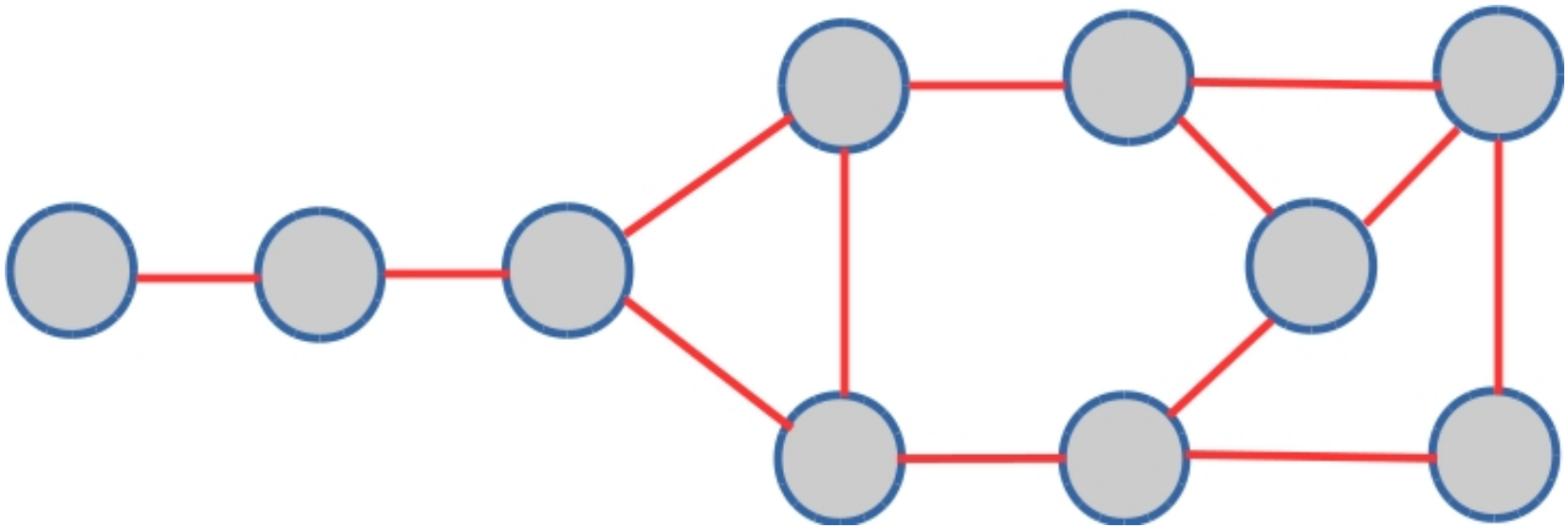
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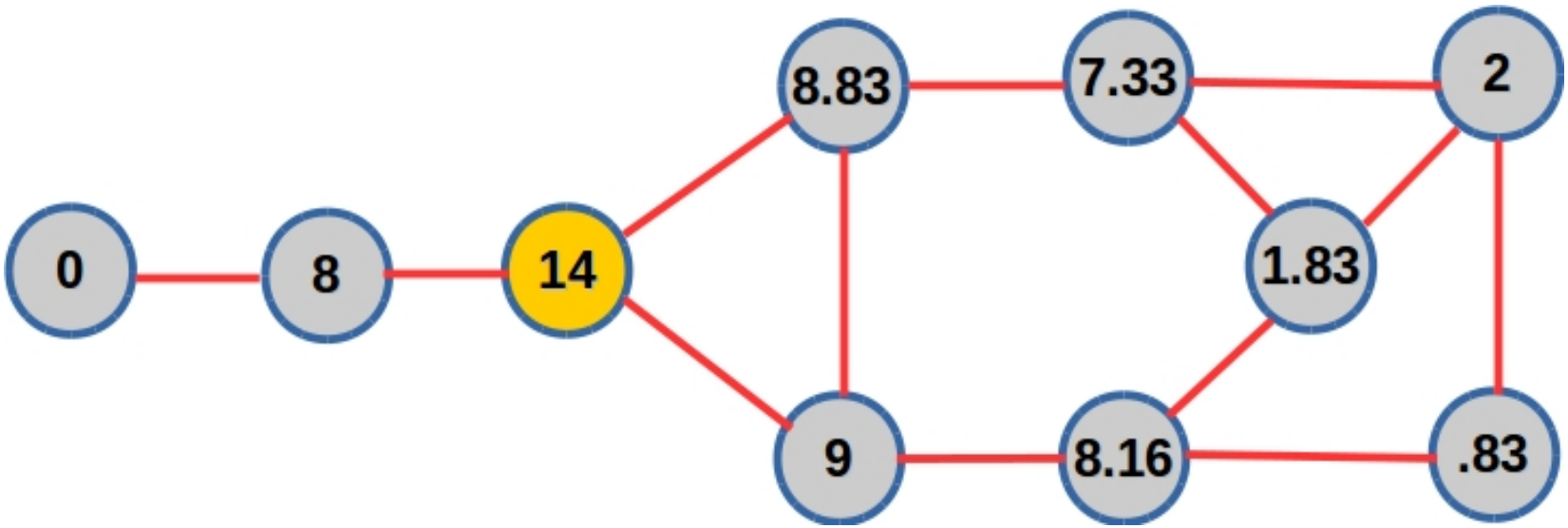
Betweenness Centrality: Example



Betweenness Centrality: Example



Betweenness Centrality: Example



Betweenness Centrality : Some Key Points

- Betweenness centrality $\mapsto$ Brokerage power, i.e. control over the flow
- Assumption: information is flowing through shortest paths
- Higher betweenness $\mapsto$ Control over major fraction of the flow
- Higher betweenness $\mapsto$ Higher load
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Application: Betweenness centrality

- Detecting communities
- Identifying sensitive nodes in
  - Electronic communication system network
  - Public transit system network
  - Gas pipeline network
  - Waste-water disposal system network etc.
- Used to identify crucial nodes for information flow in a brain network

Centralization

Centralization

- Star network is an idle example of fully centralized network.
- Freeman defined centralization of any network G in case of a centrality measure M as:

$$C_M^G = \frac{\sum_{v \in V} \left(M_{max}^G - M(v) \right)}{\sum_{v \in V} \left(M_{max}^S - M(v) \right)}$$

where
M_{max}^G: Highest M-centrality value in G
M_{max}^S: M-centrality value of the center in a star network with the same number of nodes.

Centralization

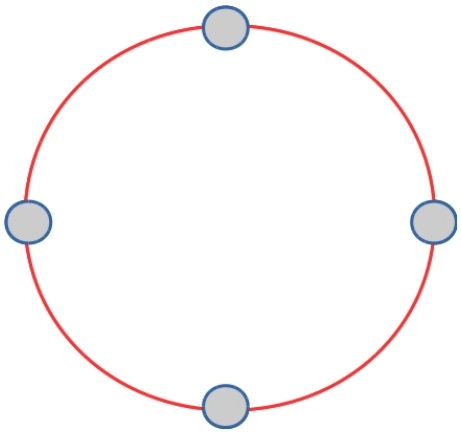
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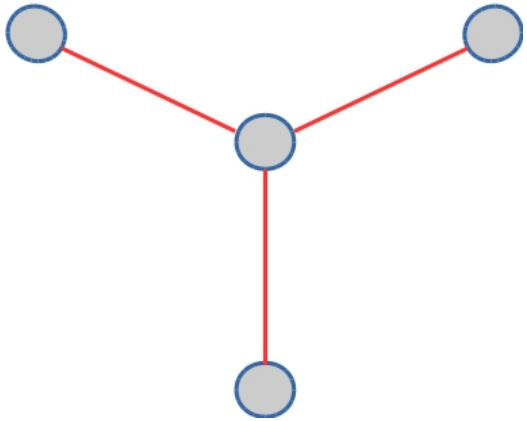
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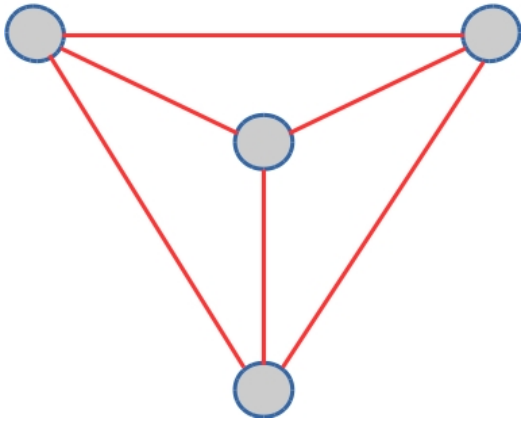
(a) Centralization=0



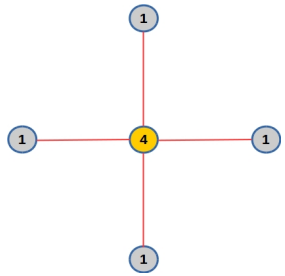
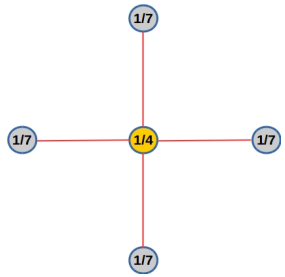
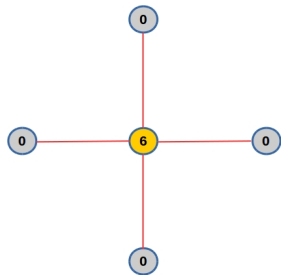
(b) Centralization=1



(c) Centralization=0



Largest Centrality Values

	Higest value	**Highest Centralization value**
Degree	$n-1$	
Closeness	$\frac{1}{n-1}$	
Betweenness	$\frac{n^2-3n+2}{2}$	

Quiz

Questions adapted from a slide by James Moody

Low Degree and High Closeness

?

Low Degree and High Closeness

Key player tied to important/active players

Low Degree and High Betweenness

?

Low Degree and High Betweenness

Node's few ties are crucial for network flow

Low Closeness and High Degree

?

Low Closeness and High Degree

Embedded in cluster that is far from the rest of the network

Low Closeness and High Betweenness

?

Low Closeness and High Betweenness

Very rare cell. Would mean that the node monopolizes the ties from a small number of people to many others.

Low Betweenness and High Degree

?

Low Betweenness and High Degree

Node's connections are redundant - communication bypasses him/her

Low Betweenness and High Closeness

?

Low Betweenness and High Closeness

Probably multiple paths in the network, Node is near many people, but so are many others

Eigenvector Centrality

- Introduces by Bonacich in 1972[17].
- Node's importance is proportional to the sum of the importance of neighboring nodes.
- Mathematically:

$$EC(u) = \frac{1}{\lambda} \sum_{v \in V \setminus \{u\}} (a_{uv} \cdot EC(v))$$

where λ is a constant.

- Popular as *Page-rank* measure

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Application:Eigenvector Centrality

- Google uses to compute the Page-rank of documents while searching
- Twitter uses to recommend users, who to follow
- More sophisticated measure (than degree) to find highly exposed nodes

Some other popular centrality measure

- Katz
- Decay
- Defusion
- h-index
- Random-walk betweenness
- Stress

Computation

Computation

- Degree Centrality
  - For one node: $O(n)$
  - For all nodes: $O(m)$
- Eccentricity, Closeness and Harmonic Centrality
  - For one node: $O(m)$
  - For all nodes: $O(mn)$

where m: Number of edges
n: Number of nodes

Computation: Betweenness Centrality

- A trivial approach:
  1. Compute table containing number of shortest paths between every pair.
  2. For each node compute the number of shortest paths passing through it.
- Time complexity analysis:
  - BFS traversal from a node: $O(m)$
  - BFS traversal from all nodes: $O(mn)$
  - For every pair (s, t), computing $\sigma_{st}(v) = \sigma_{sv} * \sigma_{vt}$: $O(n^2)$
- Overall time complexity:
  - For one node: $O(mn + n^2) = O(mn)$
  - For all nodes: $O(mn + n^3) = O(n^3)$

where m: Number of edges
n: Number of nodes

Some Basic Terminology (Brandes' Approach)

- Pair dependency of a pair of vertices (s, t) on a vertex v:

$$\delta_{st}(v) = \frac{\sigma_{st}(v)}{\sigma_{st}}.$$

- Dependency of a vertex s on vertex v is defined as:

$$\delta_{s\bullet}(v) = \sum_{t \in V} \delta_{st}(v)$$

- Betweenness centrality of a vertex v defined in terms of pair dependency and dependency as:

$$BC(v) = \sum_{s \neq v \neq t \in V} \delta_{st}(v) = \sum_{s \in V} \delta_{s\bullet}(v)$$

Terminology

- Define a set $P^s(w)$ as: $\{v : v \in V, w \text{ is a successor of } v \text{ in Breadth First SearchTree rooted at } s\}$.

Theorem 1 (Brandes' Theorem[18])

$$\delta_{s\bullet}(v) = \sum_{w:v \in P^s(w)} \frac{\sigma_{sv}}{\sigma_{sw}}(1 + \delta_{s\bullet}(w)).$$

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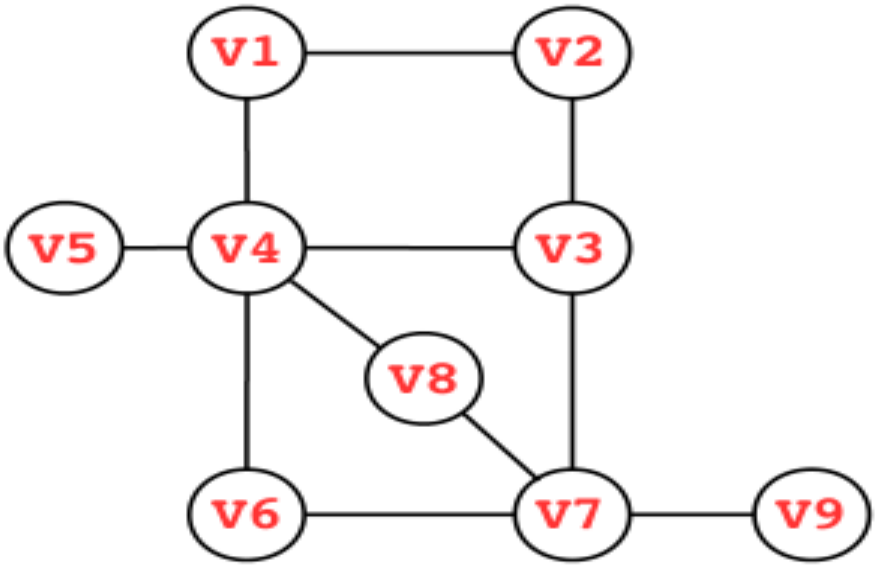


Figure 13: Graph

Figure 14: Breadth First Traversal

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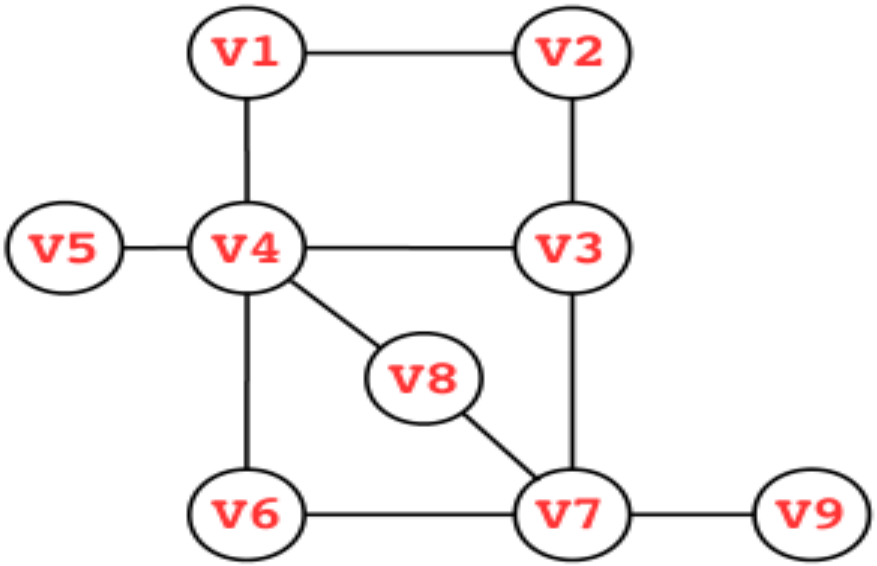


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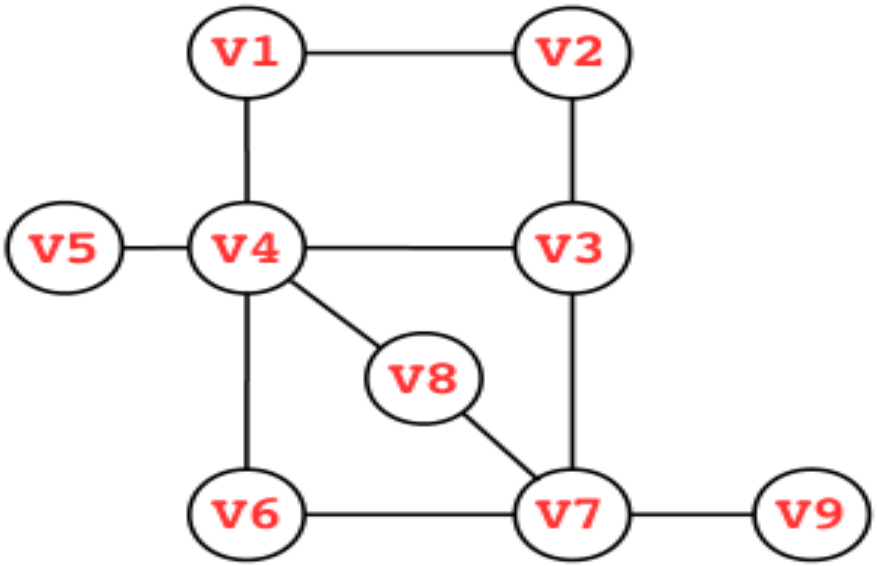


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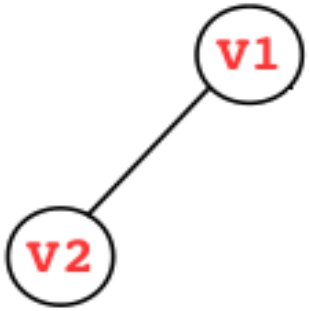


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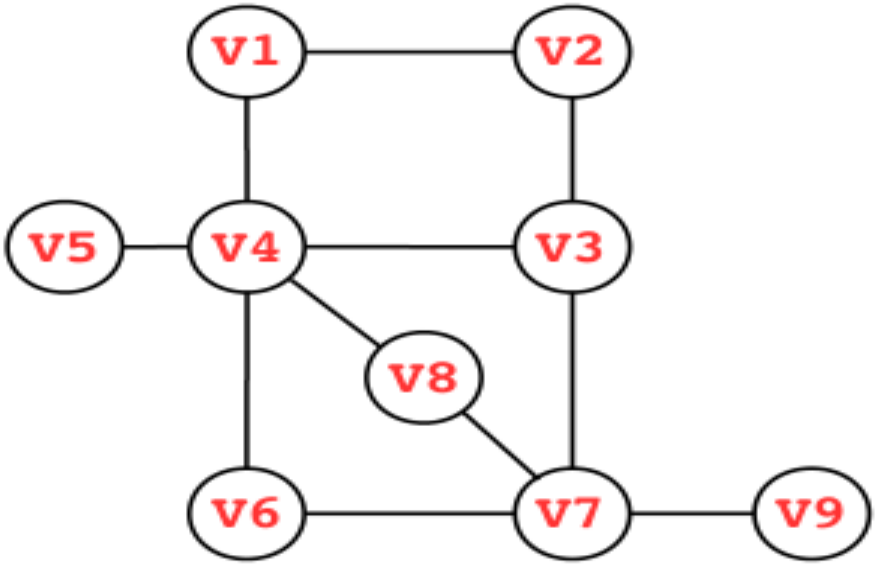


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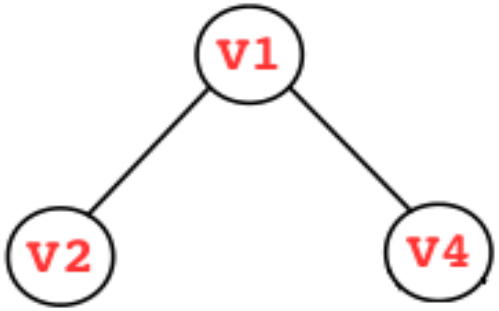


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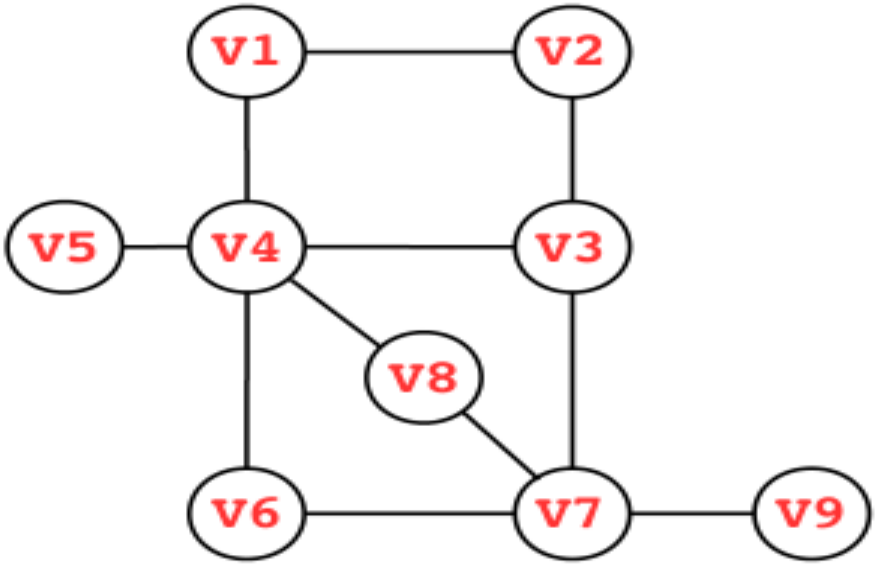


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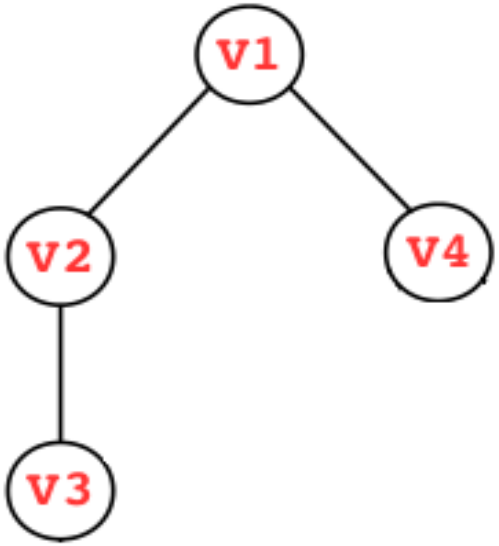


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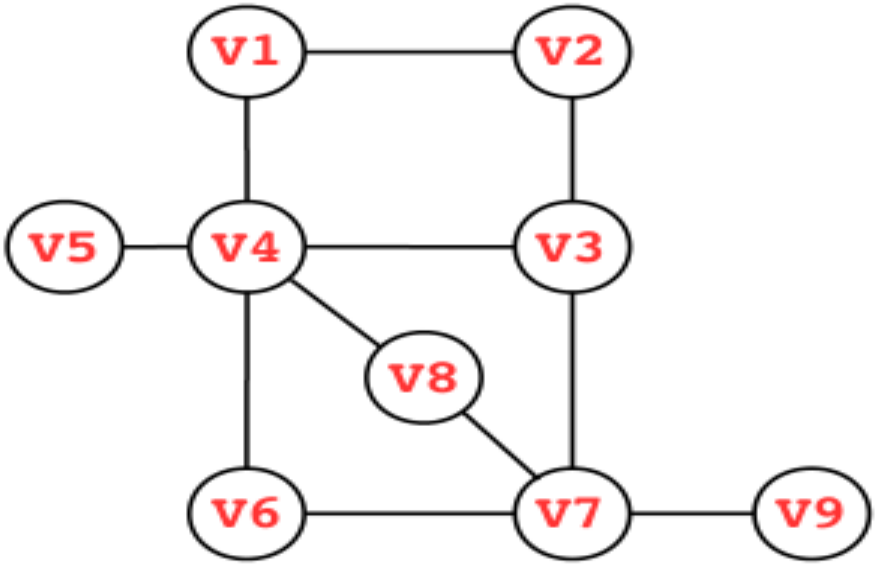


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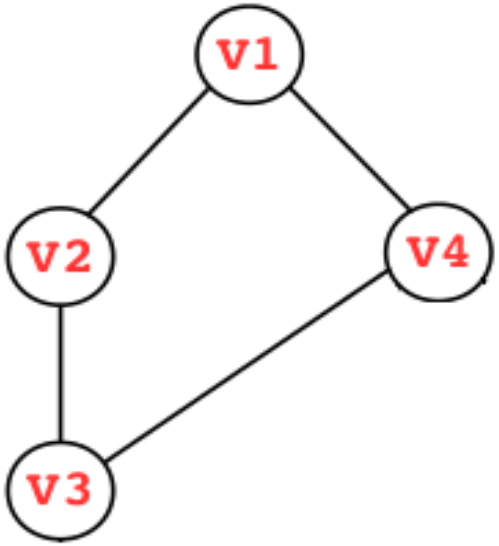


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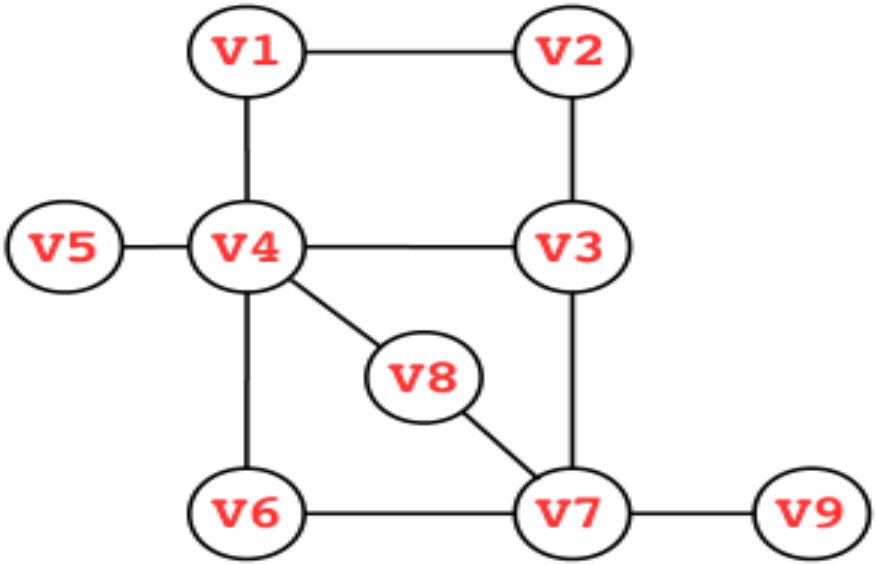


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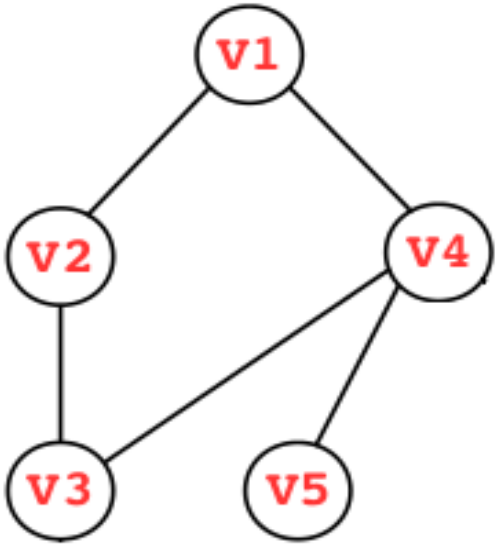


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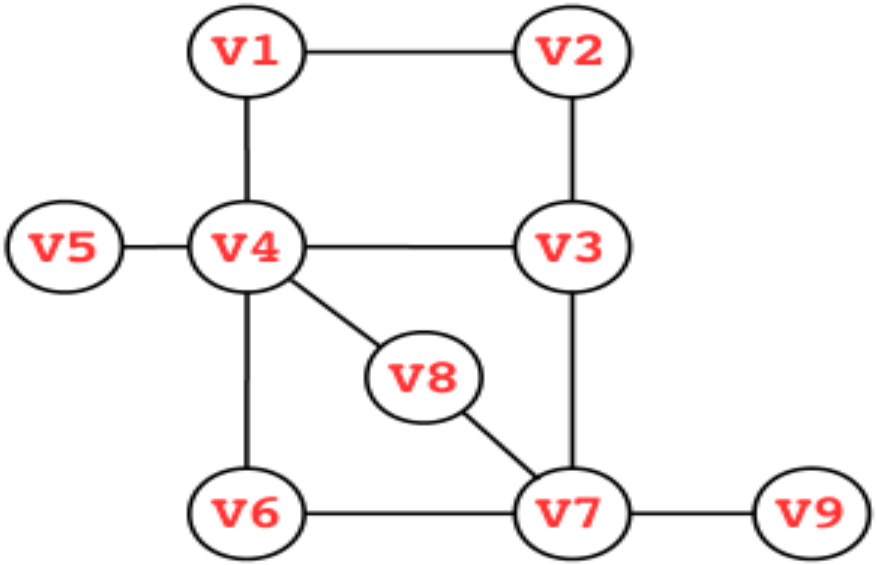


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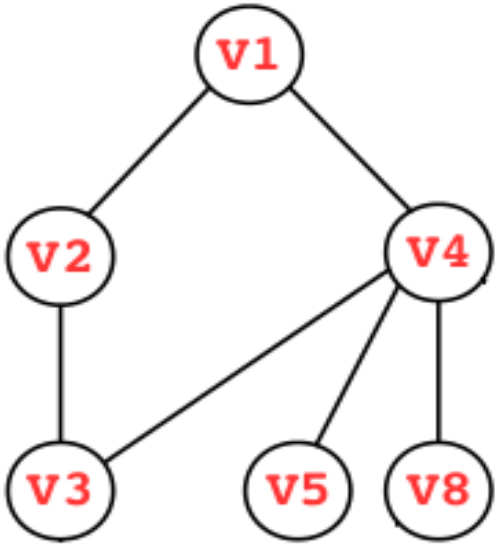


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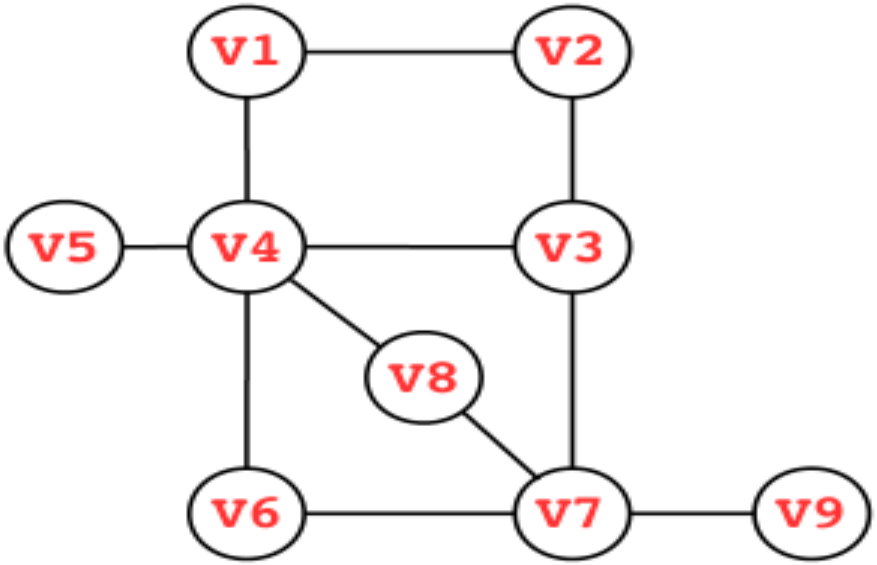


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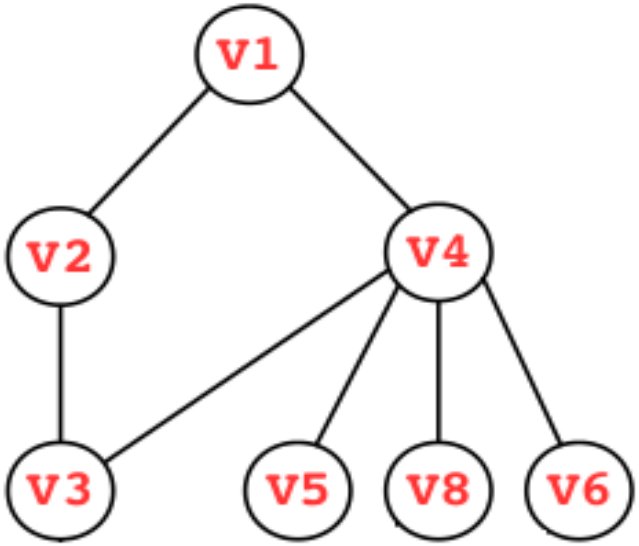


Figure 14: Breadth First Traversal

An Example Depicting Brandes' Algorithm

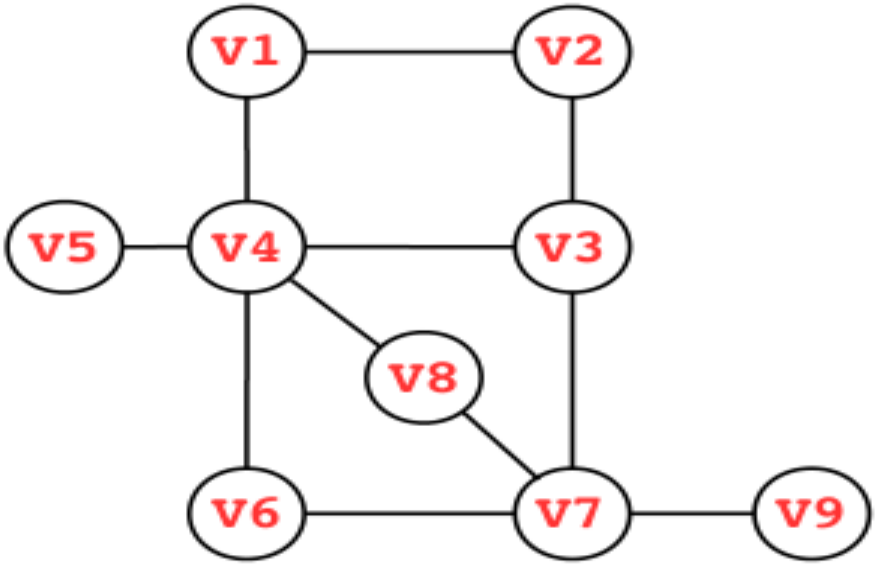


Figure 13: Graph

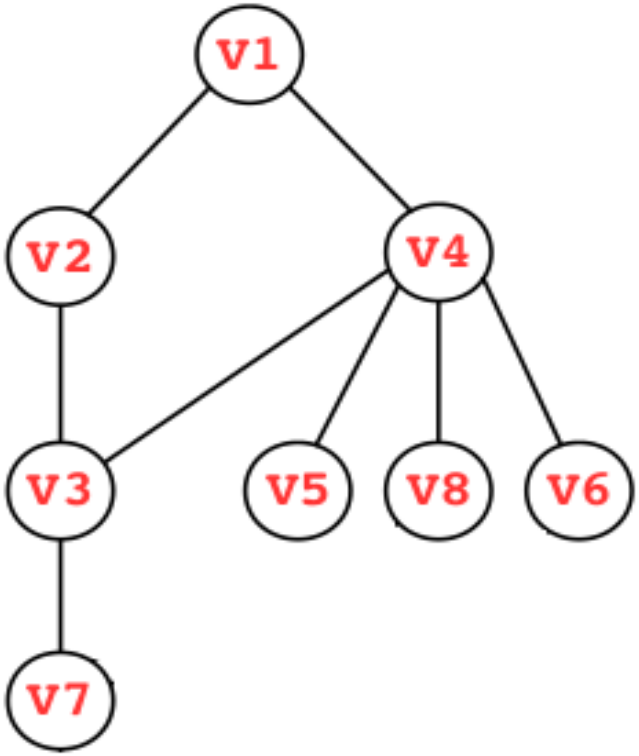


Figure 14: Breadth First Traversal

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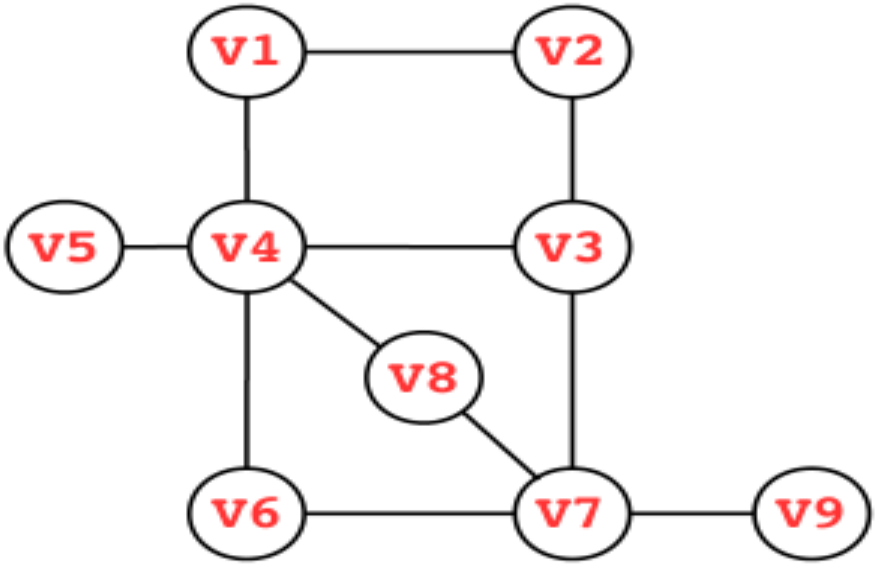


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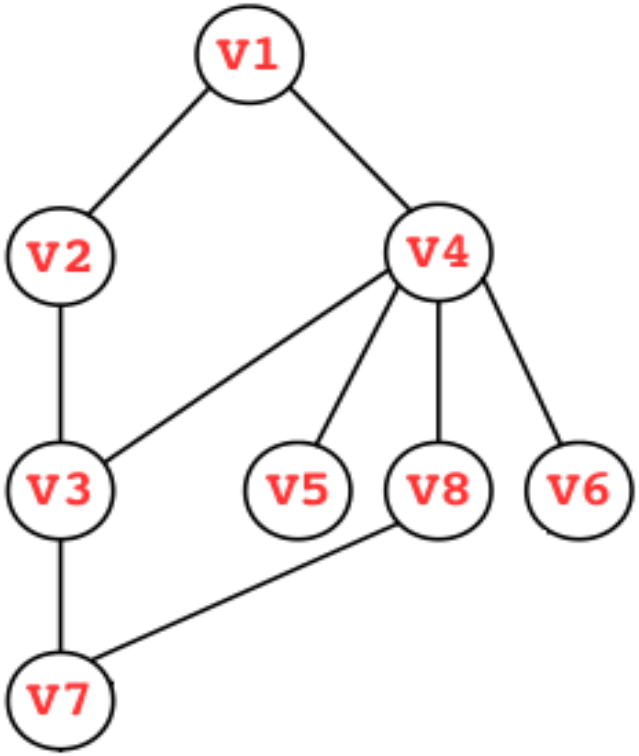


Figure 14: Breadth First Traversal

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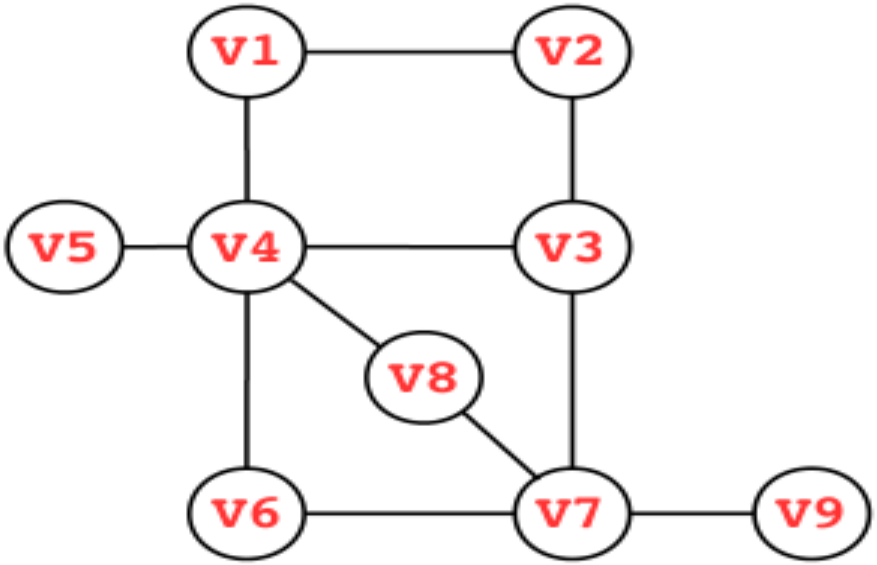


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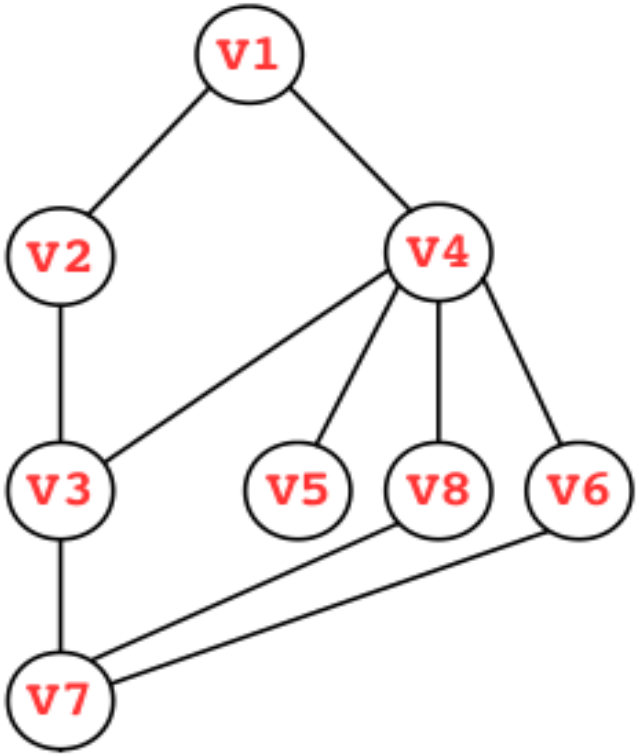


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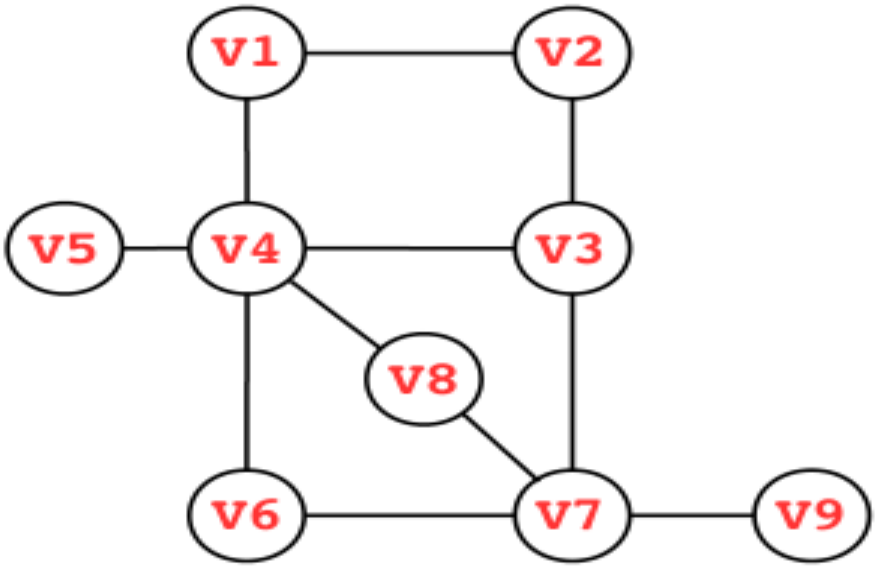


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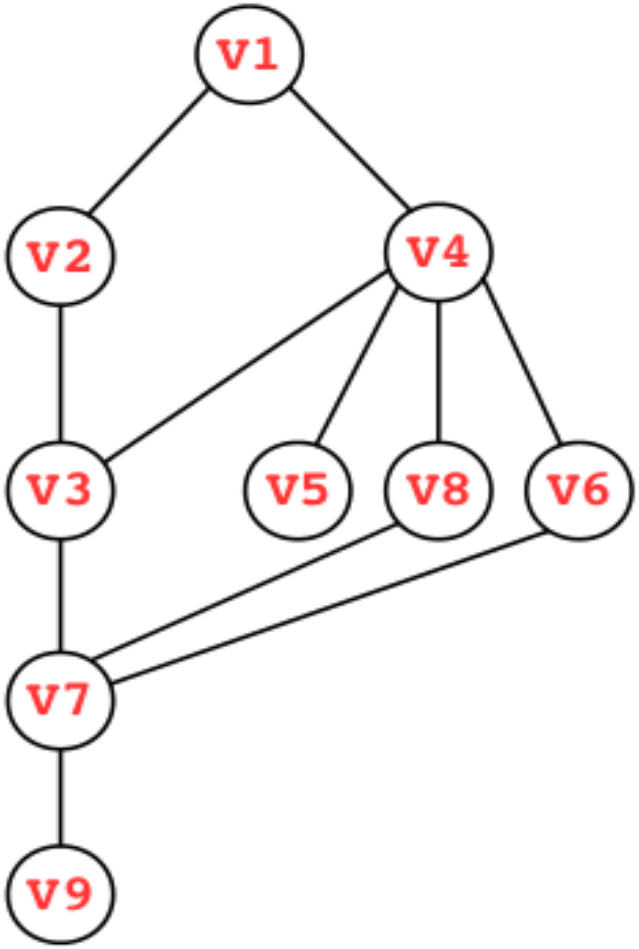


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An Example Depicting Brandes' Algorithm

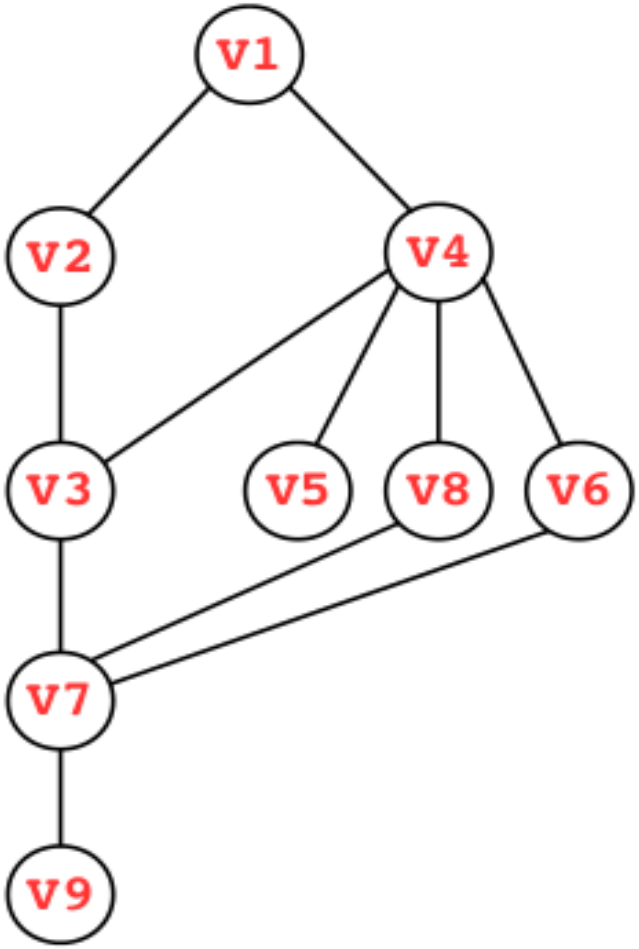


Figure 15: Breadth First Traversal

An Example Depicting Brandes' Algorithm

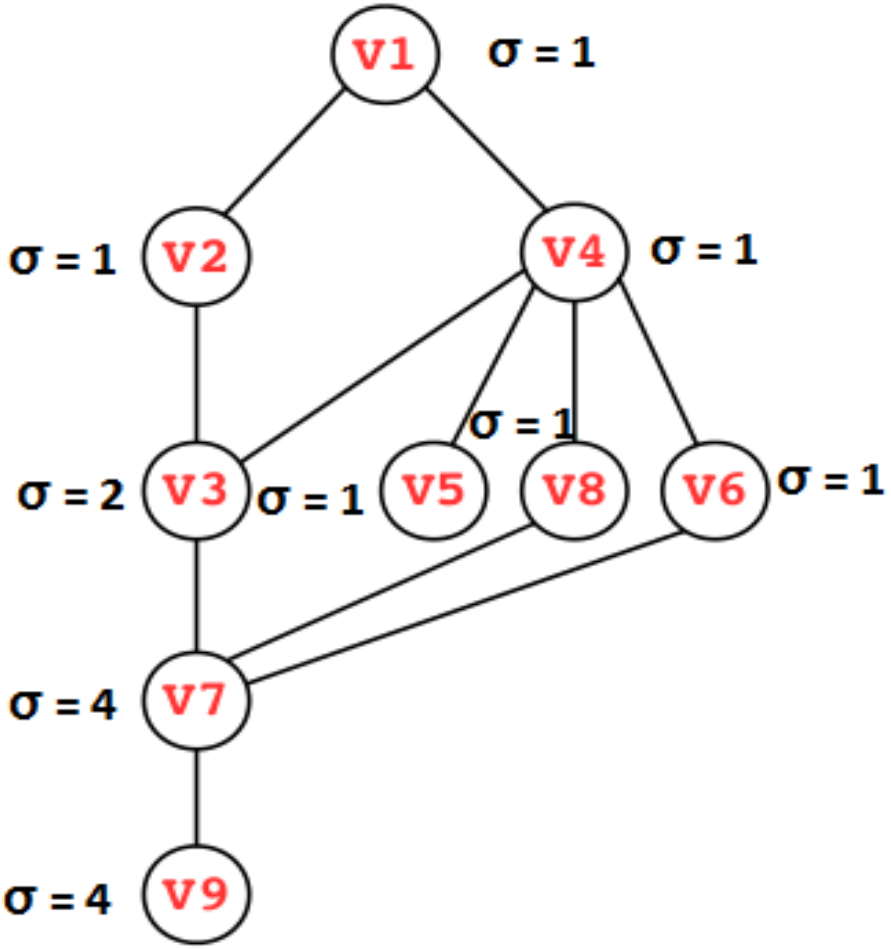


Figure 16: Breadth First Traversal

- Here, σ denotes number of shortest paths between the root (in this case $v1$) and the node under considertion.
- According to Brandes, $\delta_{s\bullet}(v) = \sum_{w:v \,\epsilon\, P(w)} \frac{\sigma_{sv}}{\sigma_{sw}}(1 + \delta_{s\bullet}(w))$

An Example Depicting Brandes' Algorithm

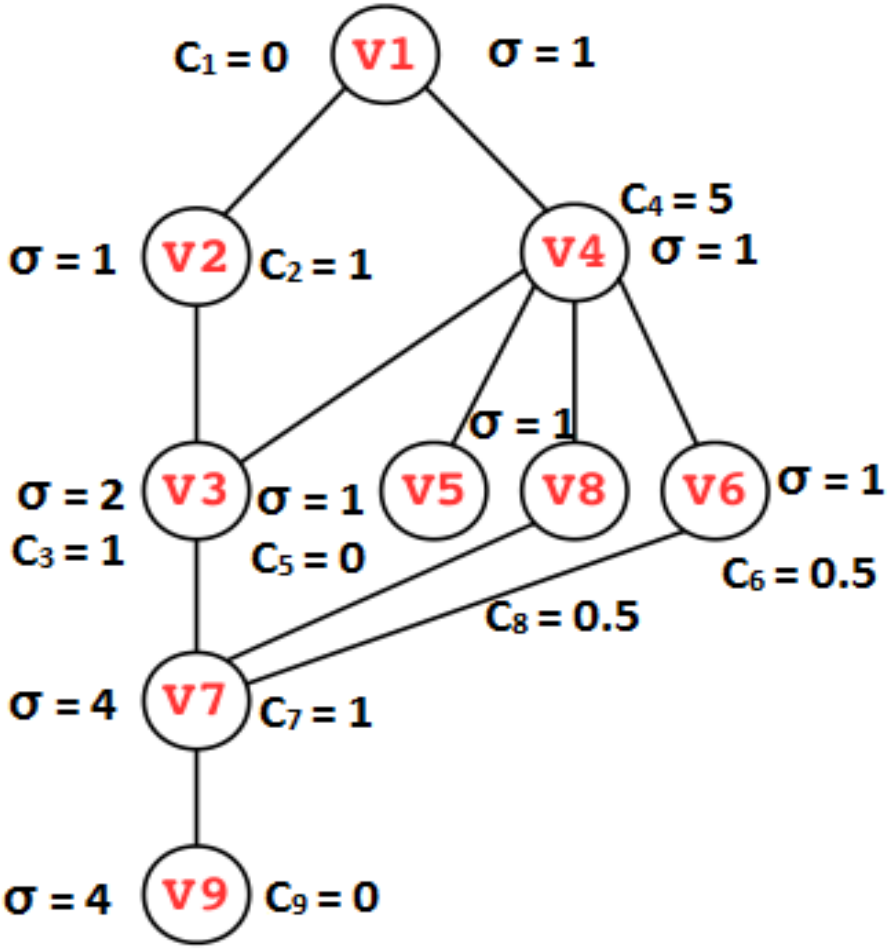


Figure 17: Breadth First Traversal

- C_v denotes the dependency $\delta_{v_1\bullet}(v)$ of the root v_1 on the vertex v.
- According to Brandes, $\delta_{v_1\bullet}(v) = \sum_{w:v \in P(w)} \frac{\sigma_{v_1 v}}{\sigma_{v_1 w}}(1 + \delta_{v_1\bullet}(w))$
- $BC(v) = \sum_{s \in V} \delta_{s\bullet}(v)$

An Example Depicting Brandes' Algorithm

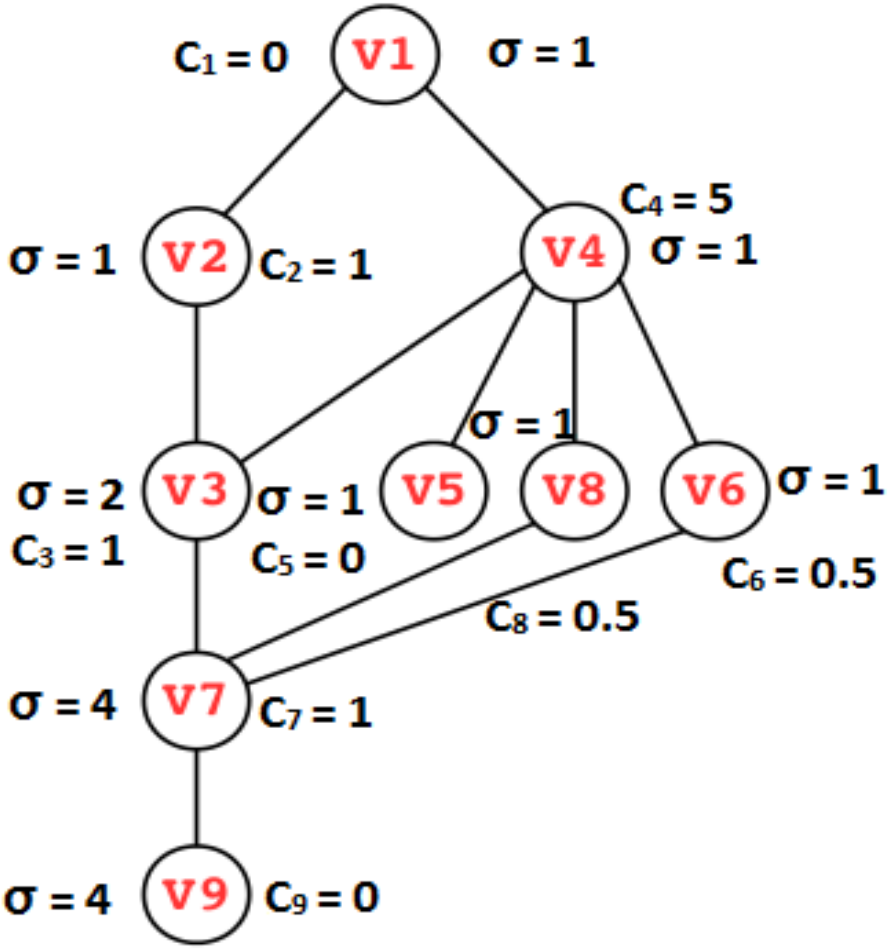


Figure 18: Breadth First Traversal

Vertex v	1
1	0
2	1
3	1
4	5
5	0
6	0.5
7	1
8	0.5
9	0

An Example Depicting Brandes' Algorithm

Vertex v	1	2	3	4	5	6	7	8	9	CB(v)
1	0	1.67	0	0.5	0.5	0.33	0	0.33	0	3.33
2	1	0	0.5	0	0	0	0.25	0	0.25	2
3	1	4.33	0	1.167	1.167	0.667	2.16	0.667	2.1667	13.334
4	5	2.33	2.5	0	7	3.667	1.75	3.667	1.75	27.667
5	0	0	0	0	0	0	0	0	0	0
6	0.5	0	0	0.667	0.667	0	0.91667	0	0.91667	3.6667
7	1	1.667	2	1	1	2.334	0	2.334	7	18.334
8	0.5	0	0	0.667	0.667	0	0.91667	0	0.91667	3.667
9	0	0	0	0	0	0	0	0	0	0

Why Computation of BC is a daunting task?

1. Global characteristics of betweenness centrality measure
   - For one node: $O(mn)$
   - For all nodes: $O(mn)$
2. Large size and the dynamic nature of networks

Power Method: Eigenvector centrality computation

Produces good approximation of the dominant eigenvector.

1. Pick a starting vector x_0.
2. Compute

$$x_{i+1} = Ax_i$$

for $i = 1, 2, \ldots$. until for some positive integer j and some real constant c, $x_{j+1} = cx_j$

- Scaling in each iteration is recommended.
- Rate of convergence is proportional to the ratio between the dominant eigenvalue and second dominant eigenvalue.

Estimating

Related Work (Centrality Estimation)

- Several approximation algorithms have been developed by.
  - Eppstein and Wang[19]
  - Brandes and Pich[20]
  - Bader et al.[21]
  - Gkorou et al.[22]
  - Riondato and Kornaropoulos[23]
- Among them, most are based on **uniform sampling technique**.
- **Node sampling** and **Path sampling** based techniques.

---

[19] David Eppstein and Joseph Wang, Fast approximation of centrality, J. Graph Algorithms Appl., 2004

[20] U. Brandes and C. Pich, *Centrality estimation in large networks*, 2007.

[21] D. A. Bader, S. Kintali, K. Madduri, and M. Mihail,*Approximating betweenness centrality*, 2007.

[22] D. Gkorou, J. Pouwelse, D. Epema, T. Kielmann, M. van Kreveld, and W. Niessen, *Efficient approximate computation of betweenness centrality*, 2010.

[23] M. Riondato and E. M. Kornaropoulos, *Fast approximation of be- tweenness centrality through sampling*, 2014.

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Updating

Updating Centrality Measures

- Nature of update:
  - Edge addition or deletion
  - Vertex addition or deletion
  - Increase or decrease in edge weight
- Types of algorithms:
  - Incremental
  - Decremental
  - Fully Dynamic

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  - Incremental
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Related Work: Updating Centrality

- Closeness:
  - Sariyüce et al. 2013[24]
  - Yen et al. 2013[25]
- Betweenness
  - Lee et al. 2012[26]
  - Green et al. 2012[27]
  - Kas et al. 2013[28]
  - Nasre et al, 2014 [29]
- Closeness and Betweenness
  - Khopkar et al. 2014[30]
  - Wei and Carley 2014[31]

[24] Incremental algorithms for closeness centrality, Big Data, 2013.

[25] "An efficient approach to updating closeness centrality and average path length in dynamic networks, 13th International Conference on Data Mining, 2013.

[26] QUBE: A quick algorithm for updating betweenness centrality, WWW, 2012

[27] A fast algorithm for streaming betweenness centrality, Privacy, Security, Risk and Trust (PASSAT), 2012

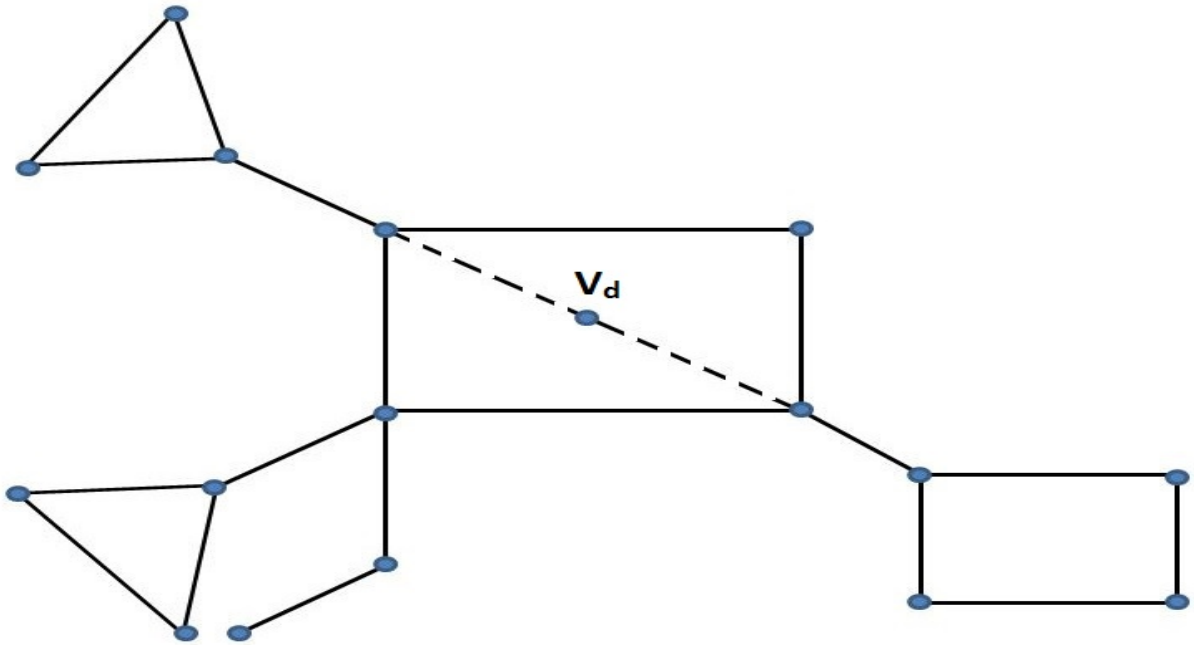
[28] Incremental algorithm for updating betweenness centrality in dynamically growing networks,ASONAM, 2013

[29] Betweenness centrality - incremental and faster, MFCS, 2014

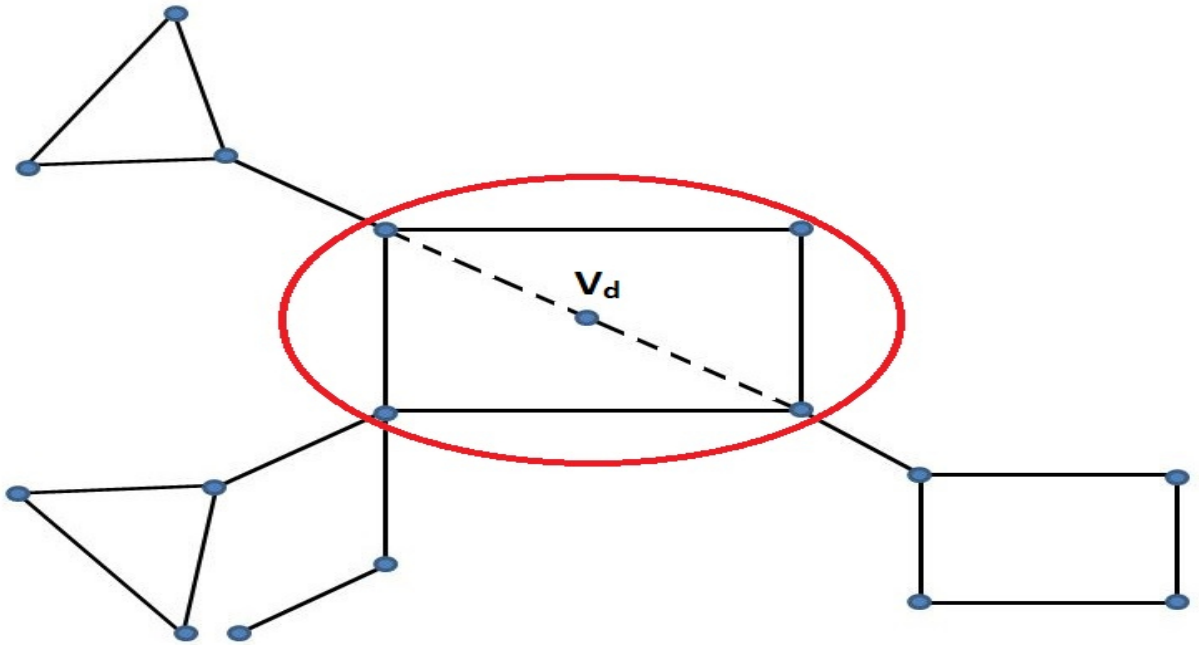
[30] Efficient algorithms for incremental all pairs shortest paths, closeness and betweenness in social network analysis, Social Network Analysis and Mining, 2014.

[31] Real time closeness and betweenness centrality calculations on streaming network data, ASE, 2014.

One Approach : [Lee et al. 2012]



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Ordering

Centrality-Ordering and Betweenness-Ordering

- *Compute between two nodes, which one is more central than the other*

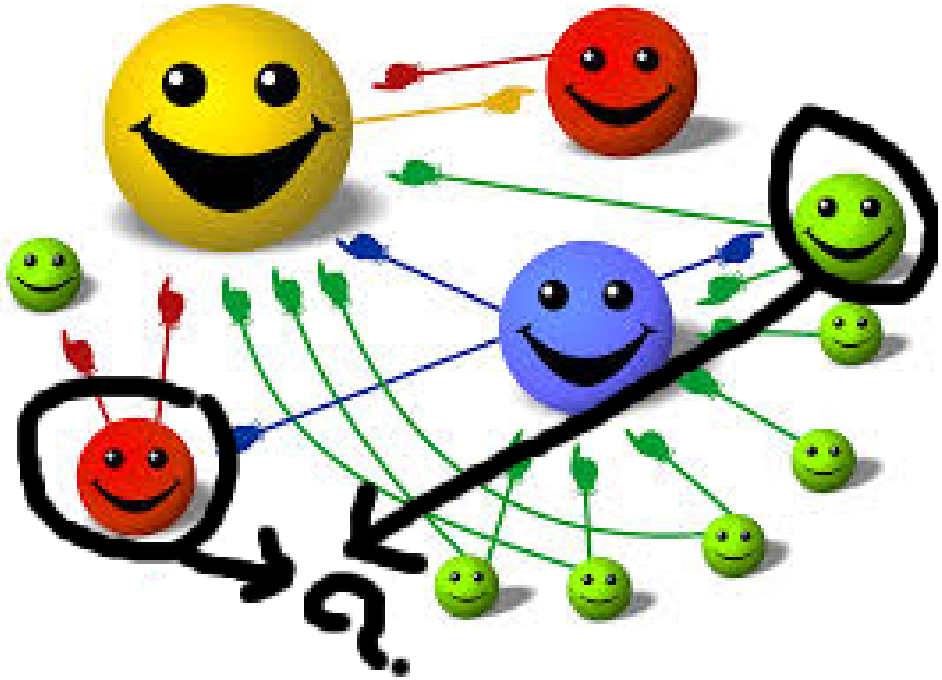
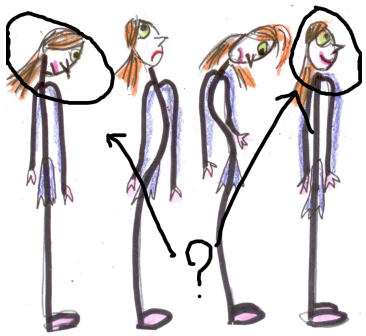


Figure 19: (Source :- Google Search)[32]

[32] https://en.wikipedia.org/wiki/PageRank

Three Different Approaches

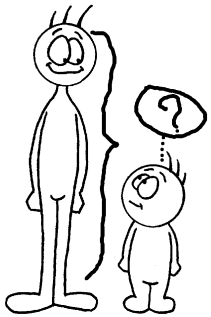
(a) Who is taller?



(b) 1: Measure their heights and compare



(c) 2: Guess their height and compare



(d) 3: Make them stand together and compare



Figure 20: An Analogy : Height Ordering(Source :- Google Search)

One Approach for Betweenness Ordering

- Efficiently estimates betweenness of given node using their algorithm based on a non-uniform sampling model[33].
- Estimation algorithm was based on a probabilistic analysis of random graphs generated by Erdos and Renyi's G(n,p) model[34].

[33] Rishi Ranjan Singh, Manas Agarwal, Shubham Chaudhary, Sudarshan Iyengar, An Efficient Heuristic for Betweenness-Ordering, Social Networks (under review)

[34] P. Erdos and A. Renyi, "On random graphs i." Publ. Math. Debrecen, 1959.

Average Iterative Ordering Performance on Real World Graphs using our approach

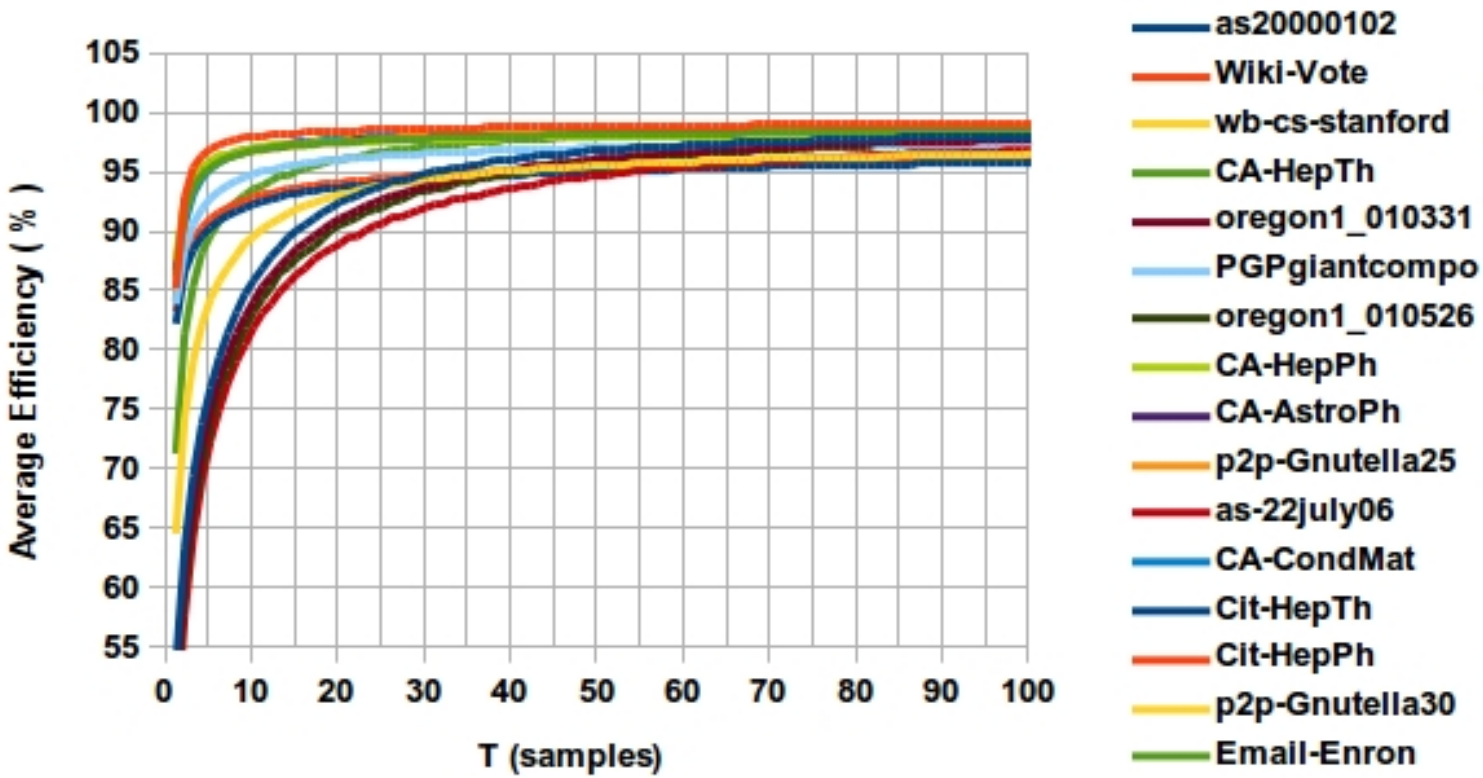


Figure 21: Average Iterative Ordering Performance on Real World Graphs[35]

[35] http://snap.stanford.edu/data/

Weighted Centrality

- Opsahl et al.[36], Barret et al.[37] extended the centrality measures to edge weighted networks.
- Degree Centrality is extended to Weighted degree
- To compute Closeness and Betweenness, weighted shortest paths are considered.

[36] Node centrality in weighted networks: Generalizing degree and shortest paths, Social Networks, 2010

[37] The architecture of complex weighted networks, PNAS, 2004

Hybridization

- Abbasi and Hossain[38] proposed a new set of hybrid centralities; Degree-Degree, Degree Closeness and Degree Betweenness.
- Zhang et al.[39] proposed a new hybrid centrality measure to measure a node's importance in satellite communication network based on the combination of closeness and betweenness centrality.
- Qiao et al.[40] proposed a page scoring algorithm based on hybridizing degree centrality, betweenness centrality, closeness centrality and the PageRank algorithm.
- Lee and Djauhari [41] proposed to take a linear combination of the degree, closeness, betweenness, and eigenvector centrality measures to find the most influential stocks.
- Wang et al.[42] proposed a hybrid centrality measure based on the combination of degree, degree of neighbours and betweenness.
- Buechel and Buskens[43] considered a hybrid centrality model as the combination of extended closeness (a variation of closeness for disconnected networks), betweenness and degree measures.

---

[38] Hybrid centrality measures for binary and weighted networks, Complex networks, Springer, 2013.

[39] Finding most vital node in satellite communication network, Applied Mechanics and Materials, 2014.

[40] Webrank: a hybrid page scoring approach based on social network analysis, Rough Set and Knowledge Technology, 2010.

[41] An overall centrality measure: The case of US stock market, International Journal of Electrical & Computer Sciences, 2012.

[42] A new measure of node importance in complex networks with tunable parameters, Wireless Communications, Networking and Mobile Computing, 2008.

[43] The dynamics of closeness and betweenness, The Journal of Mathematical Sociology, 2013.

Group Centrality

Group Centrality

- Everett and Borgatti. [44]
- Concept of centrality is extended to group of nodes.
- Can be used to find groups that can play important role in an organization.

[44] Extending centrality, Models and methods in social network analysis, 2005.

New Applications of existing measures

Defining New Measures

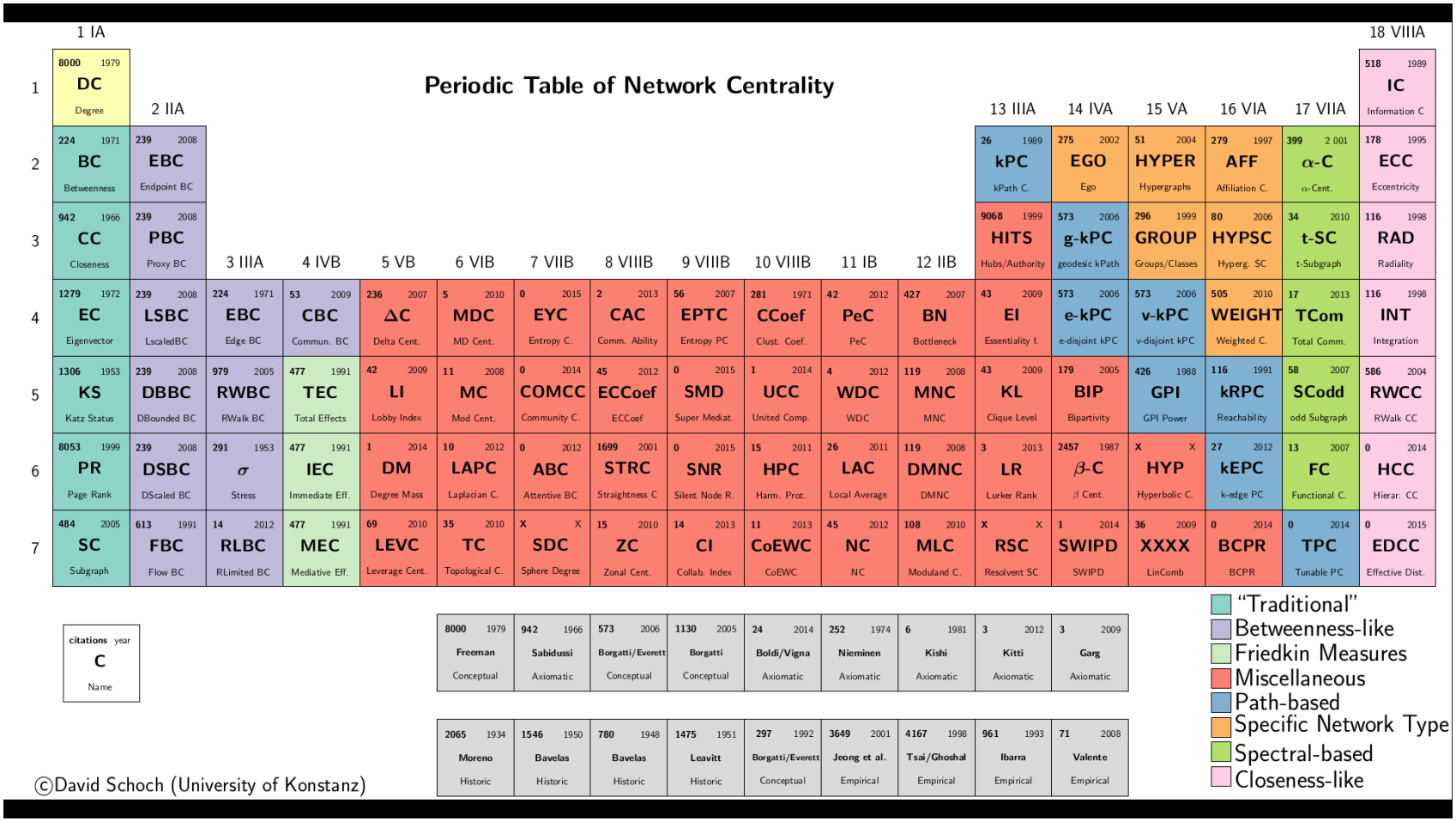


Figure 22: Interactive Version: http://schochastics.net/sna/periodic.html


Thank You!


Any
Questions